

The Metric Blind Spot in Deep Learning-Based Brain MRI Reconstruction: A Systematic Review

Dat Tat Mai^{a,*}, Thai Viet Pham^b, Thu Nguyen Thi Dang^c, James Jin Kang^a

^a*School of Science, Engineering & Technology, RMIT University Vietnam, Ho Chi Minh City, Vietnam*

^b*School of Computing Technologies, RMIT University, Melbourne, VIC, 3000, Australia*

^c*School of Health and Biomedical Science, RMIT University, Melbourne, VIC, 3000, Australia*

Abstract

Objective: Deep learning accelerates brain MRI four- to tenfold, but learned models can erase a lesion or invent tissue, and pixel-averaged scores such as PSNR and SSIM miss both. We review whether evaluation practice could detect this blind spot. **Methods:** Following PRISMA 2020 we searched seven databases without date limit, included 263 studies (1995–2026), appraised them with QUADAS-2 and matched instruments, and synthesised narratively. Categories came from titles, abstracts and controlled vocabulary; every prevalence figure is a floor. Duplicate screening achieved substantial agreement (Fleiss’ $\kappa = 0.877$); appraisal agreement was substantial (0.788; 0.390 where observable); extraction is unaudited. **Results:** Only 18 of 263 studies (6.8%) record a fidelity metric and a reader assessment on the same data, leaving the field’s central proxy unmeasured. Reader studies mostly measure reader-versus-reader agreement, which is weak: fastMRI 2020 concordance was 0.457 and 0.386 (Kendall’s W), improving only where SSIM had diverged. Erasing a 100 mm³ lacunar infarct moves global PSNR about 0.03 dB under the stated error model. The corpus grew fivefold; reader assessment fell from 32% to 18%, recovering to 21%. Among the four largest families the most hallucination-associated (generative, 39%) is among the least reader-evaluated (11.3%); self-supervised is higher still (47%) with no readers. One study in twenty both releases code and runs readers; none evaluated a model observer; no named dataset covers acute stroke or

*Corresponding author.

Email addresses: `dat.mai2@rmit.edu.vn` (Dat Tat Mai), `s4229249@student.rmit.edu.au` (Thai Viet Pham), `s4205224@student.rmit.edu.au` (Thu Nguyen Thi Dang), `james.kang@rmit.edu.vn` (James Jin Kang)

25 haemorrhage. **Conclusions:** On these floors, evaluation practice cannot certify
26 diagnostic safety. We derive five requirements safety-oriented evaluation must
27 meet.

28 *Keywords:* Deep learning, Brain MRI reconstruction, Reconstruction safety,
29 Hallucination, Image quality assessment, Systematic review

30 1. Introduction

31 Brain MRI is widely used because it can clearly distinguish different types
32 of brain tissue and reveal subtle abnormalities that are difficult to detect with
33 other imaging modalities. However, this image quality comes at the cost of long
34 acquisition times. Unlike conventional imaging methods that capture an image
35 more directly, MRI builds an image from a sequence of raw signal measure-
36 ments (samples of spatial frequencies, known as k -space) collected over time.
37 This prolonged acquisition reduces patient throughput and increases sensitivity
38 to patient motion, which can introduce artifacts and degrade image quality (Za-
39 itsev et al., 2015). To address these limitations, acceleration techniques such as
40 parallel imaging, including Sensitivity Encoding (SENSE) (Pruessmann et al.,
41 1999), and compressed sensing (Lustig et al., 2007) were developed to reduce
42 the amount of data that must be collected. More recently, deep learning has
43 enabled further acceleration by reconstructing high-quality images from incom-
44 plete MRI measurements, through cascaded convolutional networks (Schlemper
45 et al., 2017), model-based unrolled networks (Hammernik et al., 2018; Aggar-
46 wal et al., 2018), generative adversarial networks (Yang et al., 2018), and score-
47 based diffusion models (Song et al., 2021), at accelerations from fourfold to, in
48 exploratory work, an order of magnitude (Radmanesh et al., 2022).

49 The reconstructed images look convincing, and that is the problem. Small
50 changes in the input can produce large errors in the output (Antun et al., 2020;
51 Gottschling et al., 2025). Four failure types recur across the studies we re-
52 viewed: strong priors smooth away faint structure; motion causes ghosting the
53 model never saw in training (Sommer et al., 2020); a change of scanner or sam-

54 pling degrades results in ways that in-distribution testing misses (Güngör et al.,
 55 2023); and generative models can invent plausible tissue (Zhou et al., 2024).
 56 The scores used to certify these methods, peak signal-to-noise ratio (PSNR)
 57 and structural similarity (SSIM) (Wang et al., 2004), agree only partially with
 58 radiologist quality judgement even on conventional image degradations (Mason
 59 et al., 2020); and SNR, though reproducible across observers, tracks no-reference
 60 quality models only moderately (Yu et al., 2018), so an image can score well
 61 while the error that matters goes unrecorded. We call this the **metric blind**
 62 **spot**: standard image-quality metrics cannot catch these small but clinically se-
 63 rious errors, because a lesion of a few cubic millimetres barely moves an average
 64 taken over millions of voxels.

65 Parts of the field have responded, training quality metrics on radiologist
 66 rankings (Tang et al., 2025), attaching uncertainty maps to reconstructions
 67 (Khawaled and Freiman, 2024), or running reader studies that test an acceler-
 68 ated protocol against a reference one directly (Lang et al., 2024; Radmanesh
 69 et al., 2022). Causal and counterfactual methods can ask the most direct ques-
 70 tion, whether a lesion survives a controlled change of acquisition, but have
 71 mostly been applied to classification rather than reconstruction (Pawlowski
 72 et al., 2020; Castro et al., 2020). These responses share a root cause. Ac-
 73 celeration works because a learned prior fills in information the measurement
 74 does not contain, and that same filled-in information can hide a real finding or
 75 create a false one, while the field certifies the result with averaged pixel scores
 76 that cannot see either event.

77 To our knowledge no systematic review has assembled the evidence on this
 78 problem. Existing surveys cover architectures and benchmark performance
 79 (Liang et al., 2020; Montalt-Tordera et al., 2021); none asks how often con-
 80 ventional metrics miss clinically important errors, why they miss them, or what
 81 evaluation would have to look like to catch them. We address that through
 82 three questions:

83 **RQ1** *How reliably do conventional fidelity metrics track radiologist diagnostic*

84 *ratings in deep learning-based accelerated brain MRI, and how wide is the*
85 *evaluation reporting gap in the literature?*

86 **RQ2** *What underlying mathematical mechanisms cause standard metrics to*
87 *overlook localized pathology erasure and generative hallucinations in deep*
88 *learning-based reconstruction?*

89 **RQ3** *What can each evaluation paradigm in current use establish about prior-*
90 *induced diagnostic error in deep learning-based MRI reconstruction, and*
91 *what requirements for safety-oriented evaluation follow from those limits?*

92 We contribute: (C1) a synthesis of what 263 primary studies report about
93 agreement between pixel-wise metrics and radiologist judgement; (C2) four
94 mathematical mechanisms that explain the divergence, with an exact identity
95 bounding the PSNR cost of erasing a lesion; (C3) an analysis of *where* the
96 evaluation gap comes from, showing that it co-occurs with a division of evalu-
97 ation practice between research communities and with a benchmark infrastruc-
98 ture lacking acute pathology; and (C4) five requirements that safety-oriented
99 evaluation must meet. The auditing framework these requirements motivate is
100 developed and evaluated in a companion methods paper, not here.

101 Three boundaries should be stated at the outset, because they determine
102 what the findings can and cannot be asked to support. First, this is a review of
103 *evaluation practice*, not of reconstruction performance: we count what studies
104 measured, not how well their methods performed, and nothing here estimates
105 how often a deployed reconstruction actually erases a lesion. Second, the mech-
106 anisms in RQ2 are mathematical properties of the metrics, derived under a
107 stated error model; they establish that a focal error *can* pass unrecorded, not
108 that it commonly does. Third, prevalence counts rest on labels extracted from
109 what studies report, so every one of them is a lower bound on practice rather
110 than an estimate of it, and we treat the direction of that bias explicitly in Sec-
111 tion 5.2. The contribution is a diagnosis of what the field’s current instruments
112 are capable of detecting, which is logically prior to, and considerably cheaper
113 to establish than, a measurement of how often they fail.

114 2. Related Work

115 Four strands bear on this review, and the claim that none of them has
116 assembled the evidence we assemble here is a claim about each in turn.

117 *Surveys of reconstruction methods.* Reviews of deep learning MRI recon-
118 struction catalogue architectures, acceleration factors and benchmark accuracy
119 (Liang et al., 2020; Montalt-Tordera et al., 2021; Knoll et al., 2020a; Kazer-
120 ouni et al., 2023). They are organised by method, and their outcome variable
121 is reconstruction fidelity against a fully sampled reference. Safety and failure
122 modes appear as caveats in discussion sections rather than as an object of syn-
123 thesis, and none of them tabulates what evaluation each primary study actually
124 performed, which is the quantity this review extracts.

125 *Task-based image quality assessment.* The tradition that ought to have sup-
126 plied the answer established, decades before deep learning, that image quality is
127 meaningful only relative to a diagnostic task. Barrett and colleagues formalised
128 the observer as part of the measurement: an image is good if an observer per-
129 forming a specified task on it achieves a high figure of merit, typically the
130 area under an ROC curve for a detection task, and the mathematical (ideal or
131 channelised Hotelling) observer supplies a computable surrogate for the human
132 when the task and image statistics are specified (Barrett et al., 1993; Barrett
133 and Myers, 2004). ICRU Report 54 made the same argument the international
134 reference framework for assessing medical imaging quality (International Com-
135 mission on Radiation Units and Measurements, 1996). That framework already
136 contains the concept the reconstruction literature lacks: a detectability index
137 for a specified lesion, which is by construction sensitive to exactly the focal
138 change a whole-volume L_2 score averages away. It has been carried into the
139 dose- and protocol-optimisation literatures of other imaging modalities, but it
140 has not crossed into deep learning MRI reconstruction: no study in this cor-
141 pus evaluated a model observer, and our first requirement (R1, Section 4.7) is
142 in substance a restatement of the task-based position for a field that has not
143 adopted it. The reason it has not is worth naming: model observers require a

144 specified signal and a statistical description of the background, and the null-
145 space structure that makes learned reconstruction dangerous is precisely what
146 makes those two ingredients hard to specify.

147 *Failure-mode-specific work.* Artifact detection, motion correction and ro-
148 bustness under distribution shift are each addressed in depth, but in isolation,
149 by literatures that do not share an evaluation vocabulary (Esteban et al., 2017;
150 Sommer et al., 2020; Güngör et al., 2023). Analyses of hallucination in tomo-
151 graphic reconstruction establish that prior-induced structure lives in the null
152 space of the forward operator, so it is invisible to any measurement-space con-
153 sistency check (Bhadra et al., 2021; Gottschling et al., 2025), and causal and
154 counterfactual methods for medical imaging supply the machinery to intervene
155 on image content and ask whether a finding survives (Pawlowski et al., 2020;
156 Castro et al., 2020; Sanchez et al., 2022; Peng et al., 2026). Neither line has
157 been connected to review-level evidence on how reconstruction is evaluated in
158 practice.

159 *What is new in the four mechanisms of RQ2, and what is not.* Because
160 Section 4 states four mechanisms formally, their provenance should be set out
161 plainly rather than left for a reader to reconstruct. Two are elementary prop-
162 erties of the objective rather than findings: spatial error averaging follows from
163 the definition of a mean over the image volume, and loss symmetry from the
164 fact that squaring discards the sign of the residual. Two are established results
165 we restate: reference dependence is standard in the image-quality literature,
166 and the null-space account underlying the realism–reward trade-off is due to
167 Bhadra et al. (2021) and Gottschling et al. (2025). What this review adds is
168 not the mechanisms but their quantification and their assembly — the identity
169 at Equation (2), which converts any lesion volume into the PSNR movement its
170 erasure would cause, and the setting of all four against what a corpus of 263
171 studies is recorded as measuring. We claim the assembly and the bound, not
172 the underlying results.

173 *Benchmark and annotation infrastructure.* The raw k -space resource the
174 field benchmarks on is fastMRI (Knoll et al., 2020b), whose brain release fol-

lowed the knee data described in that report, and an annotation layer is emerging on top of it: fastMRI+ adds expert pathology labels to the fastMRI brain and knee data (Zhao et al., 2022), and curated stroke-lesion datasets exist for segmentation (ISLES (Hernandez Petzsche et al., 2022), ATLAS (Liew et al., 2022)). None of these supplies raw k -space paired with acute vascular pathology, which is the combination reconstruction evaluation needs, and Section 4.5 shows that this is not an incidental gap but the binding constraint on the whole evaluation problem.

What no strand supplies is the join: a systematic account of how often the field measures metric–reader agreement, why the metrics fail when it does not, and what each evaluation paradigm can therefore establish. The four strands are not merely separate; they are held apart by what each can access. The survey and hallucination-analysis literatures work on datasets with raw k -space and no clinically verified pathology; the task-based tradition needs a specified signal and a labelled task, which those datasets do not carry; and the clinical reader literature has the pathology and the readers but no access to the raw data on which the mechanisms operate. Section 4.5 shows this is not an abstraction but a measurable division in the corpus itself. Supplementary Table S4 sets out each secondary study we identified against the questions asked here.

3. Methodology

This systematic review was conducted in accordance with the PRISMA 2020 statement (Page et al., 2021); the completed checklist, search vocabulary, extraction schema, and analysis scripts are available in the supplementary material and on the Open Science Framework (<https://doi.org/10.17605/OSF.IO/EUXA8>).

Protocol and registration status. The protocol, covering the research questions, eligibility criteria, search strategy, screening procedure, classification rubric and synthesis approach, was fixed in the project repository before screening began. In the terms of PRISMA 2020 item 24a, **this review was not prospectively registered and carries no registration number:**

PROSPERO’s scope excludes reviews without direct clinical health outcomes, and no equivalent prospective registry covers methodological reviews of this kind. The Open Science Framework deposit cited above therefore timestamps the record but is retrospective, and we make no claim of prospective registration on it. Two consequences follow. An external record does not exclude undisclosed post-hoc analytic choices; and the two analyses we ourselves identify as post-hoc — the acceleration-factor extraction of Table 6 and the venue classification of Section 4.5 — are labelled as such wherever they appear.

Search strategy. Seven bibliographic databases (PubMed/MEDLINE, IEEE Xplore, Scopus, Web of Science, SpringerLink, OpenAlex, and Semantic Scholar) were searched between 23 and 24 July 2026, with no publication-date limit applied; the 1995–2026 span quoted elsewhere in this article is the publication range of the studies that were included, not a restriction imposed on the search. Queries were executed through each database’s API rather than its web interface, so the complete Boolean match set was retrieved for every term combination rather than a ranked sample, explaining why the initial identification count is large. The strategy combined three required concept groups: neuroimaging (Block B1), reconstruction and acceleration (Block B2), and safety, failure modes, and evaluation outcomes (Block B3), with MeSH headings and wildcard truncation adapted per database (Supplementary Table S5). Table 1 details the executed query structure and hit breakdown across databases.

The breadth of this search is by design. Evidence of reconstruction failure is rarely a paper’s stated subject: artifacts, hallucinations and metric–reader disagreements are most often reported in passing inside studies whose declared contribution is a new architecture, so a query tuned narrowly to safety vocabulary would miss much of this evidence. Clinical and engineering literatures also use distinct vocabularies — sequences and artifacts against inverse problems, unrolled optimization and generative priors — and neither database family indexes the other completely. We therefore tuned for sensitivity rather than precision and accepted the screening burden. The funnel from 27,327 records to

Table 1: Query structure executed in each database and record hit breakdown.

Database	Executed Query Structure	Controlled Vocabulary	Fields Searched	Hits (N)
PubMed / MEDLINE	(B1 _{ft} [TIAB] OR B1 _{mesh}) AND (B2 _{ft} [TIAB] OR B2 _{mesh}) AND (B3 _{ft} [TIAB] OR B3 _{mesh}) NOT (X[TIAB])	Yes: 9 MeSH headings OR-ed into the three blocks	Title/Abstract + MeSH	24,693
SpringerLink	(B1) AND (B2) AND (B3) NOT (X); quoted phrases, no truncation operators	No	Full record	1,598
IEEE Xplore	(B1) AND (B2) AND (B3) NOT (X); wildcard truncation retained	No	Metadata	778
Scopus	TITLE-ABS-KEY(B1) AND TITLE-ABS-KEY(B2) AND TITLE-ABS-KEY(B3) AND NOT TITLE-ABS-KEY(X)	No	TITLE-ABS-KEY	127
Semantic Scholar	(B1) + (B2) + (B3), where denotes OR and + denotes AND	No	Title/Abstract	66
OpenAlex	title_and_-abstract.search:B1, title_and_-abstract.search:B2, title_and_abstract.search:B3	No	Title/Abstract	34*
Web of Science	TS=(B1) AND TS=(B2) AND TS=(B3) NOT TS=(X)	No	Topic (TS)	31
Total identified records				27,327

Concept blocks combine as (B1) AND (B2) AND (B3) NOT (X). **B1** neuroimaging: "brain MRI", "cerebral MRI", neuroimag*, DWI, FLAIR, T1w, T2w, SWI; MeSH *Brain, Magnetic Resonance Imaging, Neuroimaging*. **B2** AI and accelerated reconstruction: "MRI reconstruction", "accelerated MRI", "undersampled reconstruction", "k-space reconstruction", "unrolled network", GAN, "diffusion prior"; MeSH *Deep Learning, Machine Learning*. **B3** safety, artifacts and evaluation: hallucinat*, "lesion suppression", reconstruction artifact*, PSNR, SSIM, "reader study"; MeSH *Artifacts, Diagnostic Errors, Reproducibility of Results*. **X** exclusion: "animal model", "case report", "in vitro". * denotes truncation. Complete term lists, MeSH headings and per-database syntax are given in Supplementary Table S5. Searches were executed between 2026-07-23 and 2026-07-24, with no publication-date limit. *OpenAlex returned 50 raw hits; 34 passed record-completeness filtering.

263 studies is the intended signature of that choice: the burden falls on initial screening, which reviewers can absorb, rather than on coverage, which no downstream step could recover.

Eligibility and selection. Records were managed in the review's own Python

239 pipeline rather than a commercial screening platform; the scripts, the per-record
 240 decision files and the per-database query logs are released, so each step below
 241 is re-executable. Deduplication by exact identifier matching (PMID/DOI) and
 242 string/TF-IDF similarity (Jaro-Winkler ≥ 0.97 , cosine ≥ 0.95) removed 403
 243 duplicates (1.47%), leaving 26,924 unique records. That rate is low enough to
 244 suggest a matching failure but is structural: PubMed’s MeSH arm returned
 245 24,693 records and the six engineering arms 2,634 between them, and sets that
 246 barely overlap leave little to deduplicate. This is not a claim of completeness,
 247 and one failure is on record: peer review identified an arXiv preprint and its
 248 journal version that both survived to the included set, carrying different DOIs,
 249 no shared PMID, and titles differing by terminal punctuation — below the
 250 thresholds above. It was removed at revision, taking the corpus from 264 reports
 251 to 263 studies (Section 4). Automated matching will not catch a preprint–
 252 journal pair whose metadata diverges in this way, and the honest inference is
 253 that a residual duplicate rate of the same order cannot be excluded.

254 Three reviewers (D.T.M., T.V.P., T.N.T.D.) screened records against the
 255 pre-specified PICOS criteria (Table 2) at both stages, under a primary-screen-
 256 plus- adjudication design: each record received one primary label of *include*,
 257 *exclude*, *flagged* or *background only*, and every record labelled *flagged* went to
 258 a second reviewer for adjudication. Labelling was deliberately conservative,
 259 so any record with an unclear PICOS dimension, or whose text we could not
 260 obtain, was flagged rather than excluded. Title and abstract screening advanced
 261 1,149 records to full-text assessment, of which 109 were labelled *include* on the
 262 abstract alone and 1,040 *flagged*; both labels carried the record forward, and
 263 the distinction records only whether the abstract was sufficient to decide. A
 264 further 25,573 were excluded and 202 set aside as background. Full texts were
 265 pursued through institutional and title-level library subscriptions, open-access
 266 repositories, preprint servers, PubMed Central and inter-library request until
 267 no route remained: 1,068 of the 1,149 (93.0%) were obtained and assessed, and
 268 the 81 that could not be obtained are broken down by recorded reason with the
 269 selection results (Section 4).

270 The three reviewers divided the 1,068 obtained reports between them un-
 271 der the same primary-screen-plus-adjudication design used at title and abstract:
 272 each report received one primary full-text label, and a report was escalated to
 273 a second reviewer wherever the primary label was *flagged*, wherever an inclu-
 274 sion decision rested on a borderline PICOS dimension, or wherever the reviewer
 275 requested a second reading. Escalated reports were resolved by consensus dis-
 276 cussion between the two assessors, with the third reviewer consulted where the
 277 two did not converge. The escalation volumes this produced are reported below,
 278 and the per-report labels, escalation marks and reasons are released in the de-
 279 cision ledger. We state the design in this form because it determines what can
 280 be computed from the ledger: one primary label per report means no chance-
 281 corrected inter-rater coefficient is recoverable from full-text screening, which is
 282 why the separate criterion-reproducibility exercise below was run. In total, 263
 283 primary studies met all inclusion criteria upon full-text verification (two ad-
 284 ditional reports were excluded as duplicate publications). Full-text exclusions
 285 ($n = 785$) were categorized by pre-specified PICOS criteria: wrong intervention
 286 ($n = 326$), out of scope ($n = 256$), wrong outcome ($n = 98$), wrong population
 287 ($n = 93$), wrong study design ($n = 4$), non-English ($n = 1$), and non-primary
 288 articles ($n = 7$). A further 18 reports were retained as background material
 289 rather than as included studies, so the 1,068 assessed reports resolve as 263
 290 included, 785 excluded on PICOS grounds, 2 excluded as duplicate publications
 291 and 18 retained as background. The complete decision ledger is released as
 292 `fulltext_screening_decisions.csv`.

293 *Publication year.* Every analysis partitioning the corpus by time uses the
 294 year of the version of record as indexed by the publisher — the issue year for
 295 journal articles, the proceedings year for conference papers — rather than first
 296 online posting or preprint deposit. The rule matters at era boundaries, and we
 297 adopt the indexed year because it is what a reader can verify from the reference
 298 list, Crossref and PubMed without online-first metadata, which is unevenly
 299 available across the 87 venues represented. The same value is recorded in the
 300 `year` field of every bibliography entry and in the `publication_year` column of

301 `included_characteristics.csv`, so the era counts in Table 9 are recomputable
302 from either. One consequence should not be read as an inconsistency: the
303 searches ran on 23–24 July 2026, yet 41 of the 263 included studies (15.6%)
304 carry a 2026 issue year, being articles published online ahead of print before the
305 searches ran but assigned by the publisher to a 2026 issue.

306 *Screening agreement, and what it does and does not cover.* The primary
307 screen described above records one label per record, so no chance-corrected co-
308 efficient is recomputable from those decisions alone. We therefore ran a separate
309 duplicate-screening exercise. A random sample of 214 records already assessed
310 at title and abstract was re-screened independently by all three reviewers, blind
311 both to the recorded decision and to one another. Reviewers judged the four
312 PICOS dimensions from title, journal and abstract, opening the DOI link only
313 where the abstract was insufficient, and returned one of three labels per record:
314 include, exclude or unsure. The sample was drawn without stratification; the
315 drawn records, the three label sets and the agreement computation are released,
316 so the sample is auditable record by record. All 642 cells were completed and
317 no record was dropped for missingness.

318 Across all 214 records Fleiss’ $\kappa = 0.877$ (95% bootstrap CI 0.823–0.925;
319 10,000 resamples, seed 20260821), with observed agreement $\bar{P} = 0.933$ against
320 $\bar{P}_e = 0.455$ expected by chance. The three reviewers agreed unanimously on
321 194 of 214 records (90.7%), and pairwise Cohen’s κ ran from 0.845 to 0.931.
322 One sensitivity analysis is worth reporting because it is the one a reader is most
323 likely to run. Restricting to the 193 records on which no reviewer recorded
324 *unsure* raises Fleiss’ κ to 0.943 (95% CI 0.900–0.979). That complete-case figure
325 discards 21 records, and it discards precisely the records at least one reviewer
326 could not decide, so it overstates agreement on the population the coefficient
327 is meant to describe. We report 0.877 as the estimate and 0.943 only as the
328 value obtained when the hard cases are removed. Two qualifications travel with
329 that figure and should not be dropped. First, it measures agreement on the
330 screening decision, not on the extraction labels from which every prevalence
331 count in this article is derived; those remain unaudited for reliability, and the

only error rate we can quote for them is the measured 4/61 (6.6%) on the design field (Section 5.2). Second, the conservative labelling policy routed almost the whole full-text workload to adjudication in any case — 1,040 of the 1,149 records advanced at title and abstract were flagged rather than included outright — so a high coefficient is evidence that the criteria were applied consistently, not that the borderline records were easy to classify.

What the primary decision records themselves support, and what we therefore also report, are the escalation volumes that policy produced. At full text, 84 reports were escalated to a second reviewer for adjudication and 29 of those were included; a further 77 reports carry a mandatory human-re-inspection mark in the released files, of which 40 were included after that review. Screening recall is bounded by the search strategy and by the criteria as applied rather than estimated against an independent reference set of known-eligible records, since no such set exists for this question; Section 5.2 treats both points as limitations. The duplicate-screened sample, both label sets and the agreement computation are released as `kappa_human_results.csv`, alongside the full decision ledger in `fulltext_screening_decisions.csv`.

Data extraction. Reviewers extracted study characteristics using the pre-specified form (Supplementary Table S6), capturing method family, reported failure modes, evaluation paradigm, field strength, pulse sequence, reference standard and reproducibility indicators, with a verbatim supporting quotation for every assigned value. Method, failure-mode and evaluation categories were assigned from each record’s title, abstract and controlled vocabulary against the vocabulary fixed in advance (Supplementary Table S5); that deliberately conservative scope is why every prevalence count here is a lower bound rather than an exhaustive tally. Field strength and pulse sequence, typically stated only in acquisition subsections, were additionally taken from full text where available. We did not contact investigators: a variable not stated in the source was recorded as *not reported* and never inferred, which is why the 54 studies without a stated field strength divide into 50 whose full text does not state it and 4 assessed on abstract or partial text.

Table 2: PICOS elements and their application in this review.

Element	Definition	Application in this Review
Population	Target imaging modalities, clinical domains, and scanner physics	Human brain MRI acquisitions (acute ischemic stroke, neurovascular lesions, intracranial haemorrhages, tumours, and structural T1w, T2w, FLAIR, DWI scans across 1.5T, 3T, and 7T field strengths).
Intervention	AI-based and accelerated reconstruction methods, and their safety/fidelity evaluation	Deep learning and accelerated reconstruction methods (unrolled physics-based networks, GANs, diffusion priors, self-supervised, denoising, super-resolution), together with methods for evaluating their fidelity, artifacts, robustness, or failure modes.
Comparison	Baseline metrics, observers, and reference standards	Conventional pixel-wise metrics (PSNR, SSIM, NRMSE), perceptual/learned metrics (LPIPS, FID), model observers, fully sampled reference images, or radiologist reader assessments.
Outcome	Reconstruction quality, fidelity, and failure characterization	Image-quality/fidelity measures, artifact and failure-mode characterization, metric-radiologist agreement, uncertainty estimates, robustness to distribution shift, and diagnostic/observer outcomes.
Study Design	Methodological criteria and empirical validation settings	Primary empirical studies, algorithmic benchmarking, counterfactual interventional stress-testing, and human or model observer evaluations.

363 *Quality appraisal, and the reader/algorithmic partition.* Methodological
364 quality was evaluated with three design-matched instruments: QUADAS-2
365 (Whiting et al., 2011) across its four domains for diagnostic accuracy and
366 radiologist reader studies; a reporting-transparency check against STARD prin-
367 ciples (Bossuyt et al., 2015), reported descriptively rather than scored, since
368 STARD yields no risk-of-bias rating; and, for algorithmic benchmarking studies,
369 a reproducibility and robustness appraisal covering code and data availability,
370 use of a named public benchmark, whether the reference standard was a fully
371 sampled acquisition, and whether robustness to acquisition or distribution
372 shift was reported. ROBINS-I (Sterne et al., 2016) was not applied, since few
373 included studies are the non-randomised interventional designs it was built for.

374 The partition between the two instruments is not the one the extraction la-
375 bels originally implied, and the correction matters for every reader-share statis-
376 tic here. Extraction recorded 61 studies as observer or reader designs; three-

reviewer appraisal of their full text found four containing no human reader at all, each using the language of observer evaluation for an automated surrogate. Those four were reclassified as algorithmic, so **the verified reader subset is 57 and the algorithmic set is 206**, and $57 + 206 = 263$ closes the corpus exactly. All 57 were appraised by all three reviewers across two rounds, 39 in the first and 18 in the second, so every QUADAS-2 tally rests on 57 studies and $57 \times 7 = 399$ domain judgements. Every reader-derived count in this article uses 57, and the released per-study file encodes the same partition with the reclassification recorded per study. One further study (S001) carries an appraisal note that a human reader could not be *confirmed*, only abstract-level text being available; it is retained in the reader subset, since non-ascertainment is not a finding of absence, and it is flagged in the deposit.

Appraisal agreement. The three reviewers rated every applicable domain Low, Unclear or High independently, each judgement carrying a verbatim supporting quotation, before joint review to harmonize thresholds. Five first-round studies were later re-appraised on recovered full text (below), so their 35 judgements no longer rest on three independent ratings and are excluded from the coefficient. Over the remaining 52 studies and 364 judgements Fleiss' $\kappa = 0.788$ (95% bootstrap CI 0.739–0.833; 10,000 resamples, seed 20260821). That figure averages two rounds that behaved differently, and the difference is more informative than the average. The 34 first-round studies retaining three independent ratings were unanimous on all 238 of their judgements, so they contribute $\kappa = 1.00$ by construction and carry no information about latent reliability. The 18 of the second round were unanimous on 60 of 126 (47.6%), and across those 126 Fleiss' $\kappa = 0.390$ (95% CI 0.293–0.484) — fair on the Landis–Koch bands, with an interval reaching into moderate. The procedure was identical across rounds; the subsets were not, the second round comprising studies the first pass deferred as not straightforward. Two readings remain open — that those studies report their methods less completely and are genuinely harder to rate, and that the first round was less independent in practice than in design — and these data do not separate them.

408 The disagreement has a structure worth recording rather than smoothing.
409 Across the 126 second-round judgements one reviewer rated Low 33.3% of the
410 time and another 61.9%, a difference of threshold rather than of reading, which
411 is why κ sits at 0.390 while raw agreement is 63.5%. Six judgements split
412 three ways with no majority; all six were recorded as *Unclear*, on the ground
413 that a panel unable to converge is itself evidence the report did not support
414 a confident judgement, which is what *Unclear* encodes in QUADAS-2. The
415 rule was applied to those six and no other case, and the per-reviewer labels are
416 released so a reader can apply a different rule and recompute. Section 5.2 states
417 what this does and does not establish. Table 3 summarizes corpus-level quality
418 indicators and the QUADAS-2 domain judgments.

Table 3: Corpus-level provenance and methodological quality indicators across the 263 included studies, with QUADAS-2 domain-wise risk of bias and applicability concerns for the full appraised diagnostic/reader subset ($n = 57$, rated across seven domains; independently by all three reviewers except for the five studies re-appraised on recovered full text). The 39 studies of the first appraisal round and the 18 of the second are pooled here; the remaining 206 studies were appraised on reproducibility and robustness instead ($57 + 206 = 263$). Six of the 399 domain judgements split three ways with no majority and are recorded as *Unclear* (Section 3).

Indicator / Domain	Value / Risk of Bias Breakdown
Publication period	1995–2026; 229/263 (87.1%) published in 2020 or later, and 174/263 (66.2%) in 2023 or later, indicating a predominantly recent evidence base.
Basis of assessment, current	Full text available for 258/263 (98.1%); partial or truncated full text for 1/263 (0.4%); title, abstract and MeSH only for 4/263 (1.5%).
Basis of assessment, at appraisal	The first appraisal round was made on the text obtainable at the time, which for 18/263 studies was abstract and MeSH only (6 of them within the QUADAS-2 subset). A later library sweep recovered full text for 14 of those 18 (including 5 of the 6), and all 14 have since been re-appraised on the recovered full text, so appraisal basis now stands at 257/263 full text, 2/263 partial text and 4/263 abstract and MeSH only — within the QUADAS-2 subset, 56/57 on full text and 1/57 (S001) on abstract-level text. The two rows now differ by a single study (S200), appraised on truncated text whose full version has since become available.
Leading venues	The eight most frequent of the 87 distinct venue titles represented in the corpus, accounting for 128/263 (48.7%) of included studies: <i>Magnetic Resonance in Medicine</i> (40), <i>IEEE Transactions on Medical Imaging</i> (17), <i>AJNR</i> (15), <i>Magnetic Resonance Imaging</i> (14), <i>JMRI</i> (12), <i>Scientific Reports</i> (12), <i>NeuroImage</i> (9), <i>Medical Image Analysis</i> (9).

(Table 3 continued)

Predominant design	Algorithmic / methodological reconstruction studies ($n = 206$), alongside a verified observer/reader subset ($n = 57$) (Radmanesh et al., 2022; Lang et al., 2024). Extraction recorded 61 reader designs; appraisal of their full text found 4 with no human reader, and those 4 were reclassified as algorithmic ($202 + 4 = 206$; $61 - 4 = 57$). Of the 57: 34 are appraised and verified by all three reviewers; 18 were appraised by all three in a second round and await final adjudication of 6 split domain judgements; and 5 (S004, S020, S027, S034, S115) were re-appraised by the reviewers on full text recovered after the first round, with domain judgements verified against the source text. Every reader-derived statistic in this article uses 57.
Reference standard	Expert reader consensus (70), an unspecified stated ground truth (62), a fully sampled acquisition (69), a conventional or standard-of-care acquisition (15); not stated in 47/263 (17.9%).
Reproducibility	Code availability stated in 97/263 (36.9%); a named public dataset or benchmark used in 109/263 (41.4%), most commonly fastMRI, HCP, ADNI and IXI. Availability is variable and remains an area for improvement.
Sampling and setting	Sampling frame not stated in 95/263 (36.1%); recruitment centre not stated in 174/263 (66.2%). Ethics approval or informed consent reported in 159/263 (60.5%).

QUADAS-2 Risk of Bias ($n = 57$ Reader / Diagnostic Accuracy Studies)

Patient Selection	Low: 15/57 (26.3%); Unclear: 19/57 (33.3%); High: 23/57 (40.4%). The weakest domain: sampling is predominantly purposive or pathology-enriched, and most of the appraised studies state no enrolment rule.
Index Test (AI Reconstruction)	Low: 26/57 (45.6%); Unclear: 9/57 (15.8%); High: 22/57 (38.6%). Low risk falls below half once the second round is included. High ratings arise where readers scored accelerated reconstructions with the reference visible, or where test parameters were set using reference-derived labels.
Reference Standard	Low: 26/57 (45.6%); Unclear: 15/57 (26.3%); High: 16/57 (28.1%). High ratings arise where no artifact-free reference exists, or where lesion truth is defined by the comparator images themselves (incorporation bias). Reading the recovered full text moved four of the re-appraised studies from Low to High here, each because the comparator images, or software applied to them, supplied the truth against which they were scored.
Flow and Timing	Low: 45/57 (78.9%); Unclear: 7/57 (12.3%); High: 5/57 (8.8%). The strongest domain, since retrospective undersampling of the same raw k -space gives a zero interval between index test and reference.

QUADAS-2 Applicability Concerns ($n = 57$)

Patient Selection	Low: 36/57 (63.2%); Unclear: 4/57 (7.0%); High: 17/57 (29.8%), chiefly healthy-volunteer, lesion-free, or non-clinical cohorts.
-------------------	---

(Table 3 continued)

Index Test	Low: 30/57 (52.6%); Unclear: 4/57 (7.0%); High: 23/57 (40.4%), where the evaluated method is image-domain post-processing, cross-field synthesis, harmonization, or a quality index rather than reconstruction from under-sampled k -space.
Reference Standard	Low: 39/57 (68.4%); Unclear: 3/57 (5.3%); High: 15/57 (26.3%).

Across all 228 risk-of-bias judgements: 112 (49.1%) Low, 50 (21.9%) Unclear, 66 (28.9%) High. Across all 171 applicability judgements: 105 (61.4%) Low, 11 (6.4%) Unclear, 55 (32.2%) High. Over the 364 judgements that rest on three independent ratings Fleiss' $\kappa = 0.788$ (95% CI 0.739–0.833), but agreement is not uniform across the two appraisal rounds: the 238 retained first-round judgements were unanimous, the 126 of the second round reach $\kappa = 0.390$ (95% CI 0.293–0.484); the 35 judgements of the 5 re-appraised studies are excluded, having been conducted subsequently on recovered full text. Section 3 gives the decomposition and Section 5.2 what it bounds. The high proportion of Unclear ratings reflects unreported blinding and sampling procedures, compounded by the one first-round study for which only abstract-level text is available. Six second-round judgements split three ways with no majority and are recorded as Unclear under the rule stated in Section 3. Per-study ratings, each with the verbatim supporting quotation, are released as `included_characteristics_supplement.csv`.

419 *Derived venue classification.* The analysis in Section 4.5 groups studies by
420 publication community. That grouping is post-hoc, was not part of the fixed
421 protocol, and is stated in full in Supplementary Table S13 so it can be re-
422 produced or contested. Each study’s journal title, and nothing else about the
423 study, is matched case-insensitively against three ordered keyword lists — en-
424 gineering and methods, then general neuroscience and neuroimaging, then clin-
425 ical imaging — with anything unmatched left *unclassified* rather than forced
426 into a group. The ordering is load-bearing, since titles such as *Medical Image*
427 *Analysis* would otherwise match two lists, and it is applied identically to all
428 263 studies. The rule as executed and the resulting per-study assignment are
429 released as `5_Code/venue_classification.py` and `venue_communityclas-`
430 `sification.csv`.

431 *Synthesis.* Due to heterogeneity in acquisition protocols, reconstruction ar-
432 chitectures, datasets, and outcome measures, evidence was synthesised narra-
433 tively in accordance with SWiM guidelines (Campbell et al., 2020). Studies were
434 structured around the primary research questions (RQ1–RQ3) and grouped by
435 reconstruction method family, field strength, pulse sequence, and failure mode.
436 Quantitative values reflect named primary studies rather than pooled meta-
437 analytic estimates; no effect measure is pooled, and no meta-analytic model

438 is fitted. Percentages are reported to one decimal place and rounded half-up
439 throughout, so 9/80 prints as 11.3% and not 11.2%. Sensitivity analysis was
440 performed by recalculating headline prevalence counts on the 258 studies verified
441 against complete full text (Section 5.2).

442 4. Results

443 All counts and prevalence statements in this section describe the 1,068 re-
444 ports assessed in full and the 263 included studies; because 81 candidate reports
445 remain unretrievable, they are lower bounds on the literature rather than esti-
446 mates of it.

447 4.1. Selection and corpus

448 From 27,327 records identified across seven databases, multi-tier dedupli-
449 cation removed 403 duplicates, leaving 26,924 unique records screened at title
450 and abstract. Of these, 1,149 proceeded to full-text eligibility, and full texts
451 were obtained for 1,068 of them (93.0%; Section 3). Each of the 81 reports still
452 out of reach was checked against its publisher platform and assigned a recorded
453 reason: 53 sit behind title-level subscriptions our institution does not hold,
454 across nineteen publishers; 13 are Springer conference chapters sold only per
455 chapter; 4 are ISMRM proceedings abstracts issued without a PDF; 4 resolve
456 to a whole proceedings volume or to front matter rather than a single report;
457 and 7 carry no DOI and no locatable source. Supplementary Table S14 gives
458 the per-publisher breakdown. Access difficulty is itself informative, though it
459 bounds the missing evidence rather than characterising it. Of the 341 assessed
460 reports obtainable only through title-level subscriptions and equivalent barriers,
461 294 (86.2%) proved ineligible on the same PICOS criteria, 2 were retained as
462 background and 45 were eligible ($294 + 2 + 45 = 341$; the 45 resolve to 44 stud-
463 ies, one being a preprint adjudicated a duplicate). Raw non-retrieval counts
464 therefore overstate the missing evidence by roughly sevenfold. But those 45
465 were concentrated in subscription clinical journals, and Section 4.5 shows they

differ systematically from the open literature in exactly the variable this review measures, so 86.2% is an upper bound on what non-retrieval costs us, not a demonstration that it costs us nothing. Of the 1,068 assessed reports, 787 were excluded (785 on PICOS grounds, 2 as duplicate reports of included studies) and 18 recorded as background, leaving **263 included primary studies** spanning 1995–2026 (Fig. 1). Each included study was verified against the fullest text obtainable for it: complete full text for 258, partial text for 1, and title, abstract and controlled vocabulary for 4 (Section 3). Their aggregate characteristics are given in Table 4, and their thematic distribution across the research questions is summarized in Supplementary Table S7; all 263 are listed, with full references, in Supplementary Tables S1–S3.

Corpus growth is recent: 34 studies appeared in 1995–2019, 55 in 2020–2022 and 174 in 2023–2026.

4.2. RQ1: the metric–reader comparison is almost never made, and where readers appear they agree only partially

Across the 84 studies reporting objective fidelity metrics and the 57 verified reader studies, the comparisons that actually exist show agreement between fidelity metrics and radiologist assessment that is partial and condition-dependent (Table 5). The size of that evidence base should be stated in the text and not left to the table caption: only two of the four rows in Table 5 were collected on reconstructions of undersampled k -space. Tang et al. (2025) collected radiologist rankings on synthetically corrupted image pairs, and Sommer et al. (2020) on clinical scans corrupted by real patient motion rather than by undersampling. The table that carries RQ1 therefore rests on two on-target rows, which is the honest size of the direct evidence for the metric–reader comparison in this setting. In the fastMRI 2020 challenge’s reader phase, six radiologists ranked submissions for quality of pathology depiction; their concordance, by Kendall’s coefficient, was 0.457 in the $4\times$ track and 0.386 in the $8\times$ track, rising to 0.781 only in the transfer track, and the organisers report that concordance improved as SSIM scores diverged, which is to say the metric separated methods

Table 4: Characteristics of the 263 included studies.

Demographic / Method Dimension	Category Breakdown	(n)	Percentage / Key Methodological Notes
Publication Period	1995–2019	$n = 34$	12.9% (Classical CS/PI & early CNNs)
	2020–2022	$n = 55$	20.9% (Unrolled physics & GAN priors)
	2023–2026	$n = 174$	66.2% (Diffusion priors & 3D/4D foundation models)
Reconstruction Paradigm (multi-label; 339 labels recorded)	Compressed Sensing / Parallel Imaging	$n = 93$	Iterative sparsity & coil sensitivity encoding
	Generative (GAN/Diffusion/VAE)	$n = 80$	Adversarial perception & posterior sampling
	Motion Correction	$n = 57$	Intra-scan motion artifact suppression
	Physics-based Unrolled	$n = 37$	Explicit k -space data consistency layers
	Denoising	$n = 30$	Image-domain noise suppression
	Super-resolution	$n = 25$	Resolution enhancement post-acquisition
MRI Pulse Sequences (multi-label; 490 labels)	Self-supervised / Unsupervised	$n = 17$	Training without fully sampled reference scans
	T1-weighted (T1w)	$n = 163$	High-resolution anatomical cortical profiling
	T2-weighted (T2w)	$n = 122$	Ventricular & CSF pathology assessment
	FLAIR	$n = 85$	Hyperintense peri-ventricular lesion detection
	Diffusion-Weighted (DWI)	$n = 72$	Acute ischemic lesion (DWI core) evaluation
	SWI / T_2^* -weighted	$n = 48$	Punctate cerebral microbleed hypointensity voids
Magnetic Field Strength (single-label; predominant field where several are used)	3.0 Tesla (3T)	$n = 153$	58.2% (Primary clinical scanner benchmark)
	7.0 Tesla (7T)	$n = 35$	13.3% (Ultra-high-field research acquisitions)
	1.5 Tesla (1.5T)	$n = 21$	8.0% (Routine clinical throughput imaging)
	Not stated	$n = 50$	19.0% (Full text obtained; no field strength stated)
	Not ascertainable	$n = 4$	1.5% (Only abstract or partial text available)
Evaluation Paradigm (multi-label; 177 labels recorded across 144 studies)	Pixel-wise (PSNR/SSIM/NRMSE)	$n = 84$	47.5% of recorded (legacy fidelity reporting)
	Observer / Reader Studies	$n = 57$	32.2% of recorded (task-based reader assessment; verified subset)
	Uncertainty Estimation	$n = 18$	10.2% of recorded (posterior variance maps)
	Learned / Perceptual	$n = 18$	10.2% of recorded (rank-trained metrics)
Co-reporting. Of the 144 studies with a recorded evaluation paradigm, 28 report more than one. Critically, only 18 studies in the entire corpus (6.8% of 263; 12.5% of the 144 recorded) report <i>both</i> a pixel-wise fidelity metric <i>and</i> a reader assessment on the same data, which is the minimum precondition for computing metric-versus-reader agreement on which safety claims implicitly depend. See Section 4.			

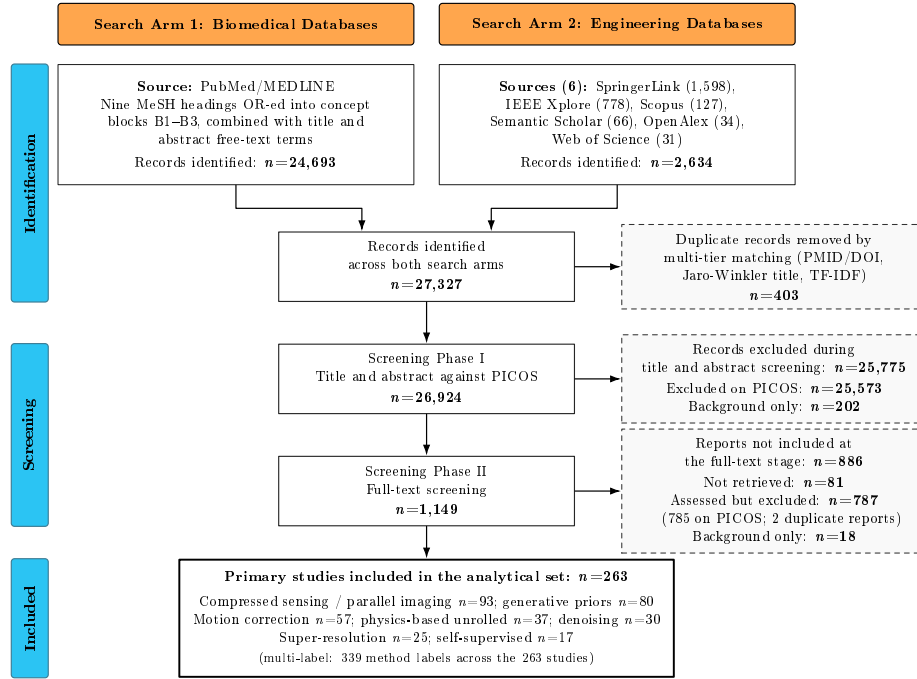


Figure 1: PRISMA 2020 flow diagram for study selection across the two search arms. Multi-tier deduplication reduced 27,327 identified records to 26,924 unique records; title and abstract screening advanced 1,149 candidates. Full texts were obtained for 1,068 of them (93.0%); 81 could not be retrieved. Following PRISMA 2020, exclusion and background retention are recorded as distinct dispositions: of the 1,068 reports assessed in full, 787 were excluded (785 on explicit PICOS grounds and 2 as duplicate reports of included studies) and a further 18 were retained as background material rather than excluded, leaving 263 primary studies in the analytical set ($263 + 787 + 18 = 1,068$).

reliably only where their outputs already differed grossly (Muckley et al., 2021). When radiologist preference is instead used as the training signal, purpose-built quality networks predict expert rankings at $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$, against radiologists’ own interobserver agreement of $61.5\% \pm 5.5\%$; the same study finds rank-based reading far more reliable than Likert scoring (70.4% versus 25% agreement between radiologists) (Tang et al., 2025). Reader tolerance of acceleration is real but bounded: in a retrospective evaluation spanning accelerations up to $100\times$, 94% of volumes (65 of 69) were rated sufficient for potential screening use at up to $14\times$ (interreader intraclass correlation 0.875), with the threshold for full diagnostic quality lower still (Radmanesh et al., 2022). The one direct clinical observation of the failure this review is about comes from an

507 emergency-department and inpatient comparison, and it is a single case. Across
 508 66 paired examinations in which a two-minute ultrafast protocol (T1w, T2/T2*,
 509 FLAIR and DWI, machine-learning-assisted reconstruction) was read against a
 510 ten-minute reference protocol, two blinded neuroradiologists agreed on the main
 511 clinical diagnosis in 65 of 66 cases (98.5%), rated overall diagnostic quality no
 512 different between protocols ($P > .05$), preferred the reference protocol for image
 513 noise and geometric distortion ($P < .05$) and preferred the accelerated protocol
 514 for reduced motion artifact ($P < .01$) (Lang et al., 2024). The single discrepant
 515 case is the one that matters here: a punctate focus of restricted diffusion, read
 516 as possible acute-to-subacute infarct on the reference DWI, was less conspicu-
 517 ous on the accelerated DWI, which the authors attribute to a combination of
 518 image-quality difference and section positioning, the lesion being smaller than
 519 the section thickness (4 mm reference against 5 mm accelerated) and so subject
 520 to partial-volume contamination. They draw the inference themselves, that the
 521 protocol’s sensitivity to small findings is an open question. One case in 66 is
 522 not a prevalence estimate and we do not offer it as one. It is a worked instance
 523 of the pattern this review predicts: an aggregate quality comparison that finds
 524 no difference, and inside it a focal finding that a reader saw and the aggregate
 525 did not. Where readers assess artifacts directly, blinded reading detects what
 526 reference-based metrics cannot express: a two-neuroradiologist blinded study of
 527 28 clinical scans with real patient motion found significant artifact-score reduc-
 528 tions after network correction ($p < .03$), a setting in which the motion-corrupted
 529 reference invalidates PSNR and SSIM by construction (Sommer et al., 2020).
 530 Motion also interacts with the sampling scheme itself, appearing as ghosting
 531 under Cartesian trajectories and as streaking or blurring under radial ones,
 532 so the visual form of the same physical corruption differs by protocol (Schau-
 533 man et al., 2026). The reference pipeline is not neutral either: conventional
 534 SWI homodyne phase filtering has been shown to create punctate artifactual
 535 microhaemorrhage-like foci, so processing can manufacture findings as well as
 536 remove them (Li et al., 2015).

537 No included study reports a rank correlation between a pixel-wise fidelity

metric and radiologist diagnostic assessment for deep learning-based brain reconstruction, and none evaluated a model observer. The second statement needs a boundary that the first does not. Our search vocabulary (Table 1, block B3) contains reader-study and metric terms but no model-observer vocabulary, so a study that used a channelised Hotelling or non-prewhitening observer and described it only in those terms could have been missed at retrieval. What we can say is that none of the 263 studies retrieved, screened and extracted here reports one, and that the term does not appear in any extracted evaluation label; whether the true field-wide count is zero or merely very small, it is not a paradigm this corpus can be said to use. That absence is itself the RQ1 finding: only 18 of 263 studies (6.8%, a screening-level lower bound; Section 3) record a fidelity metric and a reader assessment on the same data, so the field’s central proxy relationship, that high fidelity scores imply diagnostic adequacy, is largely unmeasured rather than merely disputed.

Table 6 reports how evaluation practice varies with the acceleration factor actually stated in each study. It is built only from what the corpus records: the acceleration factor was extracted post hoc from the full texts of all 263 included studies with a verbatim supporting quotation for every value, and 63 studies (24.0%) state one. Studies are grouped by the highest factor at which they evaluate their method. The table deliberately carries no fidelity threshold, reader-score difference or metric–reader rank statistic per acceleration band, because no included study reports one: that absence is the finding of Section 4, not an omission from the table.

4.3. RQ2: four mechanisms explain the divergence

Four mechanisms account for why the metrics miss what the readers see. Table 7 states each one formally alongside the clinical failure it permits, and Supplementary Table S8 gives the fuller derivations and evidence sources.

Spatial averaging. Conventional L_2 metrics average residual error uniformly over all $H \times W \times D$ voxels, so the total decomposes over lesion and background regions:

Table 5: What the reader studies in this corpus actually measured about metric–reader agreement (RQ1). Every value is quoted from the named study, with the reported coefficient and its definition stated. No included study reports a metric–reader rank correlation for deep learning-based brain reconstruction; the rows below are the closest measurements that exist, and the second column states for each row what the readers were actually shown, because two of the four rows were not collected on reconstructions of undersampled k -space.

Study	Readers and design	Comparison performed	Reported quantity (as defined in the study)
fastMRI 2020 challenge (Muckley et al., 2021)	6 radiologists ranking submissions for quality of pathology depiction	Radiologist concordance across accelerations; SSIM-based ranking vs reader ranking	Kendall's $W = 0.457$ ($4\times$), 0.386 ($8\times$), 0.781 (transfer track); concordance improved as SSIM scores diverged
Tang et al. (2025)	7 radiologists, rank-based reading; 2 also Likert scoring. Rankings were collected on synthetically corrupted image pairs, not on reconstructions of undersampled k-space	Quality networks trained on radiologist rankings vs MSE/SSIM as optimisation targets	Ranking accuracy $75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ with reference image; radiologist interobserver agreement $61.5\% \pm 5.5\%$; ranking vs Likert agreement 70.4% vs 25%
Radmanesh et al. (2022)	Retrospective; neuroradiologist scoring of diagnostic quality at accelerations up to $100\times$	Reader-assigned acceleration thresholds for diagnostic and screening quality	$65/69$ volumes (94%) sufficient for potential screening at $\leq 14\times$; interreader ICC 0.875 ; mean absolute interreader difference in threshold acceleration factor $0.7\times$ (level 1) and $5.4\times$ (level 2)
Sommer et al. (2020)	2 neuroradiologists, blinded, 28 clinical cases with real motion	Artifact severity (5-point scale) before vs after network correction	Significant reduction in mean artifact scores ($p < .03$); reference-based metrics inapplicable, as the reference itself is motion-corrupted

$$\begin{aligned} \text{MSE}(\hat{X}, X) &= \frac{V_{\text{lesion}}}{V_{\text{total}}} \text{MSE}_{\text{lesion}} \\ &+ \left(1 - \frac{V_{\text{lesion}}}{V_{\text{total}}}\right) \text{MSE}_{\text{bg}}. \end{aligned} \quad (1)$$

Table 6: Evaluation practice by the highest acceleration factor (R) each study reports, among the 63 of 263 included studies that state one. Bands are half-open, so a study reporting exactly $R = 4$ falls in the second band and one reporting exactly $R = 8$ in the third. Counts are studies; percentages are within band. *Reader*, any observer or reader assessment recorded; *M+R*, a pixel-wise fidelity metric and a reader assessment recorded on the same data; *Halluc.*, hallucination or fidelity instability among the recorded failure modes; *Artifacts*, aliasing, ghosting or Gibbs artifacts among them. Acceleration factors were extracted from full text after the registered extraction round, each with a verbatim supporting quotation and excluding matches inside reference lists and citation contexts; the 407 retained quotations come from the 63 studies that state a factor, a mean of 6.5 quotations per such study, and are released as `acceleration_factors_evidence.csv`. The extraction is screening-level, so counts are lower bounds. No study at any acceleration factor reports a rank correlation between a fidelity metric and reader judgement.

Acc. (R)	Studies	Reader	M+R	Halluc.	Artifacts	Sequences most often evaluated
$2 \leq R < 4$	32	7 (21.9%)	1 (3.1%)	7 (21.9%)	25 (78.1%)	T1w 15, T2w 12, FLAIR 11
$4 \leq R < 8$	17	5 (29.4%)	1 (5.9%)	2 (11.8%)	12 (70.6%)	T1w 9, T2w 9, FLAIR 7
$R \geq 8$	14	5 (35.7%)	4 (28.6%)	2 (14.3%)	8 (57.1%)	T1w 9, T2w 8, FLAIR 8
All	63	17 (27.0%)	6 (9.5%)	11 (17.5%)	45 (71.4%)	T1w 33, T2w 29, FLAIR 26

The R -reporting subset is close to the whole corpus on reader assessment (27.0% against 21.7% for all 263), so it is not a systematically better-evaluated stratum. Reader assessment rises with acceleration (21.9% $\rightarrow$ 29.4% $\rightarrow$ 35.7%) and paired metric-reader reporting concentrates in the most accelerated band (4/14 against 1/32 in the $2 \leq R < 4$ band), but the counts are small and most studies in every band (including 9 of the 14 with $R \geq 8$) report no reader assessment at all. Artifact description is the most frequently recorded failure mode at every acceleration level, whereas hallucination or fidelity instability is recorded in 11 of 63.

568 With lesion volume fraction $f = V_{\text{lesion}}/V_{\text{total}}$ and local error ratio $r =$
569 $\text{MSE}_{\text{lesion}}/\text{MSE}_{\text{bg}}$, the global PSNR change caused by a localized edit follows
570 directly:

$$|\Delta\text{PSNR}| = 10 \log_{10} (1 + f(r - 1)) \quad (\text{dB}). \quad (2)$$

571 The consequence is quantitative rather than rhetorical, and it is worth anchoring
572 on a lesion a neuroradiologist would expect to resolve rather than on the smallest
573 arithmetically convenient one. An acute lacunar infarct of 100 mm^3 is a sphere
574 5.8 mm across: on a clinical DWI acquired at 4–5 mm section thickness (Lang
575 et al., 2024) that is several voxels wide in-plane and spans one to two sections, so
576 it is resolved, if only just, in the through-plane direction. It occupies a volume
577 fraction $f = 6.3\text{--}8.3 \times 10^{-5}$ of a 1.2–1.6 L brain. Even at a hundredfold local

Table 7: The four mechanisms by which a whole-volume fidelity score fails to register a focal diagnostic error (RQ2). The third column names a failure the mechanism permits, not one this review has observed at a stated frequency; the fourth states why the score does not move. Numerical entries are computed from Equation (2) under the stated error model, not measured in any primary study.

Mechanism	Formal statement	Clinical failure it permits	Why the score does not move
(a) Spatial error averaging	L_2 error is averaged uniformly over all $H \times W \times D$ voxels, so a lesion contributes with weight $f = V_{\text{lesion}}/V_{\text{total}}$	Erasure of an acute lacunar infarct on DWI or FLAIR, or of a punctate microhaemorrhage on SWI or T_2^*	By (2), erasing a 100 mm^3 infarct at $r = 100$ moves global PSNR by $0.027\text{--}0.036 \text{ dB}$; a 5 mm^3 microhaemorrhage by $1.3\text{--}1.8 \times 10^{-3} \text{ dB}$
(b) Reference dependence	Every reference-based score presupposes a registered, artifact-free reference image	A valid motion-corrected reconstruction is penalised against a corrupted reference, and genuine correction is scored as error	The comparison is no longer between reconstruction and truth; PSNR and SSIM become uninterpretable rather than merely noisy
(c) Loss symmetry	$(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$: the objective is indifferent to the sign of the error	A suppressed hyperintense lesion is scored identically to added background noise of the same magnitude	Diagnosis is asymmetric where the metric is not: a missed finding and a spurious brightening carry the same penalty
(d) Realism-reward trade-off	Adversarial and perceptual objectives reward outputs typical of the training distribution, $\log D_\phi(\hat{X})$, not outputs faithful to the measurement	Synthesis of plausible vessels, sulci or texture that was never acquired	Perceptual scores can rise as measurement fidelity falls, so the objective pays for the hallucination it should penalise

error ratio ($r = 100$), Equation (2) gives $|\Delta\text{PSNR}| = 0.027\text{--}0.036 \text{ dB}$. Erasing a resolvable acute infarct outright therefore moves global PSNR by about three hundredths of a decibel, which is the same order as the third decimal place of a score conventionally reported to two.

The limiting case is worth stating precisely, because it is easy to state loosely. A punctate cerebral microhaemorrhage of 5 mm^3 is 2.1 mm across. It is *not* resolved on clinical DWI, whose voxels are roughly $12\text{--}20 \text{ mm}^3$, so no reconstruction can be said to erase it there; it is not resolved in the first place. It is several voxels across on the high-resolution SWI or T_2^* -weighted acquisition on which microhaemorrhages are actually read, and on that sequence its era-

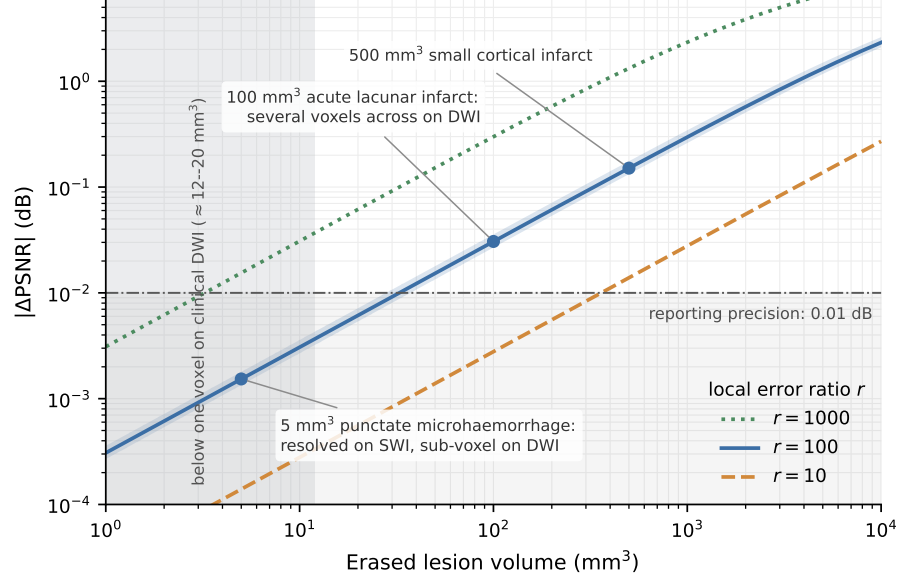


Figure 2: The PSNR cost of erasing a focal lesion, from Equation (2). Curves give $|\Delta\text{PSNR}|$ against erased lesion volume for three local error ratios r at a 1.4 L brain; the shaded band around $r = 100$ spans the 1.2–1.6 L range. Markers show the three clinical anchors used in the text. The horizontal line at 0.01 dB is the two-decimal precision at which PSNR is conventionally reported, and the shaded left-hand region is the volume below a single voxel on a clinical DWI acquisition, where erasure is not defined because the finding is not resolved. Every value is computed from the identity under the stated error model; nothing in this figure is measured.

sure costs $f = 3.1\text{--}4.2 \times 10^{-6}$ and $|\Delta\text{PSNR}| = 1.3\text{--}1.8 \times 10^{-3}$ dB, below the precision at which the score is reported at all. Scaling in the other direction, a 500 mm³ cortical infarct costs ≈ 0.15 dB (0.13–0.18 dB across the same brain-volume range). Fig. 2 plots the identity across this range, with the sub-voxel region marked. The argument does not need the extreme case: it holds by one to two orders of magnitude at a lesion size that is unambiguously resolved and unambiguously matters. These are computed bounds from the stated f and r , not measurements from any primary study, and the error model behind them, a uniform background error and a free local error ratio, is an assumption we state rather than a property of any reconstruction.

Loss-function symmetry. MSE and SSIM are indifferent to the sign of the error, since $(\hat{X}_i - X_i)^2 = (X_i - \hat{X}_i)^2$, whereas diagnosis is not: suppressing a true hyperintense lesion means the finding is missed, while adding background

noise costs nothing clinically. An evaluation respecting that asymmetry must penalise the two directions differently, for example

$$\begin{aligned}\mathcal{L}_{\text{asym}}(\hat{X}, X) = & \alpha \sum_{i \in \Omega_{\text{lesion}}} \max(0, X_i - \hat{X}_i)^2 \\ & + \beta \sum_{i \in \Omega_{\text{bg}}} (\hat{X}_i - X_i)^2, \quad (\alpha \gg \beta).\end{aligned}\tag{3}$$

Several included studies already build objectives of this shape: lesion-mask-weighted losses motivated explicitly by the dilution of lesion gradients among background voxels (Lee et al., 2026a), overlap-weighted reconstruction losses that up-weight structure against background (Jayakumar et al., 2023), and gradient-based structure-preservation terms (Zhang et al., 2026). Their existence is evidence that the symmetry problem is recognised in practice, though none of these studies pairs the correction with a reader assessment.

Realism–reward trade-off. Adversarial and perceptual objectives reward outputs that look like the training distribution, which is not the same as outputs faithful to the measurement, so a generative reconstruction can gain plausible texture that was never acquired (Zhou et al., 2024; Bhadra et al., 2021).

Reference dependence. All four metrics presuppose a registered, artifact-free reference. Intra-scan motion invalidates that assumption, and 69 of 263 studies rest their reference standard solely on a fully sampled acquisition, which is valid only where the subject did not move.

4.4. RQ3: what current evaluation paradigms can and cannot establish

Read against those mechanisms, the corpus supports a paradigm-by-paradigm account of what each evaluation approach can demonstrate about prior-induced diagnostic error. Table 8 states the account compactly; the argument behind it is given here, because two of its six rows are analytical claims we derive rather than findings we extracted, and an analysis presented only in a table has not been argued.

What the four empirical paradigms establish. Pixel-wise fidelity metrics establish aggregate agreement with a reference and nothing about focal change:

by (2) the penalty for erasing a resolvable acute infarct is of the order of the reporting precision, so a method can pass any threshold anyone would set while failing on the finding that matters. Perceptual and rank-trained metrics track reader preference better than MSE or SSIM as an optimisation target, but they inherit the variability of the readers they were trained on and, in the one study of this kind in the corpus, were calibrated on synthetically corrupted images rather than on reconstructions of undersampled k -space (Tang et al., 2025), so their transfer to the reconstruction setting is assumed rather than shown. Uncertainty estimates flag where a model is unsure, which is not the same question: a prior confident enough to remove a lesion is confident in the region where it removed it, so posterior variance can be low exactly where the diagnostic error is. Task-based observer assessment does settle the question directly, and it is the only paradigm here that does, which is why its scarcity in this corpus (57 of 263 studies, and 18 paired with a metric) is the review’s central finding rather than an incidental one.

Two limits that follow from the physics, not from the corpus. The first concerns the check that reconstruction papers most often use to argue that an output is faithful to the data: the measurement-space residual $\|\mathbf{A}\hat{X} - Y\|_2 / \|Y\|_2$ against a tolerance. This check cannot detect prior-induced edits at all, and the reason is structural rather than statistical. Write the reconstruction as the sum of its measured and unmeasured components, $\hat{X} = \mathbf{A}^+ \mathbf{A} \hat{X} + (\mathbf{I} - \mathbf{A}^+ \mathbf{A}) \hat{X}$. The second term lies in the null space $\mathcal{N}(\mathbf{A})$, and $\mathbf{A}$ annihilates it by definition, so any amount of structure written there, an erased lesion or an invented vessel alike, leaves the residual exactly unchanged. Undersampling is what creates that null space; the more aggressive the acceleration, the larger the subspace in which a learned prior can write freely without moving the number the field uses to certify data consistency. A residual check is therefore a two-sided measurement-fidelity band and not a safety test, and its apparent reassurance grows least trustworthy precisely where acceleration is highest.

The second limit is the constructive corollary. If prior-supplied content lives in $\mathcal{N}(\mathbf{A})$, then isolating $(\mathbf{I} - \mathbf{A}^+ \mathbf{A}) \hat{X}$ and inspecting it is the natural place to

look for erasure and hallucination, since that is where both must reside (Bhadra et al., 2021; Gottschling et al., 2025). This is the one paradigm in Table 8 that is not represented in the corpus at all, and its practical obstacle is the same one that limits third-party auditing generally: computing the projection requires the forward operator, hence the raw k -space and the coil sensitivities, which commercial inline pipelines do not export (Section 5). Causal and counterfactual auditing, which asks the most direct question of all, whether a specific finding survives a controlled change of acquisition, appears in a small part of the corpus and almost never for physics-constrained k -space reconstruction (Pawlowski et al., 2020; Güngör et al., 2023; Peng et al., 2026).

The answer to RQ3 is therefore not that current evaluation is weak but that it is mismatched: each paradigm in use answers a question adjacent to the safety question, and no paradigm now in use establishes reconstruction safety on its own. The two that could, task-based observer assessment and null-space inspection, are respectively rare and absent, and the requirements in Section 4.7 follow from that pairing.

4.5. Structure of the evaluation gap

The counts above establish that safety-relevant evaluation is scarce. Analysis of the same per-study records establishes where the scarcity comes from, and indicates a division of labour rather than uniform neglect.

The benchmarks cannot show the failure the field is worried about. One hundred and nine of the 263 studies (41.4%) name a public dataset, and across the whole corpus those references resolve to eleven datasets: fastMRI, the Human Connectome Project, ADNI, IXI, UK Biobank, BraTS, OASIS, Calgary–Campinas, M4Raw, the developing Human Connectome Project and TCGA. Dataset labels are multi-valued (136 cohort assignments across the 109 studies), and of the 109, the cohorts named are healthy volunteers or population samples (55, 50.5%), chronic neurodegenerative cohorts (31, 28.4%), brain tumour cohorts (16, 14.7%), or raw- k -space benchmarks not curated for pathology (34, 31.2%). These four shares are multi-label and sum to 124.8%, a study naming

Table 8: What each evaluation paradigm can and cannot establish about prior-induced diagnostic error (RQ3). Rows 1–4 summarize evaluation practice and performance as reported in the included studies; quoted values are representative results from the cited primary studies, not pooled estimates. Rows 5–6 state analytical limits derived by the authors from the acquisition physics (our own analysis; Section 4) and are labelled as such rather than attributed to any primary study.

Evaluation Paradigm	Core Mechanism	What It Can and Cannot Establish	Key References
1. Pixel-wise fidelity metrics	L_2 / SSIM whole-image voxel averaging	Aggregate agreement with a reference only. Cannot resolve focal change: by (2), erasing a resolvable 100 mm^3 lacunar infarct at $r = 100$ costs 0.03 dB. Requires a registered reference	Recorded in 84 of the 144 evaluation-labelled studies
2. Perceptual & learned metrics	Networks trained on radiologist rankings	Tracks reader preference better than MSE/SSIM as an optimisation target ($75.2\% \pm 1.3\%$ and $79.2\% \pm 1.7\%$ against readers' own $61.5\% \pm 5.5\%$). Inherits training-reader variability, and was calibrated on synthetically corrupted data rather than clinically undersampled acquisitions, so transfer is assumed (Section 4.4)	Tang et al. (2025) ($n = 18$ in corpus)
3. Uncertainty estimation	Posterior sampling / Bayesian variance $p(X Y)$	Flags out-of-distribution inputs and regions of low confidence. Does not attribute missing structure to prior vs sampling operator; high confidence in hallucinated regions is possible	Khawaled and Freiman (2024) ($n = 18$ in corpus)
4. Task-based observer assessment	Human reader performance on a diagnostic task; model observers are its automated form, unrepresented here (Barrett et al., 1993; Barrett and Myers, 2004; International Commission on Radiation Units and Measurements, 1996)	Establishes diagnostic sufficiency directly; the gold standard in assessment theory. Costly and reader-variable ($61.5\% \pm 5.5\%$, falling to 25% under absolute Likert scoring), and rarely paired with a metric: only 18/263 report both	Radmanesh et al. (2022), Lang et al. (2024) ($n = 57$ in corpus)
5. Measurement-space residual check	$\ \mathbf{A}\tilde{X} - Y\ _2 / \ Y\ _2$ against tolerance	Cannot detect prior-induced edits: the residual is invariant to anything in $\mathcal{N}(\mathbf{A})$. A two-sided measurement-fidelity band, not a safety test	Common diagnostic check in the reviewed studies
6. Null-space / hallucination mapping	Structure written into the null-space component $(\mathbf{I} - \mathbf{A}^+\mathbf{A})\hat{X}$	Isolates prior-supplied content, where erasure and hallucination must reside. Requires $\mathbf{A}$, hence raw k -space and coil sensitivities	Bhadra et al. (2021), Gottschling et al. (2025); formalization for reconstruction auditing is author-derived

688 two cohorts being counted in both; every other percentage here is single-label.
 689 *Within this corpus, no study benchmarks against a dataset curated for acute is-*
 690 *chemic stroke or intracranial haemorrhage.* The sharper version of this finding
 691 sits in the annotation layer the field has already built. fastMRI+ supplies 7,570
 692 expert bounding-box annotations across thirty brain pathology categories (Zhao
 693 et al., 2022), and not one of the thirty is a haemorrhage category of any kind,
 694 microhaemorrhage included. The apparent counterexample should be stated
 695 plainly rather than left for a reader to find, because it is the category closest
 696 to this article’s own worked example: fastMRI+ *does* carry an image-level *La-*
 697 *cunar Infarct* label (113 annotations across 32 subjects), alongside study-level
 698 *Small Vessel Chronic White Matter Ischemic Change* and a *Global Ischemia*
 699 label recorded for a single subject. It does not rescue the benchmark, and the
 700 reasons are the ones that matter for reconstruction safety. No fastMRI+ cat-
 701 egory distinguishes acute from chronic infarction, so a lacunar label carries no
 702 acuity and cannot identify the acute presentation whose erasure is at issue. The
 703 limit is in the source images as much as in the label set: the annotated se-
 704 ries are axial T2-FLAIR, T1w or contrast-enhanced T1w only, so neither DWI,
 705 on which infarct acuity is established, nor SWI or T_2^* -weighted imaging, on
 706 which microhaemorrhage is read, was annotated at all. Acute vascular pathol-
 707 ogy is therefore absent even where annotation infrastructure exists, and it is
 708 absent for a reason that adding labels alone would not repair. Supplementary
 709 Table S10 sets out the full derivation, including the sub-selection of 1,001 of
 710 the 5,847 fastMRI brain studies, the category listing, and the single-annotator
 711 limitation that travels with these annotations. Stroke-lesion datasets do exist,
 712 ISLES (Hernandez Petzsche et al., 2022) and ATLAS (Liew et al., 2022) among
 713 them, but they are segmentation benchmarks distributed as processed images
 714 without raw k -space, so they cannot serve reconstruction evaluation as released.
 715 This matters mechanically: lesion erasure cannot be measured on cohorts that
 716 contain no acute lesions, so a model can lead every public leaderboard in the
 717 field while remaining untested against the failure mode that would matter most
 718 in an emergency department. Where a pathology-labelled dataset is used at all

719 it is BraTS, whose tumours are large, high-contrast mass lesions, close to the
720 easiest possible case for lesion preservation and the least informative about the
721 lacunar and punctate findings at issue here. The benchmark infrastructure and
722 the metric therefore fail in the same direction.

723 *Rigour is divided between two literatures.* Of the 263 studies, 84 (31.9%)
724 release code but report no reader assessment, 44 (16.7%) report a reader assess-
725 ment but release no code, 122 (46.4%) do neither, and only 13 (4.9%) do both.
726 The division tracks publication venue. Applying the post-hoc classification rule
727 stated in Section 3 to all 263 venues, clinical imaging journals ($n = 131$) carry
728 reader assessment at nearly three times the rate of engineering and methods
729 venues ($n = 78$), which in turn release code at nearly twice the clinical rate;
730 general neuroscience and neuroimaging venues ($n = 37$) sit between. Supple-
731 mentary Table S12 gives the full cross-tabulation. The rule leaves 17 of the 263
732 journal titles unmatched, and those are reported as a fourth *unclassified* group
733 rather than forced into one of the three; the four groups between them account
734 for all 57 reader studies and all 97 code releases. The tendency for the two
735 practices to fall in different studies rather than the same ones gives a Fisher
736 exact odds ratio of 0.43 ($p = 0.013$). The estimate is insensitive to the one
737 labelling choice that could move it: taking the reader subset as the 61 studies
738 originally labelled at extraction rather than the 57 verified at appraisal gives an
739 odds ratio of 0.43 and $p = 0.010$. We nonetheless report the association descrip-
740 tively rather than as a hypothesis test, because the venue grouping it rests on is
741 a post-hoc keyword rule that was not part of the fixed protocol, and a p -value
742 computed on a partition chosen after seeing the data does not carry its nominal
743 meaning. The direction and magnitude are what we rely on. The magnitude
744 is the part that does not depend on the choice: computational verifiability and
745 diagnostic verifiability are largely held by different communities, only one study
746 in twenty holds both, and a safety argument requires both.

747 *Awareness of a failure mode does not produce the evaluation that would detect*
748 *it.* Generative studies record hallucination or fidelity instability in 31/80 (38.8%)
749 of cases, the highest share among the four largest method families, against

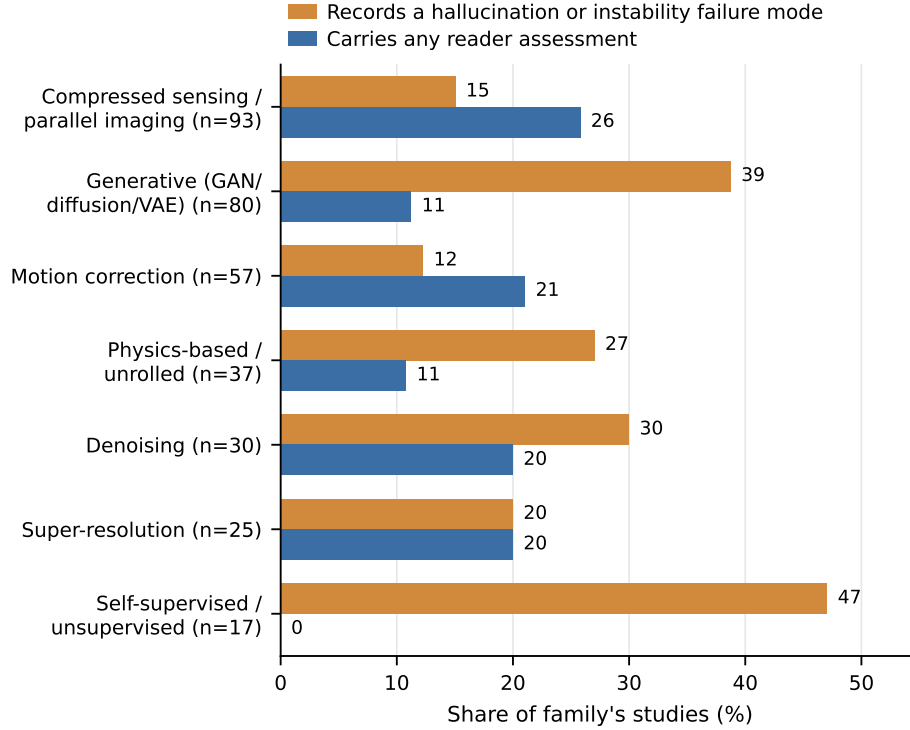


Figure 3: Reader assessment against hallucination awareness, by method family (recorded extraction labels; lower bounds). Bars are the share of each family’s studies that record a hallucination or instability failure mode (orange) and the share that carry any reader assessment (blue), using the verified reader subset ($n = 57$). Bar labels are rounded to the nearest integer; the one-decimal values quoted in the text are the same quantities (for example 15 against 15.1%, 39 against 38.8%). Generated from the released per-study file (`included_characteristics.csv`) by `5_Code/make_figures.py`; every value is recomputable from the deposit.

750 15.1% for compressed-sensing and parallel-imaging work and 12.3% for motion
751 correction. Yet at 9/80 (11.3%) they are also among the two least reader-
752 evaluated of those families, while the small self-supervised group records the
753 corpus’s highest hallucination share (8/17, 47.1%) and no reader assessment at
754 all (0/17); per-family shares are in Table 9 and Fig. 3. The sub-fields that most
755 clearly recognise that their models can synthesise structure are the least likely
756 to run the test capable of detecting it.

757 *Evaluation is not broadening as the field grows.* Reader-assessment share
758 fell from 32.4% (1995–2019) to 18.2% (2020–2022), recovering only partially to

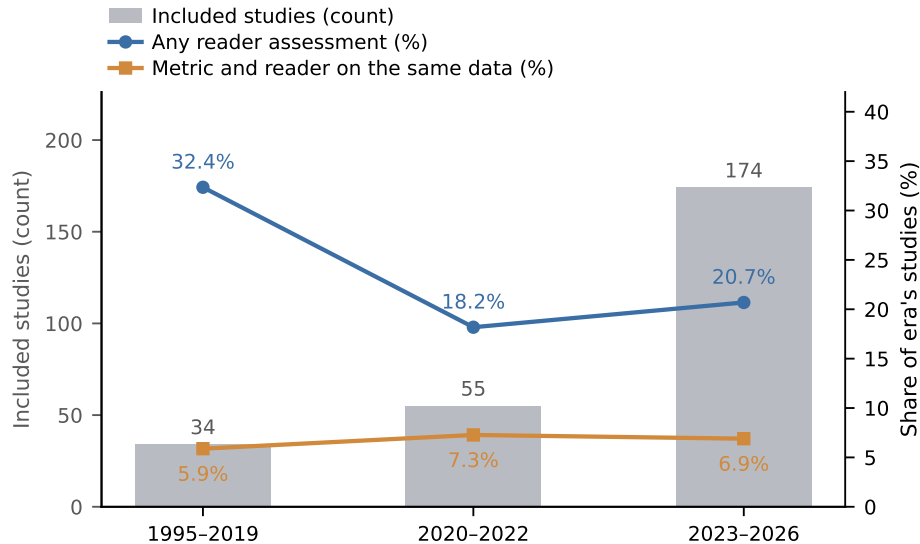


Figure 4: Evaluation practice across publication eras. Grey bars give the number of included studies per era (34, 55, 174); lines give the share with any reader assessment and the share co-reporting a fidelity metric and a reader assessment on the same data, using the verified reader subset ($n = 57$). What recovery there is in 2023–2026 comes from subscription-walled clinical journals (52.0% reader assessment in that stratum against 15.4% in the open remainder; Section 5.2). Generated from the released per-study file by `5_Code/make_figures.py`; counts are lower bounds from recorded extraction labels.

20.7% (2023–2026) while the corpus grew fivefold (Table 9 and Fig. 4). What recovery there is comes entirely from the subscription-walled clinical literature: among 2023–2026 studies obtainable only through title-level library subscriptions, reader assessment runs at 52.0% (13 of 25), against 15.4% (23 of 149) in the openly accessible remainder of that era. Reader evidence, where it exists, sits disproportionately behind paywalls, and a review that cannot retrieve those texts will understate it; Section 5.2 takes the direction of that bias seriously rather than dismissing it. Among the 144 studies recording any evaluation paradigm, four in five record exactly one, and the share recording two or more fell across the same periods (Supplementary Table S12). Of the 84 studies recording a pixel-wise metric, 60 (71.4%) record no reader, uncertainty or perceptual measure beside it.

Readers are not displaced away from the sequences where the stakes are highest. One expectation is not borne out. The comparison must be made

Table 9: Evaluation practice across publication eras and method families (recorded labels; lower bounds). Reader-assessment share fell as the corpus grew and recovers only partially in the latest era, and that recovery is carried by subscription-walled clinical journals. Among the four largest method families the generative group, the one most associated with hallucination, is one of the two least reader-evaluated and is indistinguishable from physics-based unrolled work; the small self-supervised family ($n = 17$) records no reader assessment at all. Reader shares use the verified reader subset ($n = 57$; Section 3), so counts sum to 57 across the era rows and are multi-label across the family rows.

Era	n	Reader	Uncert.	Co-report
1995–2019	34	32.4% (11/34)	5.9%	5.9%
2020–2022	55	18.2% (10/55)	5.5%	7.3%
2023–2026	174	20.7% (36/174)	7.5%	6.9%
Method family (multi-label)				
Family	n	Reader share		
Compressed sensing / parallel imaging	93	25.8% (24/93)		
Motion correction	57	21.1% (12/57)		
Super-resolution	25	20.0% (5/25)		
Denoising	30	20.0% (6/30)		
Generative (GAN/diffusion/VAE)	80	11.3% (9/80)		
Physics-based / unrolled	37	10.8% (4/37)		
Self-supervised / unsupervised	17	0.0% (0/17)		

on disjoint strata to mean anything, because pulse sequence is a multi-label field and the naive contrast double-counts the 107 studies that acquire both an acute-pathology and a structural sequence (Supplementary Table S11 gives the correction). That the two disjoint strata below sum to $23 + 84 = 107$, the same number, is a third coincidence of unrelated tallies rather than a transcription error, verified against the released per-study file as the two flagged in Section efsec:requirements are. Restricted to studies acquiring one group and not the other, reader assessment runs at 5/23 (21.7%) for DWI or FLAIR without T1w or T2w against 10/84 (11.9%) for T1w or T2w without DWI or FLAIR — a difference in the direction one would hope for, resting on 23 studies in the smaller stratum and not approaching significance (Fisher exact odds ratio 2.06, $p = 0.31$). We report it as descriptive and draw only the negative conclusion it supports: there is no evidence that reader assessment is *displaced away* from the sequences on which acute pathology is read. The deficit is one of scarcity and of distribution across research communities, not of misdirection within the clinical literature.

789 *4.6. Methodological quality and risk of bias*

790 Across all 57 QUADAS-2 appraised diagnostic and reader-
791 ological quality was highest for Flow and Timing (45/57 low risk of bias, 78.9%),
792 reflecting the standard retrospective undersampling paradigm where reconstruc-
793 tion and reference share the same acquisition timepoint without temporal degra-
794 dation. Reference Standard and Index Test both sit at 26/57 low risk (45.6%),
795 each now below half, with high ratings arising where radiologists scored acceler-
796 ated images with reference ground truth visible or where lesion truth was defined
797 by the comparator images themselves (incorporation bias). Patient Selection re-
798 mains the weakest domain (15/57 low risk, 26.3%; 19/57 unclear, 33.3%; 23/57
799 high risk, 40.4%), as cohorts were predominantly retrospective, single-centre,
800 and pathology-enriched without explicit prospective enrolment rules. Applica-
801 bility concerns follow a similar distribution, with low concerns in Patient Selec-
802 tion for 36/57 (63.2%), Index Test for 30/57 (52.6%), and Reference Standard
803 for 39/57 (68.4%).

804 The two rounds do not place the weaknesses in quite the same order, and
805 the pooled shares above differ from the first round's in every one of the seven
806 domains. Low-risk shares fell for Index Test (53.8% to 45.6%), applicability of
807 the Index Test (59.0% to 52.6%), Flow and Timing (82.1% to 78.9%), Refer-
808 ence Standard (46.2% to 45.6%) and applicability of Patient Selection (64.1% to
809 63.2%); they rose for Patient Selection risk of bias (23.1% to 26.3%) and applica-
810 bility of the Reference Standard (59.0% to 68.4%). The two largest movements
811 concern the index test and the applicability of the reference standard, and the
812 first of these runs against the appraised studies. The second-round studies are
813 therefore not a favourable addition to the appraisal, which is why the pooled
814 figures are the ones reported here. The first-round shares quoted here already
815 incorporate the five studies re-appraised on recovered full text.

816 Corpus-wide reporting transparency evaluated against STARD principles re-
817 vealed substantial omissions across the broader 263 studies: 109/263 (41.4%)
818 provided no identifiable reference standard (47 stating none at all, Table 3,
819 and a further 62 naming a ground truth they left unspecified), scanner field

strength was unreported in 54/263 (20.5%), and 166/263 (63.1%) did not make
 code openly available. Overall, 220/263 studies (83.7%) omitted at least one
 key methodological indicator (field strength, reference standard, or open code).
 Two directions must be kept apart here, and conflating them has caused con-
 fusion elsewhere in this literature. With respect to the *reconstruction methods*,
 this reporting and selection pattern is anti-conservative: retrospective single-
 centre evaluation on curated data flatters reported performance relative to what
 prospective clinical deployment would show. With respect to *this review’s own*
conclusion, the same pattern is conservative: if published performance is flat-
 tered and failure is under-reported, then the scarcity of safety-relevant evalua-
 tion we document is an understatement, and the true gap is at least as large
 as the one measured. The safety argument turns on the second direction and
 is not weakened by the first. Detailed domain-level judgments and supporting
 quotations are available in the open project repository.

4.7. Requirements for safety-oriented evaluation

Read together, the synthesis implies five requirements that any evaluation
 intended to establish reconstruction safety must meet, each stated with the cor-
 pus evidence that motivates it and none met by more than a small minority
 of included studies. **R1, focal task-based measurement:** every global fi-
 delity score must be accompanied by a region-of-interest or task-based observer
 measure, because spatial averaging is the first mechanism by which lesion-scale
 error disappears (60/84, or 71.4%, of pixel-metric studies record nothing beside
 the metric; this equals the artifact share of the *R*-reporting subset in Table 6
 (45/63) to the decimal, which is a coincidence of two unrelated tallies rather than
 a transcription error, verified against the released per-study file). **R2, paired**
metric–reader reporting: metric validity claims require a reader comparison
 on the same data (6.8% of studies provide one). **R3, reference-independent**
validity: evaluation must remain interpretable when no registered artifact-free
 reference exists (26.2% rest solely on a fully sampled acquisition). **R4, robust-**
ness under acquisition shift as a primary outcome, reported as a result

rather than a caveat; robustness or distribution shift is recorded as a failure-mode concern in 76/263 studies (28.9%), almost always in the discussion rather than as a pre-specified endpoint. **R5, hallucination-specific detection**, operating on the image side, since measurement-space residual checks are provably blind to null-space edits. To these the analysis in Section 4.5 adds a precondition: benchmarks containing the pathology whose erasure is at issue, without which R1, R2 and R5 cannot be exercised at all.

5. Discussion

Three conclusions follow. First, the comparison on which the field’s central proxy rests is almost never made: no included study reports a rank correlation between a pixel-wise fidelity metric and radiologist diagnostic assessment, and only 18 of 263 record the two on the same data at all. Where readers do appear, what is measured is mostly agreement between readers, and it is weak and condition-dependent. The finding therefore rests more on the absence of measurement than on the measurements that exist. Second, the reason is structural, through the four mechanisms of Section 4: background voxels average focal error away, the loss functions are sign-symmetric where diagnosis is not, adversarial terms pay for invented texture, and every one of these metrics assumes a registered reference that motion destroys. The mechanisms are mathematical properties of the metrics themselves; how often each matters clinically is the part the thin reader evidence leaves less certain. Third, no evaluation paradigm now in use settles the safety question by itself, a conclusion resting on the paradigm-by-paradigm synthesis of Section 4.

The analysis of where the gap comes from, which was not part of the fixed protocol and is reported as exploratory throughout, changes what would follow from these findings if it holds. The shortfall does not look like a uniform oversight; it co-occurs with two structural features. Evaluation practice is divided between a community that releases code and a community that runs readers, with only 4.9% of studies doing both (an association we interpret descriptively

879 in Section 4.5, its venue partition being post-hoc), and the benchmark datasets
880 named in this corpus contain no acute pathology, so the failure mode of great-
881 est clinical consequence is not merely under-measured but unmeasurable on the
882 datasets in common use. If that association is causal, which this design cannot
883 establish, recommendations addressed to individual authors will not close the
884 gap; what would be required is benchmark construction and reporting conven-
885 tions that force the two halves of the evidence together.

886 This matters because accelerated reconstruction has left the research setting.
887 Deep-learning acceleration, denoising and image enhancement ship inside com-
888 mercial MRI products, several of them cleared as software as a medical device:
889 one such tool, an FDA-cleared vendor-agnostic DICOM-domain enhancement
890 product, is evaluated in this corpus by a prospective multicentre multireader
891 trial (Bash et al., 2021), which is also a reminder that a cleared product can
892 be validated on one sequence family and deployed across others. Published
893 guidance in both jurisdictions increasingly asks manufacturers to demonstrate
894 performance for the device’s stated intended use and under conditions repre-
895 sentative of deployment rather than on curated development data: the FDA,
896 Health Canada and MHRA guiding principles on good machine learning practice
897 ask that datasets be representative of the intended patient population and that
898 testing be independent of training (U.S. Food and Drug Administration et al.,
899 2021), the EU AI Act imposes accuracy and robustness obligations on high-risk
900 AI systems (European Parliament and Council of the European Union, 2024),
901 and the Medical Device Regulation requires clinical evidence appropriate to the
902 device’s intended purpose (European Parliament and Council of the European
903 Union, 2017). Our reading of that guidance is that aggregate pixel-wise fidelity
904 cannot support a claim about small-lesion detection, because (2) shows the ag-
905 gregate does not move when the small lesion is removed. That reading is ours:
906 no regulator has reviewed or endorsed it, and none of the three instruments
907 names an image-quality metric.

908 A practical obstacle compounds it, and it is the reason the requirements in
909 Section 4.7 are addressed to infrastructure rather than to authors. Commer-

910 cial inline pipelines export DICOM images after undisclosed normalisation, coil
 911 combination and vendor post-processing, and the raw complex k -space typi-
 912 cally never leaves the scanner. Every method in Table 8 that could detect a
 913 prior-induced edit needs something the exported DICOM does not carry: the
 914 null-space projection needs the forward operator and the coil sensitivities, a
 915 counterfactual audit needs the ability to re-run the acquisition under a con-
 916 trolled change, and even a measurement-space residual needs Y . Independent
 917 third-party auditing of a deployed reconstruction is therefore not merely difficult
 918 but structurally impossible without an open raw-data interface, and no amount
 919 of reporting discipline by authors of research papers can supply one.

920 Three things follow that are worth stating as claims rather than as recom-
 921 mendations.

922 *The cheapest useful change costs no new data.* A reconstruction paper that
 923 claims its method preserves small lesions can state, from (2) and its own brain
 924 volume and assumed local error ratio, the change in global PSNR that erasing
 925 the smallest lesion it claims to preserve would produce. For a lacunar infarct
 926 that is about 0.03 dB, which is smaller than the difference the paper is likely to
 927 be reporting between its method and its baseline. This is arithmetic, not a new
 928 experiment, and it converts an implicit claim into an explicit one at the point
 929 where the claim is made. Nothing in this review suggests such a line would be
 930 difficult; the finding is that nobody currently reports it.

931 *The two-community split is self-reinforcing, which is why exhortation has*
 932 *not worked.* Each community’s evaluation is close to the best it can do with
 933 the data it can obtain. Engineering venues evaluate on the datasets that carry
 934 raw k -space, and those contain healthy volunteers, chronic cohorts and tumours
 935 rather than acute pathology, so a reader study run on them would not test lesion
 936 preservation even if it were run. Clinical venues have the pathology, the readers
 937 and the acute presentations, and no access to raw k -space or to the vendor’s
 938 reconstruction, so a reproducibility claim is not available to them. The 4.9% of
 939 studies doing both are not more diligent than the rest; they are the studies that
 940 happened to sit where both kinds of access coincide. An instruction to individual

941 authors to run more reader studies asks most of them to do something the data
942 available to them makes uninformative.

943 *The missing object is a dataset, and its requirements are now specifiable.*

944 The public resources discussed in this review supply each necessary compo-
945 nent separately, and the distinction between what the corpus uses and what
946 merely exists is part of the point: raw multi-coil k -space is available in three
947 of the eleven datasets the included studies actually name (fastMRI, Calgary-
948 Campinas, M4Raw); acute cerebrovascular pathology with expert segmentation
949 exists in ISLES and ATLAS, which no included study uses because they are dis-
950 tributed as processed images; and reader-level pathology annotation on brain
951 MRI exists in fastMRI+, which no included study uses either. None combines
952 them, and the combination is what every requirement in Section 4.7 presup-
953 poses: raw k -space so the forward operator is known, acute lesions confirmed
954 against a standard independent of the images being evaluated, lesion-level an-
955 notation so a focal figure of merit can be computed, and a licence permitting
956 redistribution so the benchmark can be used by people who did not collect it.
957 That object does not exist, and until it does the safety question is not merely
958 unanswered but unaskable on public data.

959 5.1. Priorities for future research

960 Three of the five requirements in Section 4.7 translate directly into research
961 priorities, and we do not restate them here: paired metric-reader reporting
962 (R2), robustness under acquisition shift reported as a primary endpoint rather
963 than a caveat (R4), and benchmarks carrying the pathology whose erasure is at
964 issue, which is the precondition on which R1, R2 and R5 all depend. Calibrated
965 voxel-level uncertainty that tracks local reconstruction error (Khawaled and
966 Freiman, 2024) belongs with R4 and is the form in which R4 is most likely to
967 be actionable for generative and self-supervised architectures.

968 A fourth priority does not follow from anything this review extracted, and we
969 separate it for that reason. Many current pipelines operate slice by slice and in-
970 troduce inter-slice discontinuities on isotropic 3D acquisitions, so reconstruction

971 and auditing frameworks should model 3D anatomical coherence directly (Peng
972 et al., 2026) while meeting scanner-side latency constraints. Slice-wise versus
973 volumetric processing was not an extraction field in this review. We state the
974 priority on mechanistic grounds and on the cited work, not on a corpus count.

975 5.2. Threats to validity

976 The strongest conclusions rest on a small subset of reader-based studies, and
977 we state that prominently rather than in passing: 57 of 263 studies carry any
978 reader assessment and 18 pair it with a metric. Four threat classes bound what
979 follows from that base.

980 *How to read the limitations below.* They are not all bounded in the same
981 way, and treating them as one undifferentiated list of caveats would misrepresent
982 what this review knows about itself. Three kinds appear. Some are *bounded*
983 *by released evidence*: the non-retrieval analysis, the paywall stratification and
984 the appraisal-agreement decomposition are each quantified from files a reader
985 can recompute. Some are *bounded by argument*: the direction of publication
986 bias and of the reporting asymmetry is reasoned rather than measured, and we
987 say which way each points. One is simply *unbounded*, and it is the sharpest:
988 extraction has no reliability estimate.

989 Three properties of the extraction layer follow from that and should be
990 weighed together. First, its depth is deliberately shallow. Method, failure-
991 mode and evaluation categories were assigned from title, abstract and controlled
992 vocabulary, so the prevalence counts are floors on what studies *report at that*
993 *level* rather than measurements of what they did. Second, the shallowness is
994 visible in the counts themselves: only 84 of 263 studies are recorded as reporting
995 a pixel-wise fidelity metric, and 39 of the 57 verified reader studies are recorded
996 as reporting none, which for this literature is implausible as a description of
997 practice and entirely plausible as a description of abstracts. A further 119 of
998 263 studies (45.2%) carry no evaluation-paradigm label at all, which is why
999 we report both the /263 and the /144 denominators wherever the distinction
1000 matters. Third, the headline 18/263 mixes two depths: the reader label was

1001 verified against full text and corrected at appraisal, while the metric label was
1002 not. That asymmetry biases the intersection downward by an unknown factor,
1003 and it is the single respect in which our headline figure is least defensible as a
1004 point estimate — though the direction is conservative, since correcting it could
1005 only widen the measured gap.

1006 What we offer in place of a reliability coefficient is auditability, and the
1007 substitution is deliberate rather than a shortfall we failed to notice. Every
1008 extracted value is released with the verbatim quotation that supports it, so a
1009 reader can check whether an assignment is *correct*, which is a stronger and more
1010 useful guarantee than a coefficient showing that two people were *consistent*. It
1011 is not a substitute for reliability in the statistical sense, and we do not claim
1012 it is: consistency and correctness bound different failure modes, and only the
1013 first is estimable from our records. A duplicate-extraction exercise on a random
1014 subsample, reported with a κ , is the obvious next step and we name it as an
1015 outstanding task rather than a completed one.

1016 *Construct validity.* We searched with broad, high-sensitivity terms for
1017 architectures, artifacts and evaluation paradigms, and the corpus spans the
1018 reconstruction-safety construct, from pixel-wise metrics through perceptual
1019 losses to reader studies. What the corpus cannot guarantee is that a metric
1020 means the same thing wherever it appears: studies differ in what they treat as
1021 the quantity of interest, so a rank correlation with reader scores in one paper
1022 is not straightforwardly the same measurement as in another, and grouping
1023 such values under one heading is an interpretation we make, not a property of
1024 the data. The venue classification underlying Section 4.5 is likewise a post-hoc
1025 keyword assignment, released with the data rather than extracted from the
1026 studies.

1027 *Internal validity.* A protocol fixed before screening began, a documented
1028 search across seven databases, and independent screening and appraisal by three
1029 reviewers provide robust protection, and every extracted value is released with
1030 the verbatim snippet supporting it. Five specific threats bound these findings.
1031 First, 81 of the 1,149 candidate reports remain without obtainable text through

1032 structural licensing barriers. Screening the nearest obtainable stratum — the
 1033 341 reports recovered only through title-level subscriptions — found 86.2% ineli-
 1034 gible on the same criteria, so at that rate the 81 would contribute on the order of
 1035 11 studies rather than 81. That bounds the missing evidence without establish-
 1036 ing that the bias is nil. Non-retrieval is demonstrably *not* random with respect
 1037 to the variable this review measures: within the same stratum, 2023–2026 stud-
 1038 ies carry reader assessment at 52.0% against 15.4% in the openly accessible
 1039 literature of that era. Both findings point the same way. Reader evidence is
 1040 over-represented behind paywalls, so a review that cannot retrieve those texts
 1041 understates how much reader assessment the field performs, and our headline
 1042 scarcity figures are conservative bounds rather than point estimates. Second,
 1043 four included studies were assessed on abstract and controlled vocabulary alone
 1044 and one on partial text; a sensitivity analysis recomputing all headline metrics
 1045 on the 258 studies verified against complete full text leaves every finding direc-
 1046 tionally stable (e.g., metric–reader co-reporting remains 7.0% vs 6.8%). Third,
 1047 every QUADAS-2 tally in this article rests on the whole verified reader subset
 1048 ($n = 57, 399$ domain judgements), so none is computed on a partial base. The
 1049 second round moved all seven domain distributions, and the movements that
 1050 most concern the index test run against the appraised studies: low risk of bias
 1051 for the Index Test falls from 53.8% to 45.6%, and high applicability concern
 1052 for the Index Test rises from 33.3% to 40.4% (Section 4). Four studies remain
 1053 abstract-only and are appraised on that basis. Fourth, reliability is measured
 1054 for screening and for appraisal but not for extraction, and extraction is the
 1055 sharpest methodological limitation of the review. The screening and appraisal
 1056 coefficients, and the decomposition that must be read with the appraisal figure,
 1057 are given in Section 3 and not restated here; the consequence for this section
 1058 is that 126 of the 399 QUADAS-2 judgements were reached at a level of agree-
 1059 ment a single rater could not be relied on to reproduce, so the domain shares
 1060 in Table 3 carry that uncertainty. It does not propagate to the prevalence find-
 1061 ings, which rest on extraction labels rather than QUADAS-2 ratings, but it
 1062 bounds how firmly the risk-of-bias picture can be stated. Neither screening nor

appraisal agreement transfers to extraction. The duplicate-screening sample is
 also smaller than the 300–400 records a referee might reasonably ask for, and
 the interval reflects that. Fifth, four studies recorded at extraction as observer
 or reader designs were found at appraisal to contain no human reader and were
 reclassified as algorithmic, moving the reader subset from 61 to 57 and the al-
 gorithmic set from 202 to 206. The effect on the headline numbers is small and
 in the direction of a larger gap: metric–reader co-reporting moves from 7.6%
 to 6.8%, the 2023–2026 reader share from 23.0% to 20.7%, the code-and-reader
 intersection from 5.3% to 4.9%, and the Fisher exact p from 0.010 to 0.013 at
 an odds ratio that moves only from 0.43 to 0.43. Those four corrections are
 also the source of the 4/61 (6.6%) error rate quoted above, and the design field
 is the only one that received three-reviewer full-text verification: a comparable
 rate elsewhere would not overturn any finding here, since the headline shares are
 separated from their comparators by more than that margin, but it is a source
 of variability measured in one place and assumed in the rest.

External validity. Among the 209 studies stating a field strength the corpus
 covers 1.5, 3 and 7 Tesla (21, 153 and 35 respectively), together with several
 vendors’ coil configurations and the major pulse sequences; the remaining 54
 (20.5%) state none, and nothing about field-strength coverage should be read
 into them. Tasks, protocols, architectures, datasets and metrics otherwise vary
 substantially, which is why we report the direction and consistency of find-
 ings rather than their magnitudes. Many algorithmic studies evaluate on retro-
 spectively undersampled single-centre data, which flatters performance relative
 to prospective multi-centre use; where the public data were themselves pre-
 processed before undersampling, that inflation is larger again (Shimron et al.,
 2022). Prospective multicentre reader evidence is thin, and the examples this re-
 view leans on each carry a boundary that must travel with them: the emergency-
 department comparison is single-centre and its one discrepant finding is a single
 case (Lang et al., 2024); the widest-ranging acceleration reader study is retro-
 spective and single-centre (Radmanesh et al., 2022); and the largest prospective
 multicentre multireader trial evaluated 3D T1-weighted acquisitions only, with

an image-domain DICOM enhancement tool rather than reconstruction from undersampled k -space, reported improved rather than degraded lesion conspicuity, and has several authors affiliated with the vendor of the evaluated tool (Bash et al., 2021) — we cite it for what it does establish, that such tools are cleared and deployed. Finally, the benchmark concentration of Section 4.5 — eleven public datasets named across the corpus, not one curated for acute ischaemic stroke or intracranial haemorrhage — caps how far any leaderboard result can be carried into emergency imaging.

Conclusion validity. Because pixel-wise metrics agree only loosely with radiologist judgement (RQ1), a study resting on PSNR and SSIM alone supports a safety claim less well than one with readers or uncertainty-aware observers, and we weight the evidence accordingly. Publication bias favouring successful reconstructions is a material limitation and works in the same direction, making the reported prevalence of failure evidence an understatement: a literature with little incentive to report failure is exactly the corpus a review about failure must draw on, and the absence of a common effect measure rules out funnel-plot and small-study asymmetry tests. We applied no certainty framework such as GRADE, since the review produces no pooled effect estimates for such a framework to rate. Prevalence counts derive from recorded extraction labels and understate true practice, and no range we quote should be read as an interval estimate rather than as values from named studies.

6. Conclusion

We reviewed 263 primary studies on the safety, artifacts and fidelity of deep learning-based brain MRI reconstruction. Wherever metrics and readers have been compared, agreement between conventional fidelity scores and radiologist judgement is weak, and at most 18 of the 263 studies record the paired metric–reader evaluation capable of exposing that divergence. Every prevalence figure here is a floor derived from screening-level labels rather than a measurement of practice, and although screening agreement was high, the extraction labels

1123 those floors rest on carry no reliability estimate of their own. The weakness
1124 is structural rather than incidental: the four mechanisms identified under RQ2
1125 explain why, and the safety question is not one that any current evaluation
1126 paradigm answers alone.

1127 The contribution we would emphasise is the account of why the gap persists,
1128 and we offer it as a hypothesis these data are consistent with rather than as
1129 an established structural fact: the venue grouping it rests on is post-hoc, so
1130 although the association between code release and reader assessment is strong
1131 and would be conventionally significant on a pre-specified partition, we do not
1132 read it as a test (Section 4.5). What does not depend on that classification
1133 is the magnitude. Only 4.9% of studies both release code and run readers, so
1134 computational verifiability and diagnostic verifiability are, as recorded, held by
1135 different studies roughly nineteen times in twenty; and no dataset named in this
1136 corpus is curated for acute stroke or intracranial haemorrhage. A reconstruction
1137 model can therefore satisfy every metric and every public benchmark in current
1138 use while the question of whether it preserves a small acute lesion has never been
1139 asked. Closing that gap requires benchmarks that contain the pathology at issue
1140 and reporting conventions under which no pixel-wise metric appears alone, and
1141 until both exist, claims that accelerated reconstruction is diagnostically safe will
1142 continue to rest on measurements that cannot see the failures that matter.

1143 One change is available immediately and needs no new data. Equation (2)
1144 converts any lesion volume into the PSNR movement its erasure would cause, so
1145 an author reporting a reconstruction at a stated acceleration can state, alongside
1146 the global score, the smallest focal change that score is capable of registering
1147 at their image size. For a 100 mm³ infarct that figure is around 0.03 dB, which
1148 is below the precision at which PSNR is conventionally reported. Publishing
1149 that bound next to the metric costs one line, requires no reader and no new
1150 acquisition, and makes explicit what the metric is silent about. It is the cheap-
1151 est useful step towards evaluation that could detect the failure this review is
1152 concerned with.

1153 **Funding Statement**

1154 This research was supported by institutional research computing infrastruc-
1155 ture and scholarship funding from RMIT University. The funder had no role in
1156 the design of the review, the search strategy, the eligibility criteria, the screening
1157 or appraisal of studies, the synthesis or interpretation of findings, the writing
1158 of this report, or the decision to submit it for publication. No external, com-
1159 mercial, or industry funding was received, and no vendor of MRI hardware or
1160 reconstruction software contributed financial or non-financial support to this
1161 work.

1162 **Ethics statement**

1163 This review synthesises secondary data from published, de-identified study
1164 reports. It involved no human participants, no animal experimentation and
1165 no collection of private health data, so institutional review board approval and
1166 patient informed consent were not required.

1167 **Data and code availability**

1168 The review protocol, executed search strategy, screening criteria, every
1169 title/abstract and full-text decision with its recorded reason, the com-
1170 pleted PRISMA 2020 checklist, the per-study extraction and QUADAS-
1171 2 worksheets for all 263 included studies, the analysis code, and the
1172 duplicate-screening sample and per-reviewer appraisal worksheets with their
1173 agreement computations (`kappa_worksheet_214_HUMAN_*.csv`, `kappa_hu-`
1174 `man_results.csv`, `reviewer_worksheet_R*.csv` and `quadas2_three_re-`
1175 `viewer_worksheet.csv`) are openly available on the Open Science Frame-
1176 work at <https://doi.org/10.17605/OSF.IO/EUXA8> and mirrored at <https://github.com/dmai287/brain-mri-recon-safety-slr>, archived at Zenodo
1177 under DOI 10.5281/zenodo.22100024 (release v1.0.3, commit 8c856ba;
1178 concept DOI 10.5281/zenodo.21836966). Every count reported here is regen-
1179 erable from `included_characteristics.csv` in that release, and the deposit
1180

1181 encodes the same reader/algorithmic partition (57/206) that this article re-
1182 ports, with the four reclassified studies flagged individually. The two figures
1183 generated from that file and the Δ PSNR identity figure are produced by
1184 `5_Code/make_figures.py` and `5_Code/make_dpsnr_figure.py`. The deposit
1185 is retrospective and is documented as such. Full-text PDFs of the included
1186 studies are not redistributed, as publisher licence terms do not permit it; the
1187 DOI list in the released bibliography (`included_studies_263.bib`) identifies
1188 the corpus in full.

1189 **CRediT authorship contribution statement**

1190 **Dat Tat Mai:** Conceptualization, Methodology, Software, Investigation,
1191 Data curation, Formal analysis, Visualization, Project administration, Writing
1192 – original draft, Writing – review & editing. **Thai Viet Pham:** Investiga-
1193 tion, Data curation, Validation, Resources, Writing – review & editing. **Thu**
1194 **Nguyen Thi Dang:** Investigation, Data curation, Validation, Writing – re-
1195 view & editing. **James Jin Kang:** Supervision, Validation, Writing – review
1196 & editing.

1197 **Declaration of competing interest**

1198 The authors declare that they have no known competing financial interests or
1199 personal relationships that could have appeared to influence the work reported
1200 in this paper.

1201 **Declaration of generative AI and AI-assisted technologies in the** 1202 **manuscript preparation process**

1203 During the preparation of this work the authors used generative AI technolo-
1204 gies solely to assist with language editing and proofreading of the manuscript
1205 text. After using these tools, the authors reviewed and edited the content as
1206 needed and take full responsibility for the content of the published article. All

study screening, data extraction, and quality appraisals reported in this review
were conducted manually by the authors.

Appendix A. The included studies and how to trace a claim to them

All 263 included primary studies appear in the reference list, so the evidence base is part of the article of record rather than of the supplement alone. Because the synthesis is narrative and most claims are grouped rather than attached to a single study, the reference list alone does not show which studies produced a given number. That mapping is carried by `included_characteristics.csv`, which holds one row per included study with its extracted method family, evaluation paradigm, failure modes, sequences, field strength, reference standard and code availability, each with the verbatim quotation supporting it, so any count, share or crosstab in Section 4 can be reproduced and the contributing studies listed by filtering that file. Supplementary Tables S1–S3 give the same 263 studies as a numbered index by publication year, and `included_characteristics_supplement.csv` adds the per-study QUADAS-2 and reproducibility ratings.

The reference list therefore contains two kinds of entry, and a reader needs a rule to tell them apart. It prints 299: the 263 included studies, carried into the list under PRISMA 2020 item 17, and 36 narrative-only works cited in argument. **An entry appearing in the numbered index of Supplementary Tables S1–S3 is a member of the reviewed corpus; an entry that does not is a narrative citation.** Equivalently, and checkable without the supplement, every narrative citation is cited at a specific point in the running text of Sections 1–6, whereas corpus entries not discussed by name appear without an in-text citation. The two lists were generated by independent mechanisms and agree exactly as sets. We would be glad to move the corpus to a separately labelled supplementary reference list if the editor prefers that production route.

1234 **Appendix B. Supplementary material**

1235 Supplementary material associated with this article (Supplementary Ta-
1236 bles S1–S12 and Supplementary Figure S1) is provided as a separate document.
1237 Tables S1–S3 list all 263 included primary studies with full bibliographic details;
1238 Tables S4–S9 give the comparison against related secondary studies, the search
1239 vocabulary and its derivation, the extraction form, the thematic evidence ar-
1240 chitecture, the four metric-failure mechanisms with their evidence sources, and
1241 the provenance of every released data file. Tables S10–S12 carry the support-
1242 ing detail for Section 4.5: the annotation-layer derivation for fastMRI+ (S10),
1243 the disjoint-strata correction for the sequence contrast (S11), and the venue
1244 cross-tabulation with evaluation-paradigm multiplicity by era (S12).

1245 **References**

- 1246 Aali, A., Arvinte, M., Kumar, S., Arefeen, Y.I., Tamir, J.I., 2025. Robust multi-coil
1247 MRI reconstruction via self-supervised denoising. *Magnetic resonance in medicine*
1248 doi:10.1002/mrm.30591. PMID: 40457510.
- 1249 Abed, M.A., Demir, S., Groteklaes, A., Germani, E., Faghihroohi, S., Sabir, H., Albar-
1250 qouni, S., 2026. Mriqt: Physics-aware diffusion model for image quality transfer in
1251 neonatal ultra-low-field MRI. 2026 IEEE 23rd International Symposium on Biomed-
1252 ical Imaging (ISBI) doi:10.1109/isbi61048.2026.11515815.
- 1253 Aggarwal, H.K., Mani, M.P., Jacob, M., 2018. MoDL: Model-based deep learning
1254 architecture for inverse problems. *IEEE Transactions on Medical Imaging* 38, 394–
1255 405.
- 1256 Al-Masni, M.A., Lee, S., Yi, J., Kim, S., Gho, S.M., Choi, Y.H., Kim, D.H., 2022.
1257 Stacked u-nets with self-assisted priors towards robust correction of rigid motion
1258 artifact in brain MRI. *NeuroImage* doi:10.1016/j.neuroimage.2022.119411.
- 1259 Alçalar, Y.U., Akçakaya, M., 2025. Sparsity-driven parallel imaging consistency for
1260 improved self-supervised MRI reconstruction. 2025 IEEE International Conference
1261 on Image Processing (ICIP) doi:10.1109/icip55913.2025.11084355.
- 1262 Altmann, S., Grauhan, N.F., Brockstedt, L., Kondova, M., Schmidtman, I., Paul,
1263 R., Clifford, B., Feiweier, T., Hosseini, Z., Uphaus, T., Groppa, S., Brockmann,
1264 M.A., Othman, A.E., 2024. Ultrafast brain MRI with deep learning reconstruction
1265 for suspected acute ischemic stroke. *Radiology* doi:10.1148/radiol.231938. PMID:
1266 38376403.
- 1267 Altmann, S., Grauhan, N.F., Mercado, M.A.A., Almansour, H., Paul, R.H., Schöffing,
1268 V.I., Ottenhausen, M., Brockmann, M.A., Othman, A.E., 2026. Ultra-fast sub-
1269 minute brain MRI with deep-learning-reconstruction for anesthesia-free emergency
1270 imaging in children. *Clinical Neuroradiology* doi:10.1007/s00062-026-01627-7.

1271 Angella, P., Balbi, L., Ferrando, F., Traverso, P., Varriale, R., Pastore, V.P., Santace-
1272 saria, M., 2025a. Dima: Diffusing motion artifacts for unsupervised correction in
1273 brain MRI images. *IEEE Access* doi:10.1109/access.2025.3634749.

1274 Angella, P., Pastore, V.P., Santacesaria, M., 2025b. Assessing the use of diffusion
1275 models for motion artifact correction in brain MRI. 2025 IEEE 22nd International
1276 Symposium on Biomedical Imaging (ISBI) doi:10.1109/isbi60581.2025.10981000.

1277 Antun, V., Renna, F., Poon, C., Adcock, B., Hansen, A.C., 2020. On instabilities
1278 of deep learning in image reconstruction and the potential costs of AI. *Proceed-*
1279 *ings of the National Academy of Sciences* 117, 30088–30095. doi:10.1073/pnas
1280 .1907377117.

1281 Barrett, H.H., Myers, K.J., 2004. *Foundations of Image Science*. Wiley, Hoboken, NJ.

1282 Barrett, H.H., Yao, J., Rolland, J.P., Myers, K.J., 1993. Model observers for as-
1283 sessment of image quality. *Proceedings of the National Academy of Sciences* 90,
1284 9758–9765.

1285 Bash, S., Wang, L., Airriess, C., Zaharchuk, G., Gong, E., Shankaranarayanan, A.,
1286 Tanenbaum, L.N., 2021. Deep learning enables 60% accelerated volumetric brain
1287 MRI while preserving quantitative performance: A prospective, multicenter, multi-
1288 reader trial. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a7358.
1289 pMID: 34824098.

1290 Bathla, G., Rai, P., Chan, V.E.Y., Nickel, M.D., Kollasch, P., Dueker, K., Fagan, A.J.,
1291 Benson, J.C., Huston, J., Messina, S.A., 2025. Deep-learning accelerated vessel wall
1292 imaging using T1-SPACE at ultra-high-field strength MRI. *American Journal of*
1293 *Neuroradiology* 47, 1424–1431. doi:10.3174/ajnr.a9097.

1294 Bauman, G., Afshari, R., Bieri, O., 2025. Ultra-high-resolution brain MRI at 0.55t:
1295 bstar and its application to magnetization transfer ratio imaging. *Zeitschrift fur*
1296 *medizinische Physik* doi:10.1016/j.zemedi.2024.12.001. pMID: 39824686.

1297 Baz, A.M., Bendella, Z., Katemann, C., Sprinkart, A.M., Weiss, K., Weber, O.M.,
1298 Peeters, J.M., Lehn, N.C., Clauberg, R., Luetkens, J.A., Radbruch, A., Wicht-
1299 mann, B.D., 2026. Deep learning improves image quality in motion-robust and
1300 sedation-free pediatric brain MRI. *European radiology* doi:10.1007/s00330-026-
1301 12482-y. pMID: 41925834.

1302 Begnoche, J.P., Schilling, K.G., Boyd, B.D., Cai, L.Y., Taylor, W.D., Landman, B.A.,
1303 2022. Epi susceptibility correction introduces significant differences far from local
1304 areas of high distortion. *Magnetic resonance imaging* doi:10.1016/j.mri.2022.05.
1305 016. pMID: 35644448.

1306 Behrendt, F., Bhattacharya, D., Mielsing, R., Maack, L., Krüger, J., Opfer, R., Schlae-
1307 fer, A., 2025. Guided reconstruction with conditioned diffusion models for un-
1308 supervised anomaly detection in brain mris. *Computers in biology and medicine*
1309 doi:10.1016/j.combiomed.2025.109660.

1310 Beljaards, L., Nagtegaal, M., Rao, C., Dong, Y., Osch, M.J.P.v., Pezzotti, N., Doneva,
1311 M., Staring, M., 2026. Deep-disorder: Motion correction in 3D MRI via segment re-
1312 construction and registration. *NMR in biomedicine* doi:10.1002/nbm.70286. pMID:
1313 41958180.

1314 Beljaards, L., Pezzotti, N., Rao, C., Doneva, M., Osch, M.J.P.v., Staring, M., 2024.
1315 AI-based motion artifact severity estimation in undersampled MRI allowing for
1316 selection of appropriate reconstruction models. *Medical physics* doi:10.1002/mp
1317 .16918. PMID: 38167996.

1318 Benjamin, A.J.V., Bano, W., Mair, G., Thompson, G., Casado, A., Perri, C.D., Davies,
1319 M., Marshall, I., 2020. Diagnostic quality assessment of ir-prepared 3D magnetic
1320 resonance neuroimaging accelerated using compressed sensing and k-space sampling
1321 order optimization. *Magnetic resonance imaging* doi:10.1016/j.mri.2020.08.025.
1322 PMID: 32890675.

1323 Bhadra, S., Kelkar, V.A., Brooks, F.J., Anastasio, M.A., 2021. On hallucinations
1324 in tomographic image reconstruction. *IEEE Transactions on Medical Imaging* 40,
1325 3249–3260. doi:10.1109/TMI.2021.3077857.

1326 Bian, W., Jang, A., Liu, F., 2024. Improving quantitative MRI using self-supervised
1327 deep learning with model reinforcement: Demonstration for rapid t1 mapping. *Mag-*
1328 *netic resonance in medicine* doi:10.1002/mrm.30045. PMID: 38342980.

1329 Bjørkeli, E.B., Geitung, J.T., Esmaeili, M., 2025. Artificial intelligence-powered four-
1330 fold upscaling of human brain synthetic metabolite maps. *The Journal of interna-*
1331 *tional medical research* doi:10.1177/03000605251330578. PMID: 40257058.

1332 Bollmann, S., Kristensen, M.H., Larsen, M.S., Olsen, M.V., Pedersen, M.J., Øster-
1333 gaard, L.R., O'Brien, K., Langkammer, C., Fazlollahi, A., Barth, M., 2019. Shar-
1334 qnet - sophisticated harmonic artifact reduction in quantitative susceptibility map-
1335 ping using a deep convolutional neural network. *Zeitschrift fur medizinische Physik*
1336 doi:10.1016/j.zemedi.2019.01.001. PMID: 30773331.

1337 Bossuyt, P.M., Reitsma, J.B., Bruns, D.E., Gatsonis, C.A., Glasziou, P.P., Irwig, L.,
1338 Lijmer, J.G., Moher, D., Rennie, D., de Vet, H.C.W., 2015. STARD 2015: An
1339 updated list of essential items for reporting diagnostic accuracy studies. *BMJ* 351,
1340 h5527.

1341 Brunátová, J., Löcke, M., Uribe, S., Bertoglio, C., 2025. Denoising of dual-venc pc-
1342 MRI with large high/low venc ratios. *Magnetic resonance in medicine* doi:10.1002/
1343 mrm.30278.

1344 Byanju, R., Klein, S., Cristobal-Huerta, A., Hernandez-Tamames, J.A., Poot, D.H.J.,
1345 2025. Myelin water imaging from accelerated 3D-grase acquisitions using subspace
1346 constrained reconstruction. *Magma (New York, N.Y.)* doi:10.1007/s10334-025-
1347 01276-w. PMID: 40679773.

1348 Campbell, M., McKenzie, J.E., Sowden, A., Katikireddi, S.V., Brennan, S.E., Ellis,
1349 S., Hartmann-Boyce, J., Ryan, R., Shepperd, S., Thomas, J., Welch, V., Thomson,
1350 H., 2020. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting
1351 guideline. *BMJ* 368, l6890.

1352 Cao, Y., Cao, J., Li, B., 2026. Loddpm: High-fidelity pseudo-healthy brain MRI
1353 synthesis via longitudinal differential guidance. *Biomedical Signal Processing and*
1354 *Control* 123, 110600. doi:10.1016/j.bspc.2026.110600.

1355 Cao, Y., Levin, D., 1995. Using an image database to constrain the acquisition and
1356 reconstruction of MR images of the human head. *IEEE Transactions on Medical*
1357 *Imaging* doi:10.1109/42.387716.

1358 Castro, D.C., Walker, I., Glocker, B., 2020. Causality matters in medical imaging.
1359 *Nature Communications* 11, 3673.

1360 Cerjanic, A.M., Diano, A.M., Johnson, C.L., 2025. Elastographic magnetization pre-
1361 pared imaging with rapid encoding. *Magnetic resonance in medicine* doi:10.1002/
1362 *mrm*.30482. PMID: 40065646.

1363 Chan, K.L., Ziegls, T., Henning, A., 2022. Improved signal-to-noise performance of
1364 multinet GRAPPA 1h FID MRSI reconstruction with semi-synthetic calibration
1365 data. *Magnetic Resonance in Medicine* 88, 1500–1515. doi:10.1002/*mrm*.29314.

1366 Chen, G., Xie, H., Rao, X., Liu, X., Otikovs, M., Frydman, L., Sun, P., Zhang, Z., Pan,
1367 F., Yang, L., Zhou, X., Liu, M., Bao, Q., Liu, C., 2025a. MRI motion correction
1368 through disentangled CycleGAN based on multi-mask k-space subsampling. *IEEE*
1369 *Transactions on Medical Imaging* 44, 1907–1921. doi:10.1109/*tmi*.2024.3523949.

1370 Chen, L., Lin, J., Fu, G., Yang, P., Liu, M., 2025b. Pw-GAN: Pseudo-warping field
1371 guided GAN for unsupervised denoising of fetal brain MRI images. *bioRxiv* doi:10.
1372 1101/2025.04.30.651598.

1373 Chen, X., Peng, Z., Valadez, G.H., 2023. Dd-cisenet: Dual-domain cross-iteration
1374 squeeze and excitation network for accelerated MRI reconstruction. *arXiv.org*
1375 doi:10.48550/*arxiv*.2305.00088.

1376 Choi, J.W., Cho, Y.J., Lee, S.B., Lee, S., Hwang, J.Y., Choi, Y.H., Cheon, J.E.,
1377 Lee, J., 2025a. Accelerated brain magnetic resonance imaging with deep learning
1378 reconstruction: a comparative study on image quality in pediatric neuroimaging.
1379 *Pediatric Radiology* 55, 1903–1914. doi:10.1007/*s00247-025-06314-2*.

1380 Choi, K.S., Park, C., Lee, J.Y., Lee, K.H., Jeon, Y.H., Hwang, I., Yoo, R.E., Yun, T.J.,
1381 Lee, M.J., Jung, K.H., Kang, K.M., 2025b. Prospective evaluation of accelerated
1382 brain MRI using deep learning-based reconstruction: Simultaneous application to
1383 2D spin-echo and 3D gradient-echo sequences. *Korean journal of radiology* doi:10.
1384 3348/*kjr*.2024.0653. PMID: 39780631.

1385 Chow, L.S., Rajagopal, H., 2017. Modified-BRISQUE as no reference image quality
1386 assessment for structural MR images. *Magnetic Resonance Imaging* 43, 74–87.
1387 doi:10.1016/*j.mri*.2017.07.016.

1388 Chung, K.J., Souza, R., Frayne, R., 2020. Restoration of lossy JPEG-compressed brain
1389 MR images using cross-domain neural networks. *IEEE Signal Processing Letters* 27,
1390 141–145. doi:10.1109/*lsp*.2019.2961072.

1391 Ciceri, T., Squarcina, L., Pigoni, A., Ferro, A., Montano, F., Bertoldo, A., Persico,
1392 N., Boito, S., Triulzi, F.M., Conte, G., Brambilla, P., Peruzzo, D., 2023. Geomet-
1393 ric reliability of super-resolution reconstructed images from clinical fetal MRI in
1394 the second trimester. *Neuroinformatics* doi:10.1007/*s12021-023-09635-5*. PMID:
1395 37284977.

1396 Conklin, J., Longo, M.G.F., Tabari, A., Filho, A.L.G., Liu, W., Splitthoff, D.N.,
 1397 Lo, W.C., Cauley, S.F., Setsompop, K., Schaefer, P.W., Kirsch, J.E., Rapalino,
 1398 O., Huang, S.Y., 2022. Clinical validation of wave-caipi susceptibility-weighted
 1399 imaging for routine brain MRI at 1.5 t. *European radiology* doi:10.1007/s00330-
 1400 022-08871-8. pMID: 35925387.

1401 Cordero-Grande, L., Ortuño-Fisac, J.E., del Hoyo, A.A., Uus, A., Deprez, M., Santos,
 1402 A., Hajnal, J.V., Ledesma-Carbayo, M.J., 2023. Fetal MRI by robust deep generative prior reconstruction and diffeomorphic registration. *IEEE Transactions on*
 1403 *Medical Imaging* doi:10.1109/tmi.2022.3217725.

1405 Dabrowski, O., Falcone, J.L., Klauser, A., Songeon, J., Kocher, M., Chopard, B.,
 1406 Lazeyras, F., Courvoisier, S., 2025. Sismik for brain MRI: Deep-learning-based
 1407 motion estimation and model-based motion correction in k-space. *IEEE transactions*
 1408 *on medical imaging* doi:10.1109/tmi.2024.3446450. pMID: 39159019.

1409 Delattre, B.M.A., Boudabbous, S., Hansen, C., Neroladaki, A., Hachulla, A.L., Vargas,
 1410 M.I., 2019. Compressed sensing MRI of different organs: ready for clinical daily
 1411 practice? *European Radiology* 30, 308–319. doi:10.1007/s00330-019-06319-0.

1412 Desalegn, L., Jifara, W., 2024. Hara-GAN: Hybrid attention and relative average
 1413 discriminator based generative adversarial network for MR image reconstruction.
 1414 *IEEE Access* doi:10.1109/access.2024.3364699.

1415 Dewey, B.E., Remedios, S.W., Sanjayan, M., Rjeily, N.B., Lee, A.Z., Wyche, C.,
 1416 Duncan, S., Prince, J.L., Calabresi, P.A., Fitzgerald, K.C., Mowry, E.M., 2025.
 1417 Super-resolution in clinically available spinal cord mris enables automated atrophy
 1418 analysis. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a8526.
 1419 pMID: 39366765.

1420 Dieckmeyer, M., Roy, A.G., Senapati, J., Wachinger, C., Grundl, L., Döpfert, J.,
 1421 Bertran, P.F., Lemke, A., Zimmer, C., Kirschke, J.S., Hedderich, D.M., 2021. Effect
 1422 of MRI acquisition acceleration via compressed sensing and parallel imaging on brain
 1423 volumetry. *Magma (New York, N.Y.)* doi:10.1007/s10334-020-00906-9. pMID:
 1424 33502667.

1425 Ding, J., Duan, Y., Zhuo, Z., Yuan, Y., Zhang, G., Song, Q., Gao, B., Zhang, B.,
 1426 Wang, M., Yang, L., Hou, Y., Yuan, J., Feng, C., Wang, J., Lin, L., Liu, Y., 2021.
 1427 Acceleration of brain tof-mra with compressed sensitivity encoding: A multicenter
 1428 clinical study. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a7091.
 1429 pMID: 33858820.

1430 Dolui, S., Wang, Z., Wolf, R.L., Nabavizadeh, A., Xie, L., Tosun, D., Nasrallah, I.M.,
 1431 Wolk, D.A., Detre, J.A., Initiative, A.D.N., 2024. Automated quality evaluation
 1432 index for arterial spin labeling derived cerebral blood flow maps. *Journal of magnetic*
 1433 *resonance imaging : JMRI* doi:10.1002/jmri.29308. pMID: 38400805.

1434 Dong, Z., Reese, T.G., Lee, H.H., Huang, S.Y., Polimeni, J.R., Wald, L.L., Wang,
 1435 F., 2025. Romer-epti: Rotating-view motion-robust super-resolution epti for snr-
 1436 efficient distortion-free in-vivo mesoscale diffusion MRI and microstructure imaging.
 1437 *Magnetic resonance in medicine* doi:10.1002/mrm.30365. pMID: 39552568.

1438 Du, J., Wang, L., Liu, Y., Zhou, Z., He, Z., Jia, Y., 2020. Brain MRI super-resolution
1439 using 3D dilated convolutional encoder-decoder network. *IEEE Access* doi:10.1109/
1440 access.2020.2968395.

1441 Duffy, B.A., Zhao, L., Sepehrband, F., Min, J., Wang, D.J., Shi, Y., Toga, A.W., Kim,
1442 H., Initiative, A.D.N., 2021. Retrospective motion artifact correction of structural
1443 MRI images using deep learning improves the quality of cortical surface reconstruc-
1444 tions. *NeuroImage* doi:10.1016/j.neuroimage.2021.117756. pMID: 33460797.

1445 Duong, S.T.M., Phung, S.L., Bouzerdoum, A., Ang, S.P., Schira, M.M., 2021. Correct-
1446 ing susceptibility artifacts of MRI sensors in brain scanning: A 3D anatomy-guided
1447 deep learning approach. *Sensors (Basel, Switzerland)* doi:10.3390/s21072314.
1448 pMID: 33810289.

1449 Eeden Risager, K., Gholamalizadeh, T., Mehdipour Ghazi, M., 2025. Non-reference
1450 quality assessment for medical imaging: Application to synthetic brain mris. *Deep*
1451 *Generative Models* doi:10.1007/978-3-031-72744-3_19.

1452 Eggl, M.F., D. Santis, S., 2026. Simulation-based inference at the theoretical limit
1453 for fast, robust microstructural MRI with minimal diffusion data. *Communications*
1454 *Medicine* doi:10.1038/s43856-026-01614-6.

1455 Elhabian, S., Vachet, C., Piven, J., Styner, M., Gerig, G., 2016. Compressive sensing
1456 based q-space resampling for handling fast bulk motion in hardi acquisitions. 2016
1457 IEEE 13th International Symposium on Biomedical Imaging (ISBI) doi:10.1109/is
1458 bi.2016.7493412.

1459 Elsaid, N.M.H., Wu, Y.C., 2019. Super-resolution diffusion tensor imaging using sr-
1460 cnn: A feasibility study. *Annual International Conference of the IEEE Engineer-
1461 ing in Medicine and Biology Society. IEEE Engineering in Medicine and Biology*
1462 *Society. Annual International Conference* doi:10.1109/embc.2019.8857125. pMID:
1463 31946482.

1464 Ernst, P., Chatterjee, S., Rose, G., Speck, O., Nürnberger, A., 2023. Sinogram upsam-
1465 pling using primal-dual unet for undersampled CT and radial MRI reconstruction.
1466 *Neural networks : the official journal of the International Neural Network Society*
1467 doi:10.1016/j.neunet.2023.08.004.

1468 Esteban, O., Birman, D., Schaer, M., Kiar, G., Gorgolewski, K.J., Poldrack, R.A.,
1469 2017. MRIQC: Advancing the automatic prediction of image quality in MRI from
1470 unseen sites. *PLoS ONE* 12, e0184661. doi:10.1371/journal.pone.0184661.

1471 European Parliament and Council of the European Union, 2017. Regulation (EU)
1472 2017/745 on medical devices (Medical Device Regulation). *Official Journal of the*
1473 *European Union*, L 117, 5 May 2017.

1474 European Parliament and Council of the European Union, 2024. Regulation (EU)
1475 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelli-
1476 gence Act). *Official Journal of the European Union*, 12 July 2024.

1477 Fadeev, A., Eltonsy, N., Tourassi, G., Elmaghraby, A., 2005. Quality evaluation of
1478 fast morphing interpolation model for 3D volume reconstruction. *Proceedings of*
1479 *the Fifth IEEE International Symposium on Signal Processing and Information*
1480 *Technology*, 2005. doi:10.1109/isspit.2005.1577201.

1481 Fantini, I., Rittner, L., Yasuda, C., Lotufo, R., 2018. Automatic detection of mo-
1482 tion artifacts on MRI using deep CNN. 2018 International Workshop on Pattern
1483 Recognition in Neuroimaging (PRNI) doi:10.1109/prni.2018.8423948.

1484 Feizollah, S., Tardif, C.L., 2025. 3D mermaid: 3D multi-shot enhanced recovery motion
1485 artifact insensitive diffusion for submillimeter, multi-shell, and snr-efficient diffusion
1486 imaging. Magnetic resonance in medicine doi:10.1002/mrm.30436. PMID: 40035173.

1487 Fujita, S., Polak, D., Nickel, D., Splitthoff, D.N., Huang, Y., Gil, N., Buathong,
1488 S., Chiang, C.H., Lo, W.C., Clifford, B., Cauley, S.F., Conklin, J., Huang, S.Y.,
1489 2025. Motion-informed 3d deep learning reconstruction in patients with cognitive
1490 impairment. American Journal of Neuroradiology 47, 446–453. doi:10.3174/ajnr
1491 .a8977.

1492 Gao, Y., Zhu, X., Moffat, B.A., Glarin, R., Wilman, A.H., Pike, G.B., Crozier, S.,
1493 Liu, F., Sun, H., 2021. xqsm: quantitative susceptibility mapping with octave
1494 convolutional and noise-regularized neural networks. NMR in biomedicine doi:10.
1495 1002/nbm.4461. PMID: 33368705.

1496 Gardner, E.A., Ellis, J.H., Hyde, R.J., Aisen, A.M., Quint, D.J., Carson, P.L., 1995.
1497 Detection of degradation of magnetic resonance (MR) images: comparison of an
1498 automated MR image-quality analysis system with trained human observers. Aca-
1499 demic radiology doi:10.1016/s1076-6332(05)80184-9. PMID: 9419562.

1500 Garma-Oehmichen, A., Acuna-Luna, K.P., Santos-Diaz, A., 2021. Evaluation of echo
1501 planar imaging (EPI) distortion correction using synb0-disco and reversed phase
1502 encoding acquisition, in: 2021 43rd Annual International Conference of the IEEE
1503 Engineering in Medicine & Biology Society (EMBC), pp. 4023–4026. doi:10.1109/
1504 embc46164.2021.9630596.

1505 Giannakopoulos, I.I., Muckley, M.J., Kim, J., Breen, M., Johnson, P.M., Lui, Y.W.,
1506 Lattanzi, R., 2024. Accelerated MRI reconstructions via variational network and
1507 feature domain learning. Scientific Reports doi:10.1038/s41598-024-59705-0.

1508 Gil, N., Tabari, A., Lo, W.C., Clifford, B., Lang, M., Awan, K., Gaudet, K., Split-
1509 thoff, D.N., Polak, D., Cauley, S., Huang, S.Y., 2024. Quantitative evaluation of
1510 scout accelerated motion estimation and reduction (samer) mprage for morpho-
1511 metric analysis of brain tissue in patients undergoing evaluation for memory loss.
1512 NeuroImage doi:10.1016/j.neuroimage.2024.120865. PMID: 39349147.

1513 Glessgen, C., Gallichan, D., Moor, M., Hainc, N., Federau, C., 2019. Evaluation
1514 of 3D fat-navigator based retrospective motion correction in the clinical setting
1515 of patients with brain tumors. Neuroradiology doi:10.1007/s00234-019-02160-w.
1516 PMID: 30671618.

1517 Goffi, F., Bianchi, A.M., Schiena, G., Brambilla, P., Maggioni, E., 2025. Multi-metric
1518 approach for the comparison of denoising techniques for resting-state fMRI. Human
1519 brain mapping doi:10.1002/hbm.70080. PMID: 40309965.

1520 Gong, E., Pauly, J.M., Wintermark, M., Zaharchuk, G., 2018. Deep learning enables
1521 reduced gadolinium dose for contrast-enhanced brain MRI. Journal of Magnetic
1522 Resonance Imaging 48, 330–340. doi:10.1002/jmri.25970.

1523 Gottschling, N.M., Antun, V., Hansen, A.C., Adcock, B., 2025. The troublesome
1524 kernel: On hallucinations, no free lunches, and the accuracy–stability trade-off in
1525 inverse problems. *SIAM Review* 67, 73–104. doi:10.1137/23M1568739.

1526 Güngör, A., Dar, S.U., Öztürk, c., Korkmaz, Y., Bedel, H.A., Elmas, G., Ozbey,
1527 M., Çukur, T., 2023. Adaptive diffusion priors for accelerated MRI reconstruction.
1528 *Medical image analysis* doi:10.1016/j.media.2023.102872. pMID: 37384951.

1529 Gurbani, S.S., Schreibmann, E., Maudsley, A.A., Cordova, J.S., Soher, B.J., Poptani,
1530 H., Verma, G., Barker, P.B., Shim, H., Cooper, L.A.D., 2018. A convolutional neural
1531 network to filter artifacts in spectroscopic MRI. *Magnetic resonance in medicine*
1532 doi:10.1002/mrm.27166. pMID: 29520831.

1533 Hagiwara, A., Otsuka, Y., Hori, M., Tachibana, Y., Yokoyama, K., Fujita, S., Andica,
1534 C., Kamagata, K., Irie, R., Koshino, S., Maekawa, T., Chougar, L., Wada, A.,
1535 Takemura, M.Y., Hattori, N., Aoki, S., 2019. Improving the quality of synthetic
1536 FLAIR images with deep learning using a conditional generative adversarial network
1537 for pixel-by-pixel image translation. *AJNR. American journal of neuroradiology*
1538 doi:10.3174/ajnr.a5927. pMID: 30630834.

1539 Hales, P.W., Pfeuffer, J., Clark, C.A., 2020. Combined denoising and suppression
1540 of transient artifacts in arterial spin labeling MRI using deep learning. *Journal of*
1541 *magnetic resonance imaging : JMRI* doi:10.1002/jmri.27255.

1542 Hammernik, K., Klatzer, T., Kobler, E., Recht, M.P., Sodickson, D.K., Pock, T.,
1543 Knoll, F., 2018. Learning a variational network for reconstruction of accelerated
1544 MRI data. *Magnetic Resonance in Medicine* 79, 3055–3071.

1545 Hampe, N., Katscher, U., Berg, C.A.T.v.d., Tha, K.K., Mandija, S., 2020. Investigat-
1546 ing the challenges and generalizability of deep learning brain conductivity mapping.
1547 *Physics in medicine and biology* doi:10.1088/1361-6560/ab9356. pMID: 32408291.

1548 Harada, S., Takatsu, Y., Murayama, K., Sano, Y., Ikedo, M., 2025. Verification of reso-
1549 lution and imaging time for high-resolution deep learning reconstruction techniques.
1550 *Magnetic Resonance Imaging* 123, 110463. doi:10.1016/j.mri.2025.110463.

1551 Harper, J.R., Cherukuri, V., O'Reilly, T., Yu, M., Mbabazi-Kabachelor, E., Mu-
1552 lando, R., Sheth, K.N., Webb, A.G., Warf, B.C., Kulkarni, A.V., Monga, V.,
1553 Schiff, S.J., 2021. Assessing the utility of low resolution brain imaging: treatment
1554 of infant hydrocephalus. *NeuroImage. Clinical* doi:10.1016/j.nicl.2021.102896.
1555 pMID: 34911199.

1556 Haskell, M.W., Cauley, S.F., Bilgic, B., Hossbach, J., Splittthoff, D.N., Pfeuffer, J., Set-
1557 sompop, K., Wald, L.L., 2019. Network accelerated motion estimation and reduction
1558 (namer): Convolutional neural network guided retrospective motion correction using
1559 a separable motion model. *Magnetic resonance in medicine* doi:10.1002/mrm.27771.
1560 pMID: 31045278.

1561 Heo, H.Y., Singh, M., Yedavalli, V., Jiang, S., Zhou, J., 2024. Cest and nuclear over-
1562 hauser enhancement imaging with deep learning-extrapolated semisolid magnetiza-
1563 tion transfer reference: Scan-rescan reproducibility and reliability studies. *Magnetic*
1564 *resonance in medicine* doi:10.1002/mrm.29937. pMID: 38009996.

1565 Hernandez Petzsche, M.R., de la Rosa, E., Hanning, U., Wiest, R., Valenzuela, W.,
1566 Reyes, M., Meyer, M., Liew, S.L., Kofler, F., Ezhov, I., et al., 2022. ISLES 2022: A
1567 multi-center magnetic resonance imaging stroke lesion segmentation dataset. *Scientific Data* 9, 762. doi:10.1038/s41597-022-01875-5.

1569 Hewlett, M., Petrov, I., Johnson, P.M., Drangova, M., 2024. Deep-learning-based
1570 motion correction using multichannel MRI data: a study using simulated artifacts
1571 in the fastmri dataset. *NMR in biomedicine* doi:10.1002/nbm.5179.

1572 Ho, B.C., Kim, D., Kumar, A., Weiss, S., Vossler, H., Mormino, E., Zaharchuk, G.,
1573 Initiative, A.D.N., 2026. Evaluation of image-level harmonization methods for multi-
1574 center MR neuroimaging. *Journal of magnetic resonance imaging : JMRI* doi:10.
1575 1002/jmri.70221. PMID: 41489091.

1576 Hossbach, J., Splitthoff, D.N., Cauley, S., Clifford, B., Polak, D., Lo, W.C., Meyer,
1577 H., Maier, A., 2023. Deep learning-based motion quantification from k-space for
1578 fast model-based magnetic resonance imaging motion correction. *Medical physics*
1579 doi:10.1002/mp.16119. PMID: 36433748.

1580 Hosseini, M.S., Aghamiri, S.M.R., Ardekani, A.F., BagheriMofidi, S.M., Safari, M.,
1581 2025. Autocornn: An unsupervised physics-aware deep learning model for geo-
1582 metric distortion correction of brain MRI images towards MR-only stereotactic ra-
1583 diosurgery. *Journal of imaging informatics in medicine* doi:10.1007/s10278-024-
1584 01171-1. PMID: 39080159.

1585 Hu, Y., Xu, Y., Tian, Q., Chen, F., Shi, X., Moran, C.J., Daniel, B.L., Hargreaves,
1586 B.A., 2021. Run-up: Accelerated multishot diffusion-weighted MRI reconstruction
1587 using an unrolled network with u-net as priors. *Magnetic resonance in medicine*
1588 doi:10.1002/mrm.28446. PMID: 32783339.

1589 Hu, Y., Zhang, X., Chen, D., Yan, Z., Shen, X., Yan, G., Ou-Yang, L., Lin, J.,
1590 Dong, J., Qu, X., 2022. Spatiotemporal flexible sparse reconstruction for rapid
1591 dynamic contrast-enhanced MRI. *IEEE transactions on bio-medical engineering*
1592 doi:10.1109/tbme.2021.3091881. PMID: 34166181.

1593 Huang, J., Zhou, G., Yu, Z., Hu, W., 2022. [deep parallel MRI reconstruction based
1594 on a complex-valued loss function]. *Nan fang yi ke da xue xue bao = Journal of*
1595 *Southern Medical University* doi:10.12122/j.issn.1673-4254.2022.12.02. PMID:
1596 36651242.

1597 Huang, Y., Wang, Z., Zhang, X., Cao, J., Tu, Z., Lin, M., Li, L., Jiang, X., Guo, D.,
1598 Qu, X., 2025. Improve robustness to mismatched sampling rate: An alternating
1599 deep low-rank approach for exponential function reconstruction and its biomedical
1600 magnetic resonance applications. *Journal of Magnetic Resonance* 376, 107898.
1601 doi:10.1016/j.jmr.2025.107898.

1602 Huber, L.R., Stirnberg, R., Morgan, A.T., Feinberg, D.A., Ehse, P., Knudsen, L.,
1603 Gulban, O.F., Koiso, K., Gephart, I., Swegle, S., Wardle, S.G., Persichetti, A.S.,
1604 Beckett, A.J.S., Stöcker, T., Boulant, N., Poser, B.A., Bandettini, P.A., 2025. Short-
1605 term gradient imperfections in high-resolution epi lead to fuzzy ripple artifacts.
1606 *Magnetic resonance in medicine* doi:10.1002/mrm.30489. PMID: 40173320.

Huijben, E.M.C., Amirrajab, S., Pluim, J.P.W., 2025. Enhancing reconstruction-based out-of-distribution detection in brain MRI with model and metric ensembles. *Computer methods and programs in biomedicine* doi:10.1016/j.cmpb.2025.109045.

International Commission on Radiation Units and Measurements, 1996. *Medical Imaging — The Assessment of Image Quality*. Technical Report ICRU Report 54. ICRU. Bethesda, MD.

Islam, K.T., Dayarathna, S., Zhong, S., Zakavi, P., Kavnoudias, H., Farquharson, S., Durbridge, G., Sun, H., Bacchi, S., Egan, G.F., Barth, M., Dwyer, A., McMahon, K.L., Parizel, P.M., Law, M., Chen, Z., 2026. Synpoc: a high-quality generative diffusion model for transforming ultra-low-field point-of-care MRI using high-field MRI representations. *Scientific reports* doi:10.1038/s41598-025-33162-9. pMID: 41580445.

Ismail, A.R., Azhary, M.Z.R., Azwan, N.A.Z.N., Ismail, A., Alsaiani, N.A., 2024. Performance evaluation of medical image denoising using convolutional autoencoders, in: *2024 3rd International Conference on Creative Communication and Innovative Technology (ICCIT)*, pp. 1–6. doi:10.1109/iccit62134.2024.10701250.

Jayadeepthi, K., Yakshitha, K., Reddy, K.P.K., Gowtham, K., Mahesh, K., 2026. Deep learning based denoising framework for high-fidelity brain MRI reconstruction, in: *2026 4th International Conference on Inventive Computing and Informatics (ICICI)*, pp. 696–702. doi:10.1109/icici68773.2026.11580971.

Jayakumar, N., Hossain, T., Zhang, M., 2023. SADIR: Shape-aware diffusion models for 3D image reconstruction. *Shape in Medical Imaging* doi:10.1007/978-3-031-46914-5_23.

Jeong, S., 2026. Evaluating reliability of automated quantitative brain morphometry from fetal t2-weighted MRI. *Frontiers in Neuroscience* doi:10.3389/fnins.2026.1817732.

Ji, Y., Wu, W., Buck, M.H.S.d., Okell, T., Jezzard, P., 2023. Highly accelerated intracranial time-of-flight magnetic resonance angiography using wave-encoding. *Magnetic resonance in medicine* doi:10.1002/mrm.29647. pMID: 37010811.

Jiang, Y., Zhang, S., Chi, J., 2023. Multi-modal brain tumor data completion based on reconstruction consistency loss. *Journal of digital imaging* doi:10.1007/s10278-022-00697-6. pMID: 36856903.

Jochmann, T., Salman, F., Dwyer, M.G., Bergsland, N., Zivadinov, R., Hauelsen, J., Schweser, F., 2025. Quantitative susceptibility mapping in magnetically inhomogeneous tissues. *Magnetic resonance in medicine* doi:10.1002/mrm.30537. pMID: 40312865.

Joshi, D., Gautam, A., Borotikar, B., 2025. Arterial spin labeling-based CBF map generation using a vector quantized variational autoencoder, in: *2025 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)*, pp. 1–5. doi:10.1109/conecct65861.2025.11306510.

1647 Jun, Y., Arefeen, Y., Cho, J., Fujita, S., Wang, X., Grant, P.E., Gagoski, B., Jaimes,
1648 C., Gee, M.S., Bilgic, B., 2024. Zero-deepsub: Zero-shot deep subspace reconstruction
1649 for rapid multiparametric quantitative MRI using 3D-qalas. *Magnetic resonance*
1650 *in medicine* doi:10.1002/mrm.30018. PMID: 38282270.

1651 Jung, W., Jeong, G., Kim, S., Hwang, I., Choi, S.H., Jeon, Y.H., Choi, K.S., Lee, J.Y.,
1652 Yoo, R.E., Yun, T.J., Kang, K.M., 2025. Reliability of brain volume measures of
1653 accelerated 3D t1-weighted images with deep learning-based reconstruction. *Neu-*
1654 *roradiology* doi:10.1007/s00234-024-03461-5. PMID: 39316090.

1655 Kamesh Iyer, S., Moon, B.F., Josselyn, N., Kurtz, R.M., Song, J.W., Ware, J.B.,
1656 Nabavizadeh, S.A., Witschey, W.R., 2022. Quantitative susceptibility mapping
1657 using plug-and-play alternating direction method of multipliers. *Scientific Reports*
1658 doi:10.1038/s41598-022-22778-w.

1659 Kanemaru, N., Takao, H., Amemiya, S., Abe, O., 2022. The effect of a post-scan
1660 processing denoising system on image quality and morphometric analysis. *Journal of*
1661 *neuroradiology = Journal de neuroradiologie* doi:10.1016/j.neurad.2021.11.007.

1662 Kang, S.H., Chung, J.Y., Lee, Y., Initiative, A.D.N., 2026. Simulation data-based dual
1663 domain network (sim-ddnet) for motion artifact reduction in MR images. *MAGNE-*
1664 *TOCHEMISTRY* doi:10.3390/magnetochemistry12010014.

1665 Kang, S.K., Shin, S.A., Seo, S., Byun, M.S., Lee, D.Y., Kim, Y.K., Lee, D.S., Lee, J.S.,
1666 2021. Deep learning-based 3D inpainting of brain MR images. *Scientific Reports*
1667 doi:10.1038/s41598-020-80930-w.

1668 Karimi, D., Jaimes, C., Machado-Rivas, F., Vasung, L., Khan, S., Warfield,
1669 S.K., Gholipour, A., 2021a. Deep learning-based parameter estimation in fetal
1670 diffusion-weighted MRI. *NeuroImage* doi:10.1016/j.neuroimage.2021.118482.
1671 PMID: 34455242.

1672 Karimi, D., Vasung, L., Machado-Rivas, F., Jaimes, C., Khan, S., Gholipour, A.,
1673 2021b. Accurate parameter estimation in fetal diffusion-weighted MRI, in: *MICCAI*
1674 *2021, LNCS 12907*. doi:10.1007/978-3-030-87234-2/46.

1675 Karthik, A., Aggarwal, K., Kapoor, A., Singh, D., Hu, L., Gandhamal, A., Kumar, D.,
1676 2024. Comprehensive assessment of imaging quality of artificial intelligence-assisted
1677 compressed sensing-based MR images in routine clinical settings. *BMC medical*
1678 *imaging* doi:10.1186/s12880-024-01463-6. PMID: 39434010.

1679 Karuna, Y., Syamala, N., Ravikumar, C.V., Thakur, P., Saladi, S., 2025. Modified
1680 energy-based GAN for intensity in homogeneity correction in brain MR images.
1681 *Scientific reports* doi:10.1038/s41598-025-08552-8. PMID: 40691452.

1682 Kayvanrad, M., Lin, A., Joshi, R., Chiu, J., Peters, T., 2016. Diagnostic quality
1683 assessment of compressed sensing accelerated magnetic resonance neuroimaging.
1684 *Journal of Magnetic Resonance Imaging* 44, 433–444. doi:10.1002/jmri.25149.

1685 Kazerooni, A., Aghdam, E.K., Heidari, M., Azad, R., Fayyaz, M., Hacihaliloglu, I.,
1686 Merhof, D., 2023. Diffusion models in medical imaging: A comprehensive survey.
1687 *Medical Image Analysis* 88, 102846.

1688 Kharaji, M., Canton, G., Guo, Y., Mosi, M.H., Zhou, Z., Balu, N., Mossa-Basha, M.,
1689 2025. Dante-caipi accelerated contrast-enhanced 3D t1: Deep learning-based image
1690 quality improvement for vessel wall MRI. *AJNR. American journal of neuroradiol-*
1691 *ogy* doi:10.3174/ajnr.a8424. pMID: 39038956.

1692 Khawaled, S., Freiman, M., 2024. NPB-rec: A non-parametric bayesian deep-learning
1693 approach for undersampled MRI reconstruction with uncertainty estimation. *Artifi-*
1694 *cial intelligence in medicine* doi:10.1016/j.artmed.2024.102798. pMID: 38462289.

1695 Kiani Kalejahi, B., 2026. GAN-based cross-modality brain MRI synthesis: Paired
1696 versus unpaired training and comparison with diffusion and transformer models.
1697 *Biomimetics* doi:10.3390/biomimetics11030175.

1698 Kim, K.H., Park, S.H., 2017. Artificial neural network for suppression of banding
1699 artifacts in balanced steady-state free precession MRI. *Magnetic Resonance Imaging*
1700 37, 139–146. doi:10.1016/j.mri.2016.11.020.

1701 Kim, S., Park, S., 2021. Simultaneous multislice brain MRI t1 mapping with im-
1702 proved low-rank modeling. *Tomography (Ann Arbor, Mich.)* doi:10.3390/tomogr
1703 aphy7040047. pMID: 34698294.

1704 Kim, S.H., Choi, Y.H., Lee, J.S., Lee, S.B., Cho, Y.J., Lee, S.H., Shin, S.M., Cheon,
1705 J.E., 2022. Deep learning reconstruction in pediatric brain MRI: comparison of
1706 image quality with conventional T2-weighted MRI. *Neuroradiology* 65, 207–214.
1707 doi:10.1007/s00234-022-03053-1.

1708 Kim, T., Ji, S., Min, K., Kim, M., Youn, J., Oh, C., Kim, J., Lee, J., 2026. Vessel
1709 segmentation for χ -separation in quantitative susceptibility mapping. *Magnetic*
1710 *resonance in medicine* doi:10.1002/mrm.70054. pMID: 40891385.

1711 Knoll, F., Hammernik, K., Zhang, C., Moeller, S., Pock, T., Sodickson, D.K.,
1712 Akçakaya, M., 2020a. Deep-learning methods for parallel magnetic resonance imag-
1713 ing reconstruction: A survey of the current approaches, trends, and issues. *IEEE*
1714 *Signal Processing Magazine* 37, 128–140. doi:10.1109/MSP.2019.2950640.

1715 Knoll, F., Zbontar, J., Sriram, A., Muckley, M.J., Bruno, M., Defazio, A., Parente,
1716 M., Geras, K.J., Katsnelson, J., Chandarana, H., Zhang, Z., Drozdal, M., Romero,
1717 A., Rabbat, M., Vincent, P., Pinkerton, J., Wang, D., Yakubova, N., Owens, E.,
1718 Zitnick, C.L., Recht, M.P., Sodickson, D.K., Lui, Y.W., 2020b. fastmri: A publicly
1719 available raw k -space and DICOM dataset of knee images for accelerated MR image
1720 reconstruction using machine learning. *Radiology: Artificial Intelligence* 2, e190007.

1721 Kofler, A., Kerkering, K.M., Göschel, L., Fillmer, A., Kolbitsch, C., 2024. Quanti-
1722 tative MR image reconstruction using parameter-specific dictionary learning with
1723 adaptive dictionary-size and sparsity-level choice. *IEEE Transactions on Biomedical*
1724 *Engineering* doi:10.1109/tbme.2023.3300090.

1725 Kulkarni, P.H., Merchant, S.N., Awate, S.P., 2021. Dictionary+wavelet model with
1726 nested-minorized vb-em for sms-caipi r-fMRI reconstruction. *IEEE Open Journal*
1727 *of Signal Processing* doi:10.1109/oj-sp.2021.3100751.

1728 Kumar, V., Sharma, M.K., Jehadeesan, R., Venkatraman, B., Suman, G., Patra, A.,
1729 Goenka, A.H., Sheet, D., 2021. Learning to generate missing pulse sequence in MRI
1730 using deep convolution neural network trained with visual turing test, in: 2021 43rd
1731 Annual International Conference of the IEEE Engineering in Medicine & Biology
1732 Society (EMBC), pp. 3419–3422. doi:10.1109/embc46164.2021.9630435.

1733 Lacerda, L.M., Sperl, J.I., Menzel, M.I., Sprenger, T., Barker, G.J., Dell'Acqua, F.,
1734 2016. Diffusion in realistic biophysical systems can lead to aliasing effects in diffusion
1735 spectrum imaging. *Magnetic resonance in medicine* doi:10.1002/mrm.26080. PMID:
1736 26714794.

1737 Lan, H., Varghese, B.A., Sheikh-Bahaei, N., Sepehrband, F., Toga, A.W., Choupan, J.,
1738 Initiative, A.D.N., 2025. Diffusion based multi-domain neuroimaging harmonization
1739 method with preservation of anatomical details. *NeuroImage* doi:10.1016/j.neur
1740 oimage.2025.121297.

1741 Lang, M., Clifford, B., Lo, W.C., Applewhite, B.P., Tabari, A., Goncalves Filho,
1742 A.L.M., Hosseini, Z., Figueiro Longo, M.G., Cauley, S.F., Setsompop, K., Bilgic,
1743 B., Feiweier, T., Lev, M.H., Schaefer, P.W., Rapalino, O., Huang, S.Y., Conklin, J.,
1744 2024. Clinical evaluation of a 2-minute ultrafast brain MR protocol for evaluation of
1745 acute pathology in the emergency and inpatient settings. *AJNR. American Journal*
1746 *of Neuroradiology* 45, 379–385. doi:10.3174/ajnr.A8143.

1747 Lang, M., Tabari, A., Polak, D., Ford, J., Clifford, B., Lo, W., Manzoor, K., Split-
1748 thoff, D.N., Wald, L.L., Rapalino, O., Schaefer, P., Conklin, J., Cauley, S., Huang,
1749 S.Y., 2023. Clinical evaluation of scout accelerated motion estimation and reduction
1750 technique for 3D MR imaging in the inpatient and emergency department set-
1751 tings. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a7777. PMID:
1752 36702502.

1753 Lau, V., Xiao, L., Zhao, Y., Su, S., Ding, Y., Man, C., Wang, X., Tsang, A., Cao, P.,
1754 Lau, G.K.K., Leung, G.K.K., Leong, A.T.L., Wu, E.X., 2023. Pushing the limits of
1755 low-cost ultra-low-field MRI by dual-acquisition deep learning 3D superresolution.
1756 *Magnetic resonance in medicine* doi:10.1002/mrm.29642.

1757 Lee, J., Jung, W., Yang, S., Park, J.H., Hwang, I., Chung, J.W., Choi, S.H., Choi,
1758 K.S., 2024. Deep learning-based super-resolution and denoising algorithm improves
1759 reliability of dynamic contrast-enhanced MRI in diffuse glioma. *Scientific reports*
1760 doi:10.1038/s41598-024-76592-7. PMID: 39455814.

1761 Lee, J., Kim, H., Chung, H., Eom, H., Jang, J., Sohn, C.H., Choi, K.S., 2026a. Lesion-
1762 aware post-training of latent diffusion models for synthesizing diffusion MRI from
1763 CT perfusion, in: *Medical Image Computing and Computer Assisted Intervention*
1764 – MICCAI 2025, Springer. doi:10.1007/978-3-032-04937-7_27.

1765 Lee, N.G., Cui, S.X., Nayak, K.S., 2026b. Cartesian maxgirt: Model-based epi recon-
1766 struction incorporating gradient nonlinearity and concomitant field effects. *Magnetic*
1767 *resonance in medicine* doi:10.1002/mrm.70113. PMID: 41039908.

1768 Li, H., Chiew, M., Dragonu, I., Jezzard, P., Okell, T.W., 2026. Few-shot learning
1769 for highly accelerated 3D time-of-flight mra reconstruction. *Magnetic resonance in*
1770 *medicine* doi:10.1002/mrm.70072. PMID: 40931540.

1771 Li, M., Chen, C., Xiong, Z., Liu, Y., Rong, P., Shan, S., Liu, F., Sun, H.,
1772 Gao, Y., 2025a. Quantitative susceptibility mapping via deep neural networks
1773 with iterative reverse concatenations and recurrent modules. *Medical physics*
1774 doi:10.1002/mp.17747.

1775 Li, N., Wang, W.T., Pham, D.L., Butman, J.A., 2015. Artifactual microhemorrhage
1776 generated by susceptibility weighted image processing. *Journal of magnetic reso-*
1777 *nance imaging : JMRI* doi:10.1002/jmri.24728. pMID: 25143262.

1778 Li, Y., Liao, Y.P., Wang, J., Lu, W., Zhang, Y., 2025b. Patient-specific MRI super-
1779 resolution via implicit neural representations and knowledge transfer. *Physics in*
1780 *medicine and biology* doi:10.1088/1361-6560/adbed4. pMID: 40064110.

1781 Li, Z., Fan, Q., Bilgic, B., Wang, G., Wu, W., Polimeni, J.R., Miller, K.L., Huang,
1782 S.Y., Tian, Q., 2023. Diffusion MRI data analysis assisted by deep learning synthe-
1783 sized anatomical images (deepanat). *Medical image analysis* doi:10.1016/j.media.
1784 2023.102744. pMID: 36867912.

1785 Liang, D., Liu, Y., Wang, H., Zhu, Y., 2020. Deep learning for magnetic resonance
1786 image reconstruction: An overview. *IEEE Transactions on Radiation and Plasma*
1787 *Medical Sciences* 4, 641–655.

1788 Liang, Z., Guo, X., Xu, W., Ibrahim, Y., Voets, N., Pretorius, P.M., Initiative, A.D.N.,
1789 Noble, J.A., Kamnitsas, K., 2026. Itermask3d: Unsupervised anomaly detection and
1790 segmentation with test-time iterative mask refinement in 3D brain MRI. *Medical*
1791 *image analysis* doi:10.1016/j.media.2025.103763.

1792 Liebrand, L.C., Karkalousos, D., Poirion, É., Emmer, B.J., Roosendaal, S.D., Mar-
1793 quering, H.A., Majoie, C.B.L.M., Savatovsky, J., Caan, M.W.A., 2025. Deep learn-
1794 ing for efficient reconstruction of highly accelerated 3D FLAIR MRI in neurologi-
1795 cal deficits. *Magma (New York, N.Y.)* doi:10.1007/s10334-024-01200-8. pMID:
1796 39212832.

1797 Liew, S.L., Tavenner, B.P., Donnelly, M.R., Zavaliangos-Petropulu, A., Jeong, J.N.,
1798 Barisano, G., Hutton, A., Simon, J.P., Juliano, J.M., Suri, A., et al., 2022. A large,
1799 curated, open-source stroke neuroimaging dataset to improve lesion segmentation
1800 algorithms. *Scientific Data* 9, 320. doi:10.1038/s41597-022-01401-7.

1801 Liu, G., Ueguchi, T., Ogawa, S., 2026a. Block-interleaved segmented echo-planar imag-
1802 ing for improved activity detection in submillimeter high-resolution functional MRI
1803 at 7 t. *Magnetic resonance in medicine* doi:10.1002/mrm.30569. pMID: 40432253.

1804 Liu, H., Chen, Z., Law, L.H., Liu, Y., Wang, Z., Wang, J., Zhang, Y., Shen, D., Huang,
1805 J., Chan, K.W.Y., 2026b. Accelerating cest MRI through complementary under-
1806 sampling and multi-offset transformer reconstruction. *Communications Engineering*
1807 doi:10.1038/s44172-025-00580-6.

1808 Liu, H., Versteeg, E., Fuderer, M., Heide, O.v.d., Schilder, M.B., Berg, C.A.T.v.d.,
1809 Sbrizzi, A., 2025a. Time-efficient, high-resolution 3t whole-brain relaxometry using
1810 cartesian 3D MR spin tomography in time-domain (MR-stat) with cerebrospinal
1811 fluid suppression. *Magnetic resonance in medicine* doi:10.1002/mrm.30384. pMID:
1812 39607873.

1813 Liu, J., Kocak, M., Supanich, M., Deng, J., 2020. Motion artifacts reduction in brain
1814 MRI by means of a deep residual network with densely connected multi-resolution
1815 blocks (DRN-DCMB). *Magnetic Resonance Imaging* 71, 69–79. doi:10.1016/j.mri.
1816 2020.05.002.

1817 Liu, J., Lin, Q., Xiong, Z., Shan, S., Liu, C., Li, M., Liu, F., Pike, G.B., Sun, H., Gao,
1818 Y., 2026c. Highly undersampled MRI reconstruction via a single posterior sampling
1819 of diffusion models. *IEEE Transactions on Medical Imaging* doi:10.1109/tmi.2026.
1820 3654585.

1821 Liu, Z., Patel, V., Zhou, X., Tao, S., Yu, T., Ma, J., Nickel, D., Liebig, P., West-
1822 erhold, E.M., Mojahed, H., Gupta, V., Middlebrooks, E.H., 2025b. Deep learn-
1823 ing reconstruction for 7t MP2RAGE and SPACE MRI: Improving image quality
1824 at high acceleration factors. *American Journal of Neuroradiology* 46, 2446–2454.
1825 doi:10.3174/ajnr.a8841.

1826 Liu, Z., Zhou, X., Tao, S., Ma, J., Nickel, D., Liebig, P., Mostapha, M., Patel, V.,
1827 Westerhold, E.M., Mojahed, H., Gupta, V., Middlebrooks, E.H., 2025c. Application
1828 of deep learning accelerated image reconstruction in t2-weighted turbo spin-echo
1829 imaging of the brain at 7t. *AJNR. American journal of neuroradiology* doi:10.
1830 3174/ajnr.a8662. PMID: 39832954.

1831 Lustig, M., Donoho, D., Pauly, J.M., 2007. Sparse MRI: The application of compressed
1832 sensing for rapid MR imaging. *Magnetic Resonance in Medicine* 58, 1182–1195.

1833 Ma, Q., Lai, Z., Wang, Z., Qiu, Y., Zhang, H., Qu, X., 2024. MRI reconstruction with
1834 enhanced self-similarity using graph convolutional network. *BMC medical imaging*
1835 doi:10.1186/s12880-024-01297-2. PMID: 38760778.

1836 Ma, X., Ugurbil, K., Wu, X., 2022. Mitigating transmit-b1 artifacts by predicting
1837 parallel transmission images with deep learning: A feasibility study using high-
1838 resolution whole-brain diffusion at 7 tesla. *Magnetic resonance in medicine* doi:10.
1839 1002/mrm.29238. PMID: 35403237.

1840 Mallardi, G., Calefato, F., Lanubile, F., Logroscino, G., Tafuri, B., 2025. Diffusion
1841 models for neuroimaging data augmentation: Assessing realism and clinical rele-
1842 vance. *Journal of Medical Systems* doi:10.1007/s10916-025-02300-1.

1843 Marchetto, E., Eichhorn, H., Gallichan, D., Schnabel, J.A., Ganz, M., 2025. Agree-
1844 ment of image quality metrics with radiological evaluation in the presence of mo-
1845 tion artifacts. *Magma (New York, N.Y.)* doi:10.1007/s10334-025-01266-y. PMID:
1846 40493331.

1847 Marchetto, E., Flassbeck, S., Mao, A., Assländer, J., 2026. Contrast-optimized ba-
1848 sis functions for self-navigated motion correction in quantitative MRI. *Magnetic*
1849 *resonance in medicine* doi:10.1002/mrm.70090. PMID: 40961378.

1850 Marie, H.S., Draz, M.M., Said, W., Elbaz, M., 2026. Bald uakari algorithm: a novel
1851 nature-inspired metaheuristic for missing pixel imputation in medical images. *Clus-*
1852 *ter Computing* doi:10.1007/s10586-026-06180-z.

1853 Mason, A., Rioux, J., Clarke, S.E., Costa, A., Schmidt, M., Keough, V., Huynh, T.,
1854 Beyea, S., 2020. Comparison of objective image quality metrics to expert radiolo-
1855 gists' scoring of diagnostic quality of MR images. *IEEE Transactions on Medical*
1856 *Imaging* 39, 1064–1072.

1857 Matsuo, K., Nakaura, T., Morita, K., Uetani, H., Nagayama, Y., Kidoh, M., Hoka-
1858 mura, M., Yamashita, Y., Shinoda, K., Ueda, M., Mukasa, A., Hirai, T., 2023.
1859 Feasibility study of super-resolution deep learning-based reconstruction using k-
1860 space data in brain diffusion-weighted images. *Neuroradiology* 65, 1619–1629.
1861 doi:10.1007/s00234-023-03212-y.

1862 McIlvain, G., 2022. Quantitative effects of off-resonance related distortion on brain
1863 mechanical property estimation with magnetic resonance elastography. *NMR in*
1864 *Biomedicine* doi:10.1002/nbm.4616.

1865 Mello, V.B.B., McKinley, R., Wiest, R., Rummel, C., 2026. Fast surface reconstruc-
1866 tion of human brain MRI: benchmarking deep-learning based morphometry tools.
1867 *Scientific reports* doi:10.1038/s41598-026-55397-w. PMID: 42286078.

1868 Meng, Y., Allen, J.W., Sharghi, V.K., Qiu, D., 2025. Motion and temporal b0-shift cor-
1869 rections for qsm and r 2 * mapping using dual-echo spiral navigators and conjugate-
1870 phase reconstruction. *Magnetic resonance in medicine* doi:10.1002/mrm.30266.

1871 Meng, Z., Guo, R., Li, Y., Guan, Y., Wang, T., Zhao, Y., Sutton, B., Li, Y., Liang, Z.,
1872 2020. Accelerating t2mapping of the brain by integrating deep learning priors with
1873 low?rank and sparse modeling. *Magnetic Resonance in Medicine* 85, 1455–1467.
1874 doi:10.1002/mrm.28526.

1875 Mercier, C., Faisan, S., Pron, A., Girard, N., Auzias, G., Chonavel, T., Rousseau, F.,
1876 2025. Intersection-based slice motion estimation for fetal brain imaging. *Computers*
1877 *in biology and medicine* doi:10.1016/j.combiomed.2025.110005. PMID: 40112563.

1878 Miao, J., Huang, F., Narayan, S., Wilson, D.L., 2013. A new perceptual difference
1879 model for diagnostically relevant quantitative image quality evaluation: a prelimi-
1880 nary study. *Magnetic resonance imaging* doi:10.1016/j.mri.2012.09.009. PMID:
1881 23218792.

1882 Miao, J., Huang, F., Wilson, D.L., 2010. Comprehensive quantitative image quality
1883 evaluation of compressed sensing MRI reconstructions using a weighted perceptual
1884 difference model (case-PDM): selective evaluation, disturbance calibration, and ag-
1885 gregative evaluation of noise, blur, aliasing, and oil-painting artifacts, in: *SPIE*
1886 *Proceedings*, p. 762709. doi:10.1117/12.845507.

1887 Milovic, C., Tejos, C., Silva, J., Shmueli, K., Irarrazaval, P., 2024. XSIM : A structural
1888 similarity index measure optimized for MRI QSM. *Magnetic Resonance in Medicine*
1889 93, 411–421. doi:10.1002/mrm.30271.

1890 Mirza, M.U., Dalmaz, O., Bedel, H.A., Elmas, G., Korkmaz, Y., Gungor, A., Dar,
1891 S.U.H., Oguz, K.K., Çukur, T., 2026. Learning fourier-constrained diffusion bridges
1892 for MRI reconstruction. *IEEE Transactions on Medical Imaging* doi:10.1109/tmi.
1893 2026.3673956.

1894 Mohsen, M., Tamer, R., Hossam, H., Ahmed, O., Sherif, S., 2025. Deep super-
1895 resolution and age prediction for brain MRI using SRCNN: A lightweight frame-
1896 work for high-fidelity reconstruction and regression, in: 2025 Twelfth International
1897 Conference on Intelligent Computing and Information Systems (ICICIS), pp. 38–45.
1898 doi:10.1109/icicis66182.2025.11313176.

1899 Montalt-Tordera, J., Muthurangu, V., Hauptmann, A., Steeden, J.A., 2021. Machine
1900 learning in magnetic resonance imaging: Image reconstruction. *Physica Medica* 83,
1901 79–87.

1902 Moshe, Y.H., Buchsweiler, Y., Teicher, M., Artzi, M., 2024. Handling missing MRI
1903 data in brain tumors classification tasks: Usage of synthetic images vs. duplicate
1904 images and empty images. *Journal of magnetic resonance imaging : JMRI* doi:10.
1905 1002/jmri.29072.

1906 Muckley, M.J., Riemenschneider, B., Radmanesh, A., Kim, S., Jeong, G., Ko, J., Jun,
1907 Y., Shin, H., Hwang, D., Mostapha, M., Arberet, S., Nickel, D., Ramzi, Z., Ciuciu,
1908 P., Starck, J.L., Teuwen, J., Karkalousos, D., Zhang, C., Sriram, A., Huang, Z.,
1909 Yakubova, N., Lui, Y.W., Knoll, F., 2021. Results of the 2020 fastMRI challenge
1910 for machine learning MR image reconstruction. *IEEE Transactions on Medical*
1911 *Imaging* 40, 2306–2317. doi:10.1109/tmi.2021.3075856.

1912 Müller, S.J., Khadhraoui, E., Voit, D., Riedel, C.H., Frahm, J., Romero, J.M., Ernst,
1913 M., 2023. Comparison of epi DWI and steam DWI in early postoperative MRI con-
1914 trols after resection of tumors of the central nervous system. *Clinical Neuroradiology*
1915 doi:10.1007/s00062-023-01261-7.

1916 Muro, I., Isoiwa, T., Shibukawa, S., Usui, K., Otsuka, Y., 2025. Using deep learning
1917 to simultaneously reduce noise and motion artifacts in brain MR imaging. *Magnetic*
1918 *resonance in medical sciences : MRMS : an official journal of Japan Society of Mag-*
1919 *netic Resonance in Medicine* doi:10.2463/mrms.mp.2024-0098. pMID: 39938896.

1920 Muro, I., Shimizu, S., Tsukamoto, H., 2022. [improvement of motion artifacts in brain
1921 MRI using deep learning by simulation training data]. *Nihon Hoshasen Gijutsu*
1922 *Gakkai zasshi* doi:10.6009/jjrt.780108. pMID: 35046218.

1923 Nagaraj, U.D., Greer, J.S., Debnath, P., Tkach, J.A., Kline-Fath, B.M., 2025a. Slice-
1924 to-volume reconstruction of fetal brain MR imaging in clinical practice. *AJNR.*
1925 *American journal of neuroradiology* doi:10.3174/ajnr.a8728.

1926 Nagaraj, U.D., Meineke, J., Sriastwa, A., Tkach, J.A., Leach, J.L., Doneva, M.,
1927 2025b. Evaluation of synthetic images derived from a neural network in pedi-
1928 atric brain magnetic resonance imaging. *Magnetic Resonance Imaging* 121, 110427.
1929 doi:10.1016/j.mri.2025.110427.

1930 Nimje, S., Artières, T., Guye, M., de Rochefort, L., 2025. Insights on scan-specific
1931 deep-learning strategies for brain MRI parallel imaging reconstruction. *NMR in*
1932 *Biomedicine* 38, e70079. doi:10.1002/nbm.70079. pMID: 40545734.

1933 Ning, L., Laun, F., Gur, Y., DiBella, E.V.R., Deslauriers-Gauthier, S., Megherbi,
1934 T., Ghosh, A., Zucchelli, M., Menegaz, G., Fick, R., St-Jean, S., Paquette, M.,
1935 Aranda, R., Descoteaux, M., Deriche, R., O'Donnell, L., Rathi, Y., 2015. Sparse
1936 reconstruction challenge for diffusion MRI: Validation on a physical phantom to

1937 determine which acquisition scheme and analysis method to use? Medical image
1938 analysis doi:10.1016/j.media.2015.10.012. PMID: 26606457.

1939 Nishioka, N., Shimizu, Y., Kaneko, Y., Shirai, T., Suzuki, A., Amemiya, T., Ochi,
1940 H., Bito, Y., Takizawa, M., Ikebe, Y., Kameda, H., Harada, T., Fujima, N., Kudo,
1941 K., 2025. Accelerating FLAIR imaging via deep learning reconstruction: potential
1942 for evaluating white matter hyperintensities. Japanese journal of radiology doi:10.
1943 1007/s11604-024-01666-5. PMID: 39316286.

1944 Nonninger, J.N., Kienast, P., Pogledic, I., Mallouhi, A., Barkhof, F., Tractnig, S.,
1945 Robinson, S.D., Kasprian, G., Haider, L., 2025. Assessment of AI-accelerated t2-
1946 weighted brain MRI, based on clinical ratings and image quality evaluation. Euro-
1947 pean journal of radiology doi:10.1016/j.ejrad.2025.112123. PMID: 40315626.

1948 Okell, T.W., Woods, J.G., Chiew, M., 2026. Combined angiography and perfusion
1949 using radial imaging and arterial spin labeling with structural contrast. Magnetic
1950 resonance in medicine doi:10.1002/mrm.70073. PMID: 40955094.

1951 Olsson, H., Millward, J.M., Starke, L., Gladysz, T., Klein, T., Fehr, J., Lai, W.C.,
1952 Lippert, C., Niendorf, T., Waiczies, S., 2024. Simulating rigid head motion artifacts
1953 on brain magnitude MRI data-outcome on image quality and segmentation of the
1954 cerebral cortex. PloS one doi:10.1371/journal.pone.0301132. PMID: 38626138.

1955 Ortiz-Gonzalez, A., Kobler, E., Schletter, L., Effland, A., 2026. Motiondps: Motion-
1956 compensated 3D brain MRI reconstruction. IEEE Transactions on Medical Imaging
1957 doi:10.1109/tmi.2026.3705847.

1958 Osadebey, M.E., Pedersen, M., Arnold, D.L., Wendel-Mitoraj, K.E., 2018. Blind blur
1959 assessment of MRI images using parallel multiscale difference of gaussian filters.
1960 Biomedical engineering online doi:10.1186/s12938-018-0514-4. PMID: 29898715.

1961 Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow,
1962 C.D., et al., 2021. The PRISMA 2020 statement: An updated guideline for reporting
1963 systematic reviews. BMJ 372, n71.

1964 Park, S., Beckett, A., Häkkinen, S., Walker, E., Ma, S.J., Kim, S., Kim, H., Feinberg,
1965 D.A., 2025a. Higher spatial resolution and sensitivity in whole brain functional MRI
1966 at 7t using 3D epi accelerated with variable density caipi sampling and temporal
1967 random walk. Magnetic resonance in medicine doi:10.1002/mrm.30512.

1968 Park, Y.W., Yoo, R.E., Shin, I., Jeon, Y.H., Singh, K.P., Lee, M.D., Kim, S., Yang,
1969 K., Jeong, G., Ryu, L., Han, K., Ahn, S.S., Lee, S.K., Jain, R., Choi, S.H., 2025b. A
1970 multinational study of deep learning-based image enhancement for multiparametric
1971 glioma MRI. Scientific Reports doi:10.1038/s41598-025-17993-0.

1972 Pawar, K., Chen, Z., Shah, N.J., Egan, G.F., 2022. Suppressing motion artefacts
1973 in MRI using an inception-resnet network with motion simulation augmentation.
1974 NMR in biomedicine doi:10.1002/nbm.4225. PMID: 31865624.

1975 Pawlowski, N., Castro, D.C., Glocker, B., 2020. Deep structural causal models for
1976 medical imaging. Advances in Neural Information Processing Systems (NeurIPS)
1977 33, 15061–15073.

1978 Peng, W., Xia, T., Ribeiro, F.D.S., Bosschieter, T., Adeli, E., Zhao, Q., Glocker, B.,
1979 Pohl, K.M., 2026. Latent causal modeling for 3D brain MRI counterfactuals. Deep
1980 Generative Models doi:10.1007/978-3-032-05472-2_19.

1981 Pirkel, C.M., Cencini, M., Kurzawski, J.W., Waldmannstetter, D., Li, H., Sekuboyina,
1982 A., Endt, S., Peretti, L., Donatelli, G., Pasquariello, R., Costagli, M., Buonincontri,
1983 G., Tosetti, M., Menzel, M.I., Menze, B.H., 2022. Learning residual motion
1984 correction for fast and robust 3D multiparametric MRI. Medical image analysis
1985 doi:10.1016/j.media.2022.102387. pMID: 35180675.

1986 Pizarro, R., Assemblal, H.E., Jegathambal, S.K.B., Jubault, T., Antel, S., Arnold, D.,
1987 Shmuel, A., 2023. Deep learning, data ramping, and uncertainty estimation for
1988 detecting artifacts in large, imbalanced databases of MRI images. Medical image
1989 analysis doi:10.1016/j.media.2023.102942. pMID: 37797482.

1990 Pruessmann, K.P., Weiger, M., Scheidegger, M.B., Boesiger, P., 1999. SENSE: Sensi-
1991 tivity encoding for fast MRI. Magnetic Resonance in Medicine 42, 952–962.

1992 Qiao, Y., Shi, Y., 2022. Unsupervised deep learning for fod-based susceptibility
1993 distortion correction in diffusion MRI. IEEE transactions on medical imaging
1994 doi:10.1109/tmi.2021.3134496. pMID: 34882551.

1995 Qiu, Z., Hu, S., Zhao, W., Sakaie, K., Sun, J.E.P., Griswold, M.A., Jones, D.K.,
1996 Ma, D., 2024. Self-calibrated subspace reconstruction for multidimensional MR
1997 fingerprinting for simultaneous relaxation and diffusion quantification. Magnetic
1998 resonance in medicine doi:10.1002/mrm.29969. pMID: 38102776.

1999 Radmanesh, A., Muckley, M.J., Murrell, T., Lindsey, E., Sriram, A., Knoll, F., Sod-
2000 ickson, D.K., Lui, Y.W., 2022. Exploring the acceleration limits of deep learning
2001 variational network-based two-dimensional brain MRI. Radiology. Artificial intelli-
2002 gence doi:10.1148/ryai.210313. pMID: 36523647.

2003 Ramos-Llordén, G., Lobos, R.A., Kim, T.H., Tian, Q., Witzel, T., Lee, H.H., Scholz,
2004 A., Keil, B., Yendiki, A., Bilgiç, B., Halder, J.P., Huang, S.Y., 2023. High-fidelity,
2005 high-spatial-resolution diffusion magnetic resonance imaging of ex vivo whole human
2006 brain at ultra-high gradient strength with structured low-rank echo-planar imaging
2007 ghost correction. NMR in biomedicine doi:10.1002/nbm.4831. pMID: 36106429.

2008 Rassmann, S., Kugler, D., Ewert, C., Reuter, M., 2026. Regression is all you need for
2009 medical image translation. IEEE transactions on medical imaging doi:10.1109/tm
2010 i.2025.3650412. pMID: 41477795.

2011 Rauland, A., Meisler, S.L., Alexander-Bloch, A.F., Bagautdinova, J., Baller, E.B.,
2012 Gur, R.E., Gur, R.C., Luo, A.C., Moore, T.M., Popovych, O.V., Reetz, K., Roalf,
2013 D.R., Shinohara, R.T., Sotardi, S., Sydnor, V.J., Vossough, A., Eickhoff, S.B.,
2014 Cieslak, M., Satterthwaite, T.D., 2025. White matter bundle reconstruction from
2015 single-shell diffusion magnetic resonance imaging: Test-retest reliability and pre-
2016 dictive capability across orientation distribution function reconstruction methods.
2017 Human brain mapping doi:10.1002/hbm.70429. pMID: 41376618.

2018 Raymond, C., Yao, J., Clifford, B., Feiweier, T., Oshima, S., Telesca, D., Zhong, X.,
2019 Meyer, H., Everson, R.G., Salamon, N., Cloughesy, T.F., Ellingson, B.M., 2025.

2020 Leveraging physics-based synthetic MR images and deep transfer learning for arti-
2021 fact reduction in echo-planar imaging. *AJNR. American journal of neuroradiology*
2022 doi:10.3174/ajnr.a8566. PMID: 39947682.

2023 Rezvani, S., Moazami, S., Azevedo, C.J., Oberai, A.A., 2025. Probabilistic brain
2024 MR image transformation using generative models. *Scientific Reports* doi:10.1038/
2025 s41598-025-31958-3.

2026 Risk, B.B., Kociuba, M.C., Rowe, D.B., 2018. Impacts of simultaneous multislice
2027 acquisition on sensitivity and specificity in fMRI. *NeuroImage* doi:10.1016/j.neur
2028 oimage.2018.01.078. PMID: 29408461.

2029 Rizzuti, G., Schakel, T., Huttinga, N.R.F., Dankbaar, J.W., Leeuwen, T.v., Sbrizzi,
2030 A., 2024. Towards retrospective motion correction and reconstruction for clinical 3D
2031 brain MRI protocols with a reference contrast. *Magma (New York, N.Y.)* doi:10.
2032 1007/s10334-024-01161-y. PMID: 38758490.

2033 Ryu, K., Nam, Y., Gho, S.M., Jang, J., Lee, H.J., Cha, J., Baek, H.J., Park, J., Kim,
2034 D.H., 2019. Data-driven synthetic MRI FLAIR artifact correction via deep neural
2035 network. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri.26712.

2036 Saeed, S., Nickel, D., Weiland, E., Frohwein, L.J., Niehoff, J.H., Reiman, G., Schreck,
2037 J., Haag, N., Boriesosdick, J., Schoenbeck, D., Katz, M., Shahzadi, I., Woeltjen,
2038 M.M., Kroeger, J.R., Borggreffe, J., Moenninghoff, C., 2026. Improved image quality
2039 and reduced acquisition time in brain MRI using deep learning-based reconstruction:
2040 A quantitative and subjective assessment compared to standard mprage in 0.55 t
2041 MRI. *Magnetic resonance imaging* doi:10.1016/j.mri.2026.110656.

2042 Safari, M., Wang, S., Eidex, Z., Qiu, R.L.J., Chang, C.W., Yu, D.S., Yang, X., 2025a.
2043 A physics-informed deep learning model for MRI brain motion correction. *Medical*
2044 *physics* doi:10.1002/mp.70197. PMID: 41395855.

2045 Safari, M., Wang, S., Li, Q., Eidex, Z., Qiu, R.L.J., Chang, C.W., Mao, H., Yang, X.,
2046 2025b. Res-mocodiff: residual-guided diffusion models for motion artifact correction
2047 in brain MRI. *Physics in medicine and biology* doi:10.1088/1361-6560/ae1110.
2048 PMID: 41061726.

2049 Sagawa, H., Fushimi, Y., Nakajima, S., Fujimoto, K., Miyake, K.K., Numamoto,
2050 H., Koizumi, K., Nambu, M., Kataoka, H., Nakamoto, Y., Saga, T., 2021. Deep
2051 learning-based noise reduction for fast volume diffusion tensor imaging: Assessing
2052 the noise reduction effect and reliability of diffusion metrics. *Magnetic resonance*
2053 *in medical sciences : MRMS : an official journal of Japan Society of Magnetic*
2054 *Resonance in Medicine* doi:10.2463/mrms.tn.2020-0061. PMID: 32963184.

2055 Sakitis, C.J., Rowe, D.B., 2025. Bayesian merged utilization of GRAPPA and SENSE
2056 (bmugs) for in-plane accelerated reconstruction increases fMRI detection power.
2057 *Magnetic resonance imaging* doi:10.1016/j.mri.2024.110252.

2058 Sanchez, P., Kascenas, A., Liu, X., O'Neil, A.Q., Tsiftaris, S.A., 2022. What is
2059 healthy? generative counterfactual diffusion for lesion localization, in: *Deep Gener-*
2060 *ative Models (DGM4MICCAI), MICCAI Workshop, Springer.* pp. 34-44.

2061 Sanchez, T., Zalevskyi, V., Mihailov, A., Martí Juan, G., Eixarch, E., Jakab, A.,
2062 Dunet, V., Koob, M., Auzias, G., Bach Cuadra, M., 2026. Automatic quality
2063 control in multi-centric fetal brain MRI super-resolution reconstruction. *Perinatal,*
2064 *Preterm and Paediatric Image Analysis* doi:10.1007/978-3-032-05997-0_1.

2065 Sannananja, B., Zhu, C., Colip, C.G., Somasundaram, A., Ibrahim, M., Khrisat, T.,
2066 Mossa-Basha, M., 2022. Image-quality assessment of 3D intracranial vessel wall MRI
2067 using dante or dante-caipi for blood suppression and imaging acceleration. *AJNR.*
2068 *American journal of neuroradiology* doi:10.3174/ajnr.a7531. PMID: 35618420.

2069 Schauman, S., Niekerk, A.v., Norbeck, O., Rydén, H., Avventi, E., Skare, S., 2026. An
2070 exploration of motion-sampling interactions in 3D MRI for neuroimaging. *Magnetic*
2071 *resonance in medicine* doi:10.1002/mrm.70132. PMID: 41204061.

2072 Schlemper, J., Caballero, J., Hajnal, J.V., Price, A.N., Rueckert, D., 2017. A deep
2073 cascade of convolutional neural networks for dynamic MR image reconstruction.
2074 *IEEE Transactions on Medical Imaging* 37, 491–503.

2075 Shan, S., Gao, Y., Liu, P.Z.Y., Whelan, B., Sun, H., Dong, B., Liu, F., Wadding-
2076 ton, D.E.J., 2023. Distortion-corrected image reconstruction with deep learning
2077 on an MRI-linac. *Magnetic resonance in medicine* doi:10.1002/mrm.29684. PMID:
2078 37125656.

2079 Shan, S., Gao, Y., Waddington, D., Chen, H., Whelan, B., Liu, P., Wang, Y., Liu,
2080 C., Gan, H., Gao, M., Liu, F., 2024. Image reconstruction with b² inhomogeneity
2081 using a deep unrolled network on an open-bore MRI-linac. *IEEE Transactions on*
2082 *Instrumentation and Measurement* 73, 1–9. doi:10.1109/tim.2024.3481545.

2083 Sheng, H., Wang, X., Jiang, M., Zhang, Z., 2022. Deep learning-based diffusion-
2084 weighted magnetic resonance imaging in the diagnosis of ischemic penumbra in
2085 early cerebral infarction. *Contrast media & molecular imaging* doi:10.1155/2022/
2086 6270700. PMID: 35291425.

2087 Shimron, E., Tamir, J.I., Wang, K., Lustig, M., 2022. Implicit data crimes: Ma-
2088 chine learning bias arising from misuse of public data. *Proceedings of the National*
2089 *Academy of Sciences* 119, e2117203119. doi:10.1073/pnas.2117203119.

2090 Simkó, A., Ruiter, S., Löfstedt, T., Garpebring, A., Nyholm, T., Bylund, M., Jonsson,
2091 J., 2023. Improving MR image quality with a multi-task model, using convolutional
2092 losses. *BMC medical imaging* doi:10.1186/s12880-023-01109-z. PMID: 37784039.

2093 Singh, M., Mahmud, S.Z., Yedavalli, V., Zhou, J., Kamson, D.O., Zijl, P.v., Heo,
2094 H.Y., 2025. Learning-based motion artifact correction in the z-spectral domain
2095 for chemical exchange saturation transfer MRI. *Magnetic resonance in medicine*
2096 doi:10.1002/mrm.30440. PMID: 39834104.

2097 Solak, M., Tören, M., Asan, B., Kaba, E., Beyazal, M., Çeliker, F.B., 2026. Braingan
2098 framework for generating synthetic contrast-enhanced multisequence MRI in brain
2099 metastases. *Journal of Imaging Informatics in Medicine* doi:10.1007/s10278-026-
2100 02059-y.

2101 Sommer, K., Saalbach, A., Brosch, T., Hall, C., Cross, N.M., Andre, J.B., 2020.
2102 Correction of motion artifacts using a multiscale fully convolutional neural net-
2103 work. *AJNR. American journal of neuroradiology* doi:10.3174/ajnr.a6436. PMID:
2104 32054615.

2105 Song, W., Zeng, C., Zhang, X., Wang, Z., Huang, Y., Lin, J., Wei, W., Qu, X.,
2106 2022. Jointly estimating bias field and reconstructing uniform MRI image by deep
2107 learning. *Journal of Magnetic Resonance* 343, 107301. doi:10.1016/j.jmr.2022.
2108 107301.

2109 Song, Y., Sohl-Dickstein, J., Kingma, D.P., Kumar, A., Ermon, S., Poole, B., 2021.
2110 Score-based generative modeling through stochastic differential equations, in: In-
2111 ternational Conference on Learning Representations (ICLR).

2112 Starck, S., Sideri-Lampretsa, V., Kainz, B., Menten, M.J., Mueller, T.T., Rueckert,
2113 D., 2026. Diff-def: Diffusion-generated deformation fields for conditional atlases.
2114 *IEEE transactions on medical imaging* doi:10.1109/tmi.2025.3595421.

2115 Sterne, J.A.C., Hernán, M.A., Reeves, B.C., Savović, J., Berkman, N.D., Viswanathan,
2116 M., Henry, D., White, I.R., Higgins, J.P.T., 2016. Robins-i: A tool for assessing
2117 risk of bias in non-randomised studies of interventions. *BMJ* 355, i4919.

2118 Sui, Y., Afacan, O., Jaimes, C., Gholipour, A., Warfield, S.K., 2022. Scan-specific gen-
2119 erative neural network for MRI super-resolution reconstruction. *IEEE Transactions*
2120 *on Medical Imaging* doi:10.1109/tmi.2022.3142610.

2121 Sujit, S.J., Coronado, I., Kamali, A., Narayana, P.A., Gabr, R.E., 2019. Automated
2122 image quality evaluation of structural brain MRI using an ensemble of deep learning
2123 networks. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri.26693.
2124 PMID: 30811739.

2125 Tabari, A., Lang, M., Awan, K., Liu, W., Clifford, B., Lo, W.C., Splitthoff, D.N.,
2126 Cauley, S., Rapalino, O., Schaefer, P., Huang, S.Y., Conklin, J., 2023. Optimized
2127 flow compensation for contrast-enhanced t1-weighted wave-caipi 3D mprage imaging
2128 of the brain. *European radiology experimental* doi:10.1186/s41747-023-00351-y.
2129 PMID: 37394534.

2130 Tabari, A., Vejdani-Jahromi, M., Lang, M., Nickel, D., Lo, W.C., Clifford, B., Con-
2131 klin, J., Huang, S.Y., 2026. Clinical validation of deep learning-accelerated versus
2132 wave-CAIPI postcontrast 3d T1-MPRAGE for evaluation of intracranial enhancing
2133 lesions. *American Journal of Neuroradiology* 47, 394–400. doi:10.3174/ajnr.a8992.

2134 Takada, S., Nakaura, T., Yoshida, N., Uetani, H., Shiraishi, K., Kobayashi, N., Mat-
2135 suo, K., Morita, K., Nagayama, Y., Kidoh, M., Yamashita, Y., Takayanagi, R.,
2136 Hirai, T., 2025. Impact of super-resolution deep learning-based reconstruction for
2137 hippocampal MRI: A volunteer and phantom study. *European Journal of Radiology*
2138 191, 112289. doi:10.1016/j.ejrad.2025.112289.

2139 Tang, C., Eisenmenger, L.B., Rivera-Rivera, L., Huo, E., Junn, J.C., Kuner, A.D.,
2140 Oechtering, T.H., Peret, A., Starekova, J., Johnson, K.M., 2025. Incorporating
2141 radiologist knowledge into MRI quality metrics for machine learning using rank-
2142 based ratings. *Journal of magnetic resonance imaging : JMRI* doi:10.1002/jmri
2143 .29672. PMID: 39690114.

2144 Tang, H., Hu, N., Yuan, Y., Xia, C., Liu, X., Zuo, P., Stalder, A.F., Schmidt, M., Zhou,
2145 X., Song, B., Sun, J., 2019. Accelerated time-of-flight magnetic resonance angiog-
2146 raphy with sparse undersampling and iterative reconstruction for the evaluation
2147 of intracranial arteries. Korean journal of radiology doi:10.3348/kjr.2017.0634.
2148 pMID: 30672166.

2149 Tanno, R., Worrall, D.E., Kaden, E., Ghosh, A., Grussu, F., Bizzi, A., Sotiropoulos,
2150 S.N., Criminisi, A., Alexander, D.C., 2021. Uncertainty modelling in deep learning
2151 for safer neuroimage enhancement: Demonstration in diffusion MRI. NeuroImage
2152 doi:10.1016/j.neuroimage.2020.117366.

2153 Taymourtash, A., Schwartz, E., Nenning, K.H., Licandro, R., Kienast, P., Hiele,
2154 V., Prayer, D., Kasprian, G., Langs, G., 2025. Measuring the effects of motion
2155 corruption in fetal fMRI. Human brain mapping doi:10.1002/hbm.26806. pMID:
2156 39846325.

2157 Tessema, A.W., Jin, S., Cho, H., Han, S., Cho, H., 2026. Technical feasibility of
2158 an efficient deep learning-based super-resolution method for 3D MR angiography.
2159 IEEE Access doi:10.1109/access.2026.3712275.

2160 Thaler, C., Sedlacik, J., Forkert, N.D., Stellmann, J.P., Schön, G., Fiehler, J., Gel-
2161 lißen, S., 2023. Effect of geometric distortion correction on thickness and volume
2162 measurements of cortical parcellations in 3D t1w gradient echo sequences. PloS one
2163 doi:10.1371/journal.pone.0284440. pMID: 37058493.

2164 Tisdall, M.D., Atkins, M.S., 2006. Using human and model performance to com-
2165 pare MRI reconstructions. IEEE transactions on medical imaging doi:10.1109/tm
2166 i.2006.881374. pMID: 17117780.

2167 Trivedi, V.K., Sharma, B., Balamurugan, P., 2024. Mcddpm: Multichannel conditional
2168 denoising diffusion model for unsupervised anomaly detection in brain MRI. 2024
2169 17th International Congress on Image and Signal Processing, BioMedical Engineer-
2170 ing and Informatics (CISP-BMEI) doi:10.1109/cisp-bmei64163.2024.10906217.

2171 Uher, D., Drenthen, G.S., Poser, B.A., Hofman, P.A.M., Wagner, L.G., Lanen,
2172 R.H.G.J.v., Hoeberigs, C.M., Colon, A.J., Schijns, O.E.M.G., Jansen, J.F.A.,
2173 Backes, W.H., 2024. Deepflair: A neural network approach to mitigate signal and
2174 contrast loss in temporal lobes at 7 tesla FLAIR images. Magnetic resonance imag-
2175 ing doi:10.1016/j.mri.2024.04.013. pMID: 38621552.

2176 U.S. Food and Drug Administration, Health Canada, Medicines and Healthcare
2177 products Regulatory Agency, 2021. Good machine learning practice for medical
2178 device development: Guiding principles. Joint guiding principles, October 2021.
2179 URL: [https://www.fda.gov/medical-devices/software-medical-device-samd](https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles)
2180 [/good-machine-learning-practice-medical-device-development-guiding-pri](https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles)
2181 [nciples](https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles).

2182 Usui, K., Muro, I., Shibukawa, S., Goto, M., Ogawa, K., Sakano, Y., Kyogoku, S.,
2183 Daida, H., 2023. Evaluation of motion artefact reduction depending on the artefacts'
2184 directions in head MRI using conditional generative adversarial networks. Scientific
2185 reports doi:10.1038/s41598-023-35794-1. pMID: 37237139.

2186 Vakli, P., Weiss, B., Szalma, J., Barsi, P., Gyuricza, I., Kemenczky, P., Somogyi, E.,
2187 Ádám Nárai, Gál, V., Hermann, P., Vidnyánszky, Z., 2023. Automatic brain MRI
2188 motion artifact detection based on end-to-end deep learning is similarly effective
2189 as traditional machine learning trained on image quality metrics. Medical image
2190 analysis doi:10.1016/j.media.2023.102850.

2191 Vanderhasselt, T., Naeyaert, M., Watté, N., Allemeersch, G., Raeymaeckers, S.,
2192 Dudink, J., Mey, J.d., Raeymaekers, H., 2020. Synthetic MRI of preterm in-
2193 fants at term-equivalent age: Evaluation of diagnostic image quality and auto-
2194 mated brain volume segmentation. AJNR. American journal of neuroradiology
2195 doi:10.3174/ajnr.a6533. PMID: 32299803.

2196 Verclytte, S., Beaugrard, G., Nickel, D., Muñoz-Ramírez, V., Norberciak, L., Boissel, M., Chaton, V., Kwiatkowski, A., 2026. Deep learning-accelerated 3d FLAIR enables reliable MS lesion detection. American Journal of Neuroradiology 47, 1560–1568. doi:10.3174/ajnr.a9140.

2200 Verdicchio, M., Grassi, F., Esposito, G., Cavaliere, C., Alfano, V., Cerrone, M., Pappatà, S., Salvatore, E., Cecca, A.D., Salvatore, M., Aiello, M., 2025. Reliability assessment of accelerated MR sequences for the estimation of structural and vascular markers in dementia. Scientific reports doi:10.1038/s41598-025-87224-z. PMID: 40240759.

2205 Vranic, J.E., Cross, N.M., Wang, Y., Hippe, D.S., Weerd, E.d., Mossa-Basha, M., 2019. Compressed sensing-sensitivity encoding (cs-SENSE) accelerated brain imaging: Reduced scan time without reduced image quality. AJNR. American journal of neuroradiology doi:10.3174/ajnr.a5905. PMID: 30523142.

2209 Wada, A., Akashi, T., Hagiwara, A., Nishizawa, M., Shimoji, K., Kikuta, J., Maekawa, T., Sano, K., Kamagata, K., Nakanishi, A., Aoki, S., 2024. Deep learning-driven transformation: A novel approach for mitigating batch effects in diffusion MRI beyond traditional harmonization. Journal of magnetic resonance imaging : JMIR doi:10.1002/jmri.29088.

2214 Wang, C.J., Rost, N.S., Golland, P., 2023. Spatial-intensity transforms for medical image-to-image translation. IEEE transactions on medical imaging doi:10.1109/tmi.2023.3283948. PMID: 37285247.

2217 Wang, G., Gong, E., Banerjee, S., Martin, D., Tong, E., Choi, J., Chen, H., Wintermark, M., Pauly, J.M., Zaharchuk, G., 2020. Synthesize high-quality multi-contrast magnetic resonance imaging from multi-echo acquisition using multi-task deep generative model. IEEE Transactions on Medical Imaging 39, 3089–3099. doi:10.1109/tmi.2020.2987026.

2222 Wang, N., Liao, C., Cao, X., Nishimura, M., Brackner, Y.W.E., Yurt, M., Gao, M., Abraham, D., Alkan, C., Iyer, S.S., Zhou, Z., Jeong, H., Kerr, A., Haldar, J.P., Setsompop, K., 2025a. Spherical echo-planar time-resolved imaging (septi) for rapid 3D quantitative t_2^* and susceptibility imaging. Magnetic resonance in medicine doi:10.1002/mrm.30255. PMID: 39250435.

2227 Wang, X., Cabezas, M., Tang, Z., D'Souza, A., Liu, D., Wang, C., Cai, W., 2025b. Estimating uncertainty on deep learning-driven fibre orientation distribution enhancement, in: ICONIP 2024, CCIS 2285. doi:10.1007/978-981-96-6957-8/25.

2230 Wang, X., Fan, H., Tan, Z., Vasylechko, S., Yang, E., Didier, R., Afacan, O., Uecker,
2231 M., Warfield, S.K., Gholipour, A., 2025c. Rapid, high-resolution and distortion-free
2232 R_2^* mapping of fetal brain using multi-echo radial flash and model-based reconstruction.
2233 Magnetic resonance in medicine doi:10.1002/mrm.30604. pMID: 40534142.

2234 Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P., 2004. Image quality assessment:
2235 from error visibility to structural similarity. IEEE Transactions on Image Processing
2236 13, 600–612. doi:10.1109/TIP.2003.819861.

2237 Wang, Z., Ramasawmy, R., Javed, A., Mugler, J.P., Meyer, C.H., Campbell-Washburn,
2238 A.E., 2025d. Variable-flip-angle 3D spiral-in-out turbo spin-echo imaging using con-
2239 comitant gradient compensation and echo reordering at 0.55 t. Magnetic resonance
2240 in medicine doi:10.1002/mrm.30380. pMID: 39567364.

2241 Wen, Y., Ma, H., Xiang, S., Feng, Z., Guan, C., Li, X., 2025. Accelerating brain t2-
2242 weighted imaging using artificial intelligence-assisted compressed sensing combined
2243 with deep learning-based reconstruction: a feasibility study at 5.0t MRI. BMC
2244 medical imaging doi:10.1186/s12880-025-01763-5. pMID: 40596927.

2245 Whiting, P.F., Rutjes, A.W.S., Westwood, M.E., Mallett, S., Deeks, J.J., Reitsma,
2246 J.B., Leeflang, M.M.G., Sterne, J.A.C., Bossuyt, P.M.M., 2011. QUADAS-2: A
2247 revised tool for the quality assessment of diagnostic accuracy studies. Annals of
2248 Internal Medicine 155, 529–536.

2249 Wichtmann, B.D., Katemann, C., Kadrija, M., Layer, Y.C., Bischoff, L.M., Scheuver,
2250 Y., Mezger, M., Weber, O.M., Luetkens, J.A., Attenberger, U.I., Radbruch, A.,
2251 Paech, D., 2025. Rapid and motion-robust pediatric brain imaging: T2-weighted
2252 turbo-spin-echo propeller acquisition with compressed sensing. Pediatric radiology
2253 doi:10.1007/s00247-024-06088-z. pMID: 39589491.

2254 Wismüller, A., Vosoughi, M.A., 2022. Self-supervised learning in medical imaging:
2255 anomaly detection in MRI using autoencoders, in: Real-Time Image Processing
2256 and Deep Learning 2022, p. 9. doi:10.1117/12.2622233.

2257 Wu, F., Luo, H., Wang, X., Yang, Q., Zhuang, Y., Lin, L., Dong, Y., Tulupov, A.,
2258 Zhang, Y., Cai, S., Chen, Z., Cai, C., Bao, J., Cheng, J., 2025a. Application of
2259 anti-motion ultra-fast quantitative MRI in neurological disorder imaging: Insights
2260 from huntington's disease. Journal of magnetic resonance imaging : JMRI doi:10.
2261 1002/jmri.29682.

2262 Wu, J., Chen, L., Li, Z., Li, X., Sun, T., Wang, L., Wang, R., Wei, H., Zhang,
2263 Y., 2025b. 3d isotropic high-resolution fetal brain MRI reconstruction from motion
2264 corrupted thick data based on physical-informed unsupervised learning. IEEE Jour-
2265 nal of Biomedical and Health Informatics 29, 8256–8267. doi:10.1109/jbhi.2025.
2266 3586049.

2267 Xia, L., Qi, X., Liu, J., Yuan, Y., Wang, Y., Xu, N., Zhang, J., 2026. Accurate multi-b-
2268 value DWI generation using two-stage deep learning: multicenter study. European
2269 Journal of Radiology 194, 112497. doi:10.1016/j.ejrad.2025.112497.

2270 Xie, K., Gao, L., Zhang, H., Zhang, S., Xi, Q., Zhang, F., Sun, J., Lin, T., Sui, J., Ni,
2271 X., 2024. GAN-based metal artifacts region inpainting in brain MRI imaging with
2272 reflective registration. Medical physics doi:10.1002/mp.16724.

2273 Xiong, Z., Gao, Y., Liu, F., Emery, D., Butcher, K., Wilman, A.H., Sun, H., 2026.
2274 Advancing quantitative susceptibility mapping with 2.5d diffusion models for rapid
2275 intracranial hemorrhage quantification. *Magnetic resonance in medicine* doi:10.
2276 1002/mrm.70358. pMID: 41873533.

2277 Xu, J., Zu, T., Hsu, Y.C., Wang, X., Chan, K.W.Y., Zhang, Y., 2025. DEISM: Deep
2278 reconstruction framework with self-calibration mechanisms for accelerated chemical
2279 exchange saturation transfer imaging. *IEEE Transactions on Biomedical Engineering*
2280 72, 2413–2424. doi:10.1109/tbme.2025.3543403.

2281 Yang, G., Yu, S., Dong, H., Slabaugh, G., Dragotti, P.L., Ye, X., Liu, F., Arridge, S.,
2282 Keegan, J., Guo, Y., Firmin, D., 2018. DAGAN: Deep de-aliasing generative adver-
2283 sarial networks for fast compressed sensing MRI reconstruction. *IEEE Transactions*
2284 *on Medical Imaging* 37, 1310–1321.

2285 Yang, Q., Lin, Y., Wang, J., Bao, J., Wang, X., Ma, L., Zhou, Z., Yang, Q., Cai,
2286 S., He, H., Cai, C., Dong, J., Cheng, J., Chen, Z., Zhong, J., 2022. Model-based
2287 synthetic data-driven learning (most-dl): Application in single-shot t2 mapping
2288 with severe head motion using overlapping-echo acquisition. *IEEE transactions on*
2289 *medical imaging* doi:10.1109/tmi.2022.3179981. pMID: 35657830.

2290 Yarach, U., Chatnuntawech, I., Liao, C., Teerapittayanon, S., Iyer, S.S., Kim, T.H.,
2291 Haldar, J., Cho, J., Bilgic, B., Hu, Y., Hargreaves, B., Setsompop, K., 2025.
2292 Blip-up blip-down circular epi (buda-cepi) for distortion-free dmri with rapid un-
2293 rolled deep learning reconstruction. *Magnetic resonance imaging* doi:10.1016/j.mri.
2294 2024.110277. pMID: 39566835.

2295 Yarach, U., Chatnuntawech, I., Setsompop, K., Suwannasak, A., Angkurawaranon,
2296 S., Madla, C., Hanprasertpong, C., Sangpin, P., 2024. Improved reconstruction
2297 for highly accelerated propeller diffusion 1.5t clinical MRI. *Magnetic Resonance*
2298 *Materials in Physics, Biology and Medicine* 37, 283–294. doi:10.1007/s10334-023-
2299 01142-7.

2300 Yasaka, K., Akai, H., Kato, S., Tajima, T., Yoshioka, N., Furuta, T., Kageyama, H.,
2301 Toda, Y., Akahane, M., Ohtomo, K., Abe, O., Kiryu, S., 2024. Iterative motion
2302 correction technique with deep learning reconstruction for brain MRI: A volunteer
2303 and patient study. *Journal of imaging informatics in medicine* doi:10.1007/s10278-
2304 024-01184-w. pMID: 38942939.

2305 Yasaka, K., Akai, H., Sugawara, H., Tajima, T., Akahane, M., Yoshioka, N., Kabasawa,
2306 H., Miyo, R., Ohtomo, K., Abe, O., Kiryu, S., 2022. Impact of deep learning recon-
2307 struction on intracranial 1.5 t magnetic resonance angiography. *Japanese journal of*
2308 *radiology* doi:10.1007/s11604-021-01225-2. pMID: 34851499.

2309 Yilmaz, K., Çukur, T., Patel, V.M., 2023. Self-supervised MRI reconstruction with
2310 unrolled diffusion models. *Medical Image Computing and Computer Assisted Inter-*
2311 *vention – MICCAI 2023* doi:10.1007/978-3-031-43999-5_47.

2312 Yoo, H., Moon, H.E., Kim, S., Kim, D.H., Choi, Y.H., Cheon, J.E., Lee, J.S., Lee,
2313 S., 2025. Evaluation of image quality and scan time efficiency in accelerated 3D
2314 t1-weighted pediatric brain MRI using deep learning-based reconstruction. *Korean*
2315 *journal of radiology* doi:10.3348/kjr.2024.0701. pMID: 39898398.

2316 Yoshida, N., Kageyama, H., Akai, H., Sasaki, K., Sakurai, N., Koori, N., Yamamoto,
2317 S., Kodama, N., 2026. Deep learning approach to super-resolution correction of
2318 brain MRI motion artifacts for accurate hippocampal volumetry. Scientific reports
2319 doi:10.1038/s41598-026-44834-5. pMID: 41866419.

2320 Yoshida, N., Kageyama, H., Akai, H., Yasaka, K., Sugawara, H., Okada, Y., Kuni-
2321 matsu, A., 2022. Motion correction in MR image for analysis of vsrad using gener-
2322 ative adversarial network. PloS one doi:10.1371/journal.pone.0274576. pMID:
2323 36103561.

2324 You, S., Gondova, A., Izaguirre, C.S.A., Milo, G.T., Jeong, S., Lee, H.J., Tarui, T.,
2325 Rollins, C.K., Yun, H.J., Grant, P.E., Im, K., 2025. Conditional deep generative
2326 normative modeling for structural and developmental anomaly detection in the fetal
2327 brain. NeuroImage doi:10.1016/j.neuroimage.2025.121442. pMID: 40912664.

2328 Yu, S., Dai, G., Wang, Z., Li, L., Wei, X., Xie, Y., 2018. A consistency evaluation of
2329 signal-to-noise ratio in the quality assessment of human brain magnetic resonance
2330 images. BMC medical imaging doi:10.1186/s12880-018-0256-6. pMID: 29769079.

2331 Yuan, S., Liu, Y., Wei, R., Zhu, J., Men, K., Dai, J., 2024. A novel loss function to
2332 reproduce texture features for deep learning-based MRI-to-CT synthesis. Medical
2333 physics doi:10.1002/mp.16850.

2334 Zahneisen, B., Keating, B., Singh, A., Herbst, M., Ernst, T., 2016. Reverse retrospec-
2335 tive motion correction. Magnetic resonance in medicine doi:10.1002/mrm.25830.
2336 pMID: 26140504.

2337 Zaitsev, M., Maclaren, J., Herbst, M., 2015. Motion artifacts in MRI: A review. NMR
2338 in Biomedicine 28, 911–935.

2339 Zhang, C., Karkalousos, D., Bazin, P.L., Coolen, B.F., Vrenken, H., Sonke, J.J.,
2340 Forstmann, B.U., Poot, D.H.J., Cnaan, M.W.A., 2022. A unified model for recon-
2341 struction and r2* mapping of accelerated 7t data using the quantitative recurrent
2342 inference machine. NeuroImage doi:10.1016/j.neuroimage.2022.119680. pMID:
2343 36240989.

2344 Zhang, C., Klein, S., Cristobal-Huerta, A., Hernandez-Tamames, J.A., Poot, D.H.J.,
2345 2021. Apir4emc: Autocalibrated parallel imaging reconstruction for extended multi-
2346 contrast imaging. Magnetic resonance imaging doi:10.1016/j.mri.2021.02.002.
2347 pMID: 33592248.

2348 Zhang, J., Wu, J., Chen, S., Zhang, Z., Cai, S., Cai, C., Chen, Z., 2019a. Robust
2349 single-shot T2 mapping via multiple overlapping-echo acquisition and deep neural
2350 network. IEEE Transactions on Medical Imaging 38, 1801–1811. doi:10.1109/tmi.
2351 2019.2896085.

2352 Zhang, L., Wu, M., Wang, L., Steffens, D.C., Potter, G.G., Liu, M., 2026. Iterative
2353 learning for joint image denoising and motion artifact correction of 3D brain MRI.
2354 Medical image analysis doi:10.1016/j.media.2025.103872. pMID: 41274083.

2355 Zhang, Q., Ruan, G., Yang, W., Liu, Y., Zhao, K., Feng, Q., Chen, W., Wu, E.X.,
2356 Feng, Y., 2019b. MRI gibbs?ring artifact reduction by means of machine learning
2357 using convolutional neural networks. Magnetic Resonance in Medicine 82, 2133–
2358 2145. doi:10.1002/mrm.27894.

2359 Zhao, C., Shao, M., Carass, A., Li, H., Dewey, B.E., Ellingsen, L.M., Woo, J.,
2360 Guttman, M.A., Blitz, A.M., Stone, M., Calabresi, P.A., Halperin, H., Prince,
2361 J.L., 2019. Applications of a deep learning method for anti-aliasing and super-
2362 resolution in MRI. *Magnetic resonance imaging* doi:10.1016/j.mri.2019.05.038.
2363 pMID: 31247254.

2364 Zhao, R., Yaman, B., Zhang, Y., Stewart, R., Dixon, A., Knoll, F., Huang, Z., Lui,
2365 Y.W., Hansen, M.S., Lungren, M.P., 2022. fastMRI+, clinical pathology annota-
2366 tions for knee and brain fully sampled magnetic resonance imaging data. *Scientific*
2367 *Data* 9, 152. doi:10.1038/s41597-022-01255-z.

2368 Zhou, L., Ni, X., Kong, Y., Zeng, H., Xu, M., Zhou, J., Wang, Q., Liu, C., 2023. Miti-
2369 gating misalignment in MRI-to-CT synthesis for improved synthetic CT generation:
2370 an iterative refinement and knowledge distillation approach. *Physics in medicine*
2371 *and biology* doi:10.1088/1361-6560/ad0ddc.

2372 Zhou, X., Balachandra, A.R., Romano, M.F., Chin, S.P., Au, R., Kolachalama, V.B.,
2373 2024. Adversarial learning for MRI reconstruction and classification of cognitively
2374 impaired individuals. *IEEE Access* doi:10.1109/access.2024.3408840.

2375 Zhu, D., Xu, F., Liu, D., Qin, Q., 2025a. A straightforward approach for 3D single-
2376 shot arterial spin labeling-based brain perfusion imaging: Preventing artifacts due to
2377 signal fluctuations. *Magnetic resonance in medicine* doi:10.1002/mrm.30439. pMID:
2378 39887515.

2379 Zhu, P., Liu, C., Fu, Y., Chen, N., Qiu, A., 2025b. Cycle-conditional diffusion model
2380 for noise correction of diffusion-weighted images using unpaired data. *Medical image*
2381 *analysis* doi:10.1016/j.media.2025.103579. pMID: 40273728.

2382 Zhu, S., Jin, Y., Spears, T., Zawar, I., Fletcher, P.T., 2026. Realdeal: Enhancing
2383 realism and details in brain image generation via image-to-image diffusion models.
2384 *Deep Generative Models* doi:10.1007/978-3-032-05472-2_15.

2385 Zibetti, M.V.W., Knoll, F., Regatte, R.R., 2022. Alternating learning approach for
2386 variational networks and undersampling pattern in parallel MRI applications. *IEEE*
2387 *Transactions on Computational Imaging* doi:10.1109/tci.2022.3176129.